



SUB-ICT SIMPLIFIED

Information and Communications Technology



Pass Sub-ICT with a Distinction

FOR S5 AND S6

FIRST EDITION

BY

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CHAPTER ONE

INTRODUCTION TO COMPUTER STUDIES

Introduction to Computers (ICT)

A **Computer** is one the most important tool to be used in Day-today life. It has led to a number of changes in the modern world today.

Definition of a computer

A computer can be defined as an electronic device manipulates information or data.

OR

Is an automatic machine made up of electronic and electromechanical devices which manipulates and process data.

OR

A computer is an electronic device which can input data, process, store and output meaningful information.

Characteristics of Modern Computers

- **Speed.** Modern computers process data at fast speed. Their speed of operation is measured in Millions of Instructions Per Second (MIPS).
- **Accuracy.** Computers are capable of detecting and correcting any mistakes made. And they are capable of performing calculations to the extent of 24 decimal accuracy.
- **Storage.** Computers are capable of storing large volumes of data and information for future use.
- **Diligence.** Computers have the ability of performing the same task 'over and over' for a long period of time without getting tired or bored.
- **Artificial Intelligence.** Computers can respond to requests given to them and provide solutions due to their programmability.
- **Automatic.** Computers work automatically hence they don't need any supervision to perform programmed routines.

Data and Information

Data raw (unprocessed) facts fed/entered in the computer system. Examples of Data include: Alphabets, numbers, symbols and musical notes etc.

Information is the processed data which makes meaning to the user. Examples of Information include: Words, songs, comments etc.

Devices that aid in transforming data into information

- Keyboard for inputting data
- Digital camera for inputting data
- Microphone for inputting data
- Barcode reader for inputting data
- Central processing unit for processing data
- Speakers for output of information
- Printers for output of hard copies
- LCD monitors for outputting text, graphics & video information
- Flash memory for storage of information
- CD-R for storage of information
- Hard disks for storage of information

THE WORLD OF ICTs

Definition: ICT (Information and Communication Technology) is the integration of computers and other telecommunication facilities for the purpose of communication.

OR

ICTs (Information and Communication Technologies) are a set of technological tools and resources used to communicate, create, disseminate, store and manage information.

MAJOR REASONS FOR STUDYING ICTs IN SCHOOLS TODAY

- To encourage learners to develop ICT skills to enhance their work in a variety of subject areas.
- To help the learner develop and consolidate his/her knowledge of ICT and be aware of new and emerging technologies.
- To help the learner to develop as an independent user.
- To equip the learner with skills for lifelong learning.

MAIN COMPONENTS/ PARTS OF A COMPUTER SYSTEM

- i) **Hardware:** These are physical/ tangible parts of a computer which you can see, touch and feel. These include: Keyboard, mouse, Central Processing Unit(CPU), monitor, printer, flash disks, modems, etc
- ii) **Software:** Is a set of instructions or programs that tell the computer what to do and how to do it. It is an interface between hardware and its user. E.g. System software and Application software.

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- iii) **Human ware (User):** This refers to the skilled personnel available to perform the operation on the computer, such as secretaries, network administrators, programmers, system administrators, computer architects, etc
- iv) **Data and Communication:** This involves the use of computer networks hence enabling the sharing of information from all around the world.
- v) **Procedure:** These refer to documentation materials which contain the guide for using the hardware and software.

STAGES INVOLVED IN TRANSFORMING DATA INTO INFORMATION

- i) Collection and preparation of data
- ii) Input of data
- iii) Processing of data
- iv) Output of information

WAYS HOW ICT IS APPLIED IN DIFFERENT AREAS

i) Business

- ICTs are helpful in mobile banking such as use of mobile money services.
- It helps in efficient management and running of businesses. Activities such as data management, data processing are enabled.
- ICTs enable business sectors to carry out advertisements and marketing of their goods & services.
- Business premises use ICT equipments to enhance their security. For examples use of CCTV cameras, sensors, Voice recognition equipments (VRE), Smart cards, etc.
- Businesses can use the internet to carry out research on quality and genuineness of products.

ii) Health

- The use of ICTs in health has enabled many procedures such as liver, throat surgeries.
- ICT high end machines such as CT scan, Ultra sound devices, Magnetic Imaging have enabled the diagnosis and cure of many diseases.
- ICTs have enabled online consultations by medical professionals.
- The security of the hospital, property and the patients is ensured by high end ICT devices such as CCTV cameras, sensors, etc.

iii) Education

- Enables research by students. The internet is a huge resource of information with online libraries and dictionaries such as Wikipedia, Google answers.
- Online universities enable distance learning (cyber classes)
- ICT aided facilities like Audio-visual ways ease the students' understanding on complex topics.
- The use of storage devices such as CD-RW/DVD-RW & flash memory, online storage have eased the storage of data and also offering online backups.
- Enables edutainment that combines education with entertainment such as chess and titans.

iv) Security

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- Use of street CCTVs, figure print detection technology, scan equipments are used by security operators.
- Use of phone and internet tapping technique enable the tracking down of crime by authorities.
- Use of ICT gadgets in modernizing the army such as radar system, warning systems, military lasers, etc.
- Use of sophisticated technology such as use of Drones in spying on enemies and analyzing possible targets.

v) Art, leisure and entertainment

- the social and leisure world of Chat rooms and social sites have enabled many people to connect and socialize globally.
- ICT innovations such as cartoons, graphical imaging have made the entertainment experience hundred times better.
- ICTs have enabled movie making, Audio and music recording in the entertainment industry.
- There are web sites which carry news and other matters related to entertainment such as YouTube, Fan fridge, etc
- ICTs have enabled the digitalization of photo print outs, stickers, movies recordings, etc.

POSITIVE IMPLICATION OF ICTs ON THE WORLD TODAY

(a) Social/ ethical field

- ✓ Ease of communication
With the use of mobile phones, internet services and media houses that have made communication easier between individuals.
- ✓ Improvement on peoples standards of living
The use of ICTs in areas such as entertainment, communications, banking, etc has improved peoples standards of living.
- ✓ Enhancement of personal security.
People can now afford having automatic remote control gates or doors. Installation of CCTV cameras and sensors. There are security locks or even subscription with security agencies who offer online and satellite security surveillance of property.
- ✓ Easy education with Online cyber classes
There are online universities that offers online degrees and also distance learning (cyber classes). So this makes learning easier and convenient for individuals who may be busy and unable to attend lectures physically.

(b) Economy

- ✓ Infrastructural development
Capital deepening through ICT investment development is important for economic growth. It establishes infrastructure for the use of ICT networks and provide productive equipment and software to business.
- ✓ There is growth of labor productivity

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With the continued use of ICTs, many workers in firms have improved on their personal skills and productivity. A work force which has been trained or has ICT skills may bring about innovative skills to the company.

- ✓ Enhancement of advertisement of goods and services
Many sectors, industries and individuals use internet for advertising their services and products.
- ✓ There is reduction in transport and communication cost
With the use of internet facilities such as E-commerce, E-banking(use of credit cards), there is reduction to barriers to international trade.

(c)Environment

- ✓ ICTs help to improve the environment through use of improved practices in agriculture and forestry. That aid in research on better farming techniques, mechanization.
- ✓ Environmental agencies are able to monitor state of air and water pollution by use of remote sensors.
- ✓ Government agencies responsible for the environment use ICTs to signal disaster warnings and relief supplies.
- ✓ The continuous manufacture of less pollutant manufacture of hardware tools has eased the conservation of the environment.
- ✓

(d)Politics

- ✓ ICTs are a means by which the public is informed or educated about democratic matters(civic education) e.g. by use of SMS platform, the internet(social sites such as twitter, Facebook, my space)
- ✓ ICTs are a means of promoting communications(strong democracy) e.g. by use of opinion polls pre to the real election exercise.
- ✓ There is high computerization on referendum on all matters of public importance

(e)Security

- ✓ ICTs have eased the security mechanism of nations by use of high technology in defense system. Such as use of remote controlled missiles, radars, satellite.
- ✓ With computerized registration of details about every citizen, the police can now easily track down crime.
- ✓ ICTs have proved to be a basis towards modernizing the army by employing ICT skills in its defensive mechanism.

NEGATIVE IMPLICATIONS OF ICT IN DIFFERENT AREAS IN THE WORLD TODAY

i) Social/ethical field

- ✓ **Social isolation:** With the continued use of ICTs such as the internet based way of communication, social interaction with peers has been reduced. So face to face interaction has greatly been degraded.
- ✓ **Pornographic and unwanted material:** The internet has a large amount of information in the form of pornography and other unwanted material like pornographic pictures and videos.

- ✓ **Stealing of money online is now rampant:** Thieves are always active in obtaining stolen credit and details to purchase goods and services over the internet.

ii) Economy

- ✓ **There's rise in unemployment in firms:** The use of ICTs kills labor intensive sectors of the economy e.g. the use of mobile telephones, email systems has killed the postal system.
- ✓ **There is high cost of living because of use of ICTs:** With the use of ICTs such as Mobile phones, internet services, digital TV series. Users have to keep on paying or subscribe monthly for these services which greatly hinder individual/family finances and incomes.

iii) Environment

- ✓ The continuous disposal of worn out or unwanted ICT gadgets like computer hardware components may lead to the pollution of the environment.
- ✓ Some of the ICT gadgets contain many hazardous chemicals which when exposed to humans may be harmful to their health, e.g. CRT monitors, CT Scan contain emissions which risk to human health.
- ✓ Many of ICT gadgets use a lot of energy in their functionality.

iv) Politics

- ✓ The use of ICTs as a means of communication to masses of people has sometimes been used as a means of inciting violence e.g. In the incidence when the former president of Egypt(Hossin Mubarak) was topped, his opponents used sites such as face book, twitter to mobilize resistance against his government.

v) Security

- ✓ The use of High end ICTs such as drones, satellite in spying on other nations may bring about a serious security concern.

WAYS THROUGH WHICH ICT HAVE IMPROVED COMMUNICATION TODAY

- ✓ The use mobile phones enable people in distant areas to communicate instantly.
- ✓ With the internet services many people can communicate any time anywhere.
- ✓ It is possible to have real time interaction through the use of Instant Messaging, chat rooms and social websites.
- ✓ People can send a receive fax messages using telephone services.
- ✓ People can gain access to information, news by use of many multimedia services utilizing ICT technology.

WAYS IN WHICH COMPUTERS CAN BE USED

(a) In a Bank

- ✓ Automatic Teller Machines (ATM) transactions such as withdraws, deposits of money.
- ✓ Processing of customer cheques by use of Magnetic Character Recognition Reader(MICR)
- ✓ Creating a data base and keeping customer records.
- ✓ Use of notes(money) counting devices
- ✓ Use of CCTV cameras & sensors in ensuring security in the bank environment.

(b) In a Super Market

- ✓ Use of bar code readers in reading serial numbers on packaged products
- ✓ At EPOS(Electronic Point Of Sale) when receipting for customer goods
- ✓ Creating a data base and keeping customer records

(c) In a school

- ✓ Teachers can use computers to prepare and store records about their students, such as attendance, record of progressive assessment.
- ✓ The work of school accountant / bursar in storage of financial records about students and prepare staff pay roll is made easy.
- ✓ Can be used to set students examinations as well as preparing mark sheets, grading and positioning.

(d) By a farmer

- ✓ Keeping record of his farm, i.e. customer details, sales etc.
- ✓ Use of internet services for research about new technology in farming.
- ✓ Use of mobile money services in paying and receiving payments for goods and services.
- ✓ Automatically regulating the environment conditions in green houses using sensors.

(e) In a factory

- ✓ In inventory control off goods stock and sales
- ✓ Use of bar code readers in reading serial numbers on packaged products.
- ✓ Creating data base and keeping customer records
- ✓ Used during automatic control of running machines
- ✓ In maintaining the security of the premises of the factory.

GREEN COMPUTING

Green computing refers to the environmentally responsible usage of computers and related resources. It may involve use of energy efficient hardware equipments, reduced energy consumption and proper disposal of E-waste.

GREEN COMPUTING PRACTICES

- ✓ Turning off the monitor when it's not in use or using more energy efficient monitors like LCDs instead of traditional CRT monitors.
- ✓ Volunteer computing or file sharing practices
- ✓ Virtualization of servers
- ✓ Using more energy efficient and less noisy cooling systems (like using liquid cooling systems instead of the conventional heat sinks and fans)
- ✓ Temperature maintenance and regulation to reduce thermal shock wear and tear to computer parts.
- ✓ Increased online security measures through the use of firewalls, anti spyware and antivirus programs to reduce the increased amount of e-waste on the internet and on other networks.

Briefly explain two ways in which ICT related equipment may cause environmental degradation.

- **Through environmental pollution:** Improper disposal of e-waste causes pollution since some ICT equipments contain toxic substances which are harmful
- **Through Electronic Magnetic Emission:** Some ICT gadgets may contain emissions that are harmful to human health.

POSSIBLE MEASURES FOR GREEN COMPUTING

Suggest the possible measures to ensure that the use of computer technology does not affect the environment.

- **Lower power hardware:** This involves the use of PCs that consume less power i.e. use of lower power processor.
- **Virtualization:** Involves the use of computer software to stimulate hardware. Stand-alone physical servers are replaced by virtual servers that run as software on a small number of computers
- **Cloud computing:** This is a technological advancement where software applications, processing power, data and even artificial intelligence are accessed over the internet.

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- **Energy efficient coding:** This involves saving of power by getting software to make less use of the hardware. The use of small and efficient energy conscious software.
- **Improved repair, re-use, recycling and disposal:** The emphasis on hardware repair, the recycling of old computer hardware into a second use situation, the re-use of components from PCs that are beyond repair.
- **Less pollutant manufacture:** The reduction in the use of many hazardous chemicals (such as lead, mercury, cadmium) in the manufacture of hardware equipment could prevent people being exposed to them, as well as enabling more E-waste to be safely recycled.

State two reasons why emerging computer technology is causing environmental challenges.

- Most Hardware equipment are non-recyclable.
- Computer power doubles every year

CLOUD COMPUTING

What is meant by Cloud computing?

This is a technological advancement where software applications, processing power, data and even artificial intelligence are accessed over the internet.

OR

Cloud computing refers to outsourcing of services such as those of computer hardware, software and other platforms over the internet.

BENEFITS OF CLOUD COMPUTING

Less maintenance: Hardware, applications and bandwidth are managed by the service provider.

Scalability: You pay only for the applications and data storage you need.

Expert service: Services are continuously monitored and maintained by the service provider.

Continues availability: Public cloud services are available whenever you are located.

Companies may potentially reduce their carbon foot print

It removes the need for most users to run high power PCs and laptops.

Enables anybody to obtain the environmental benefits of virtualization (use of computer software to stimulate hardware)

EXAMPLES OF TOXIC SUBSTANCES THAT ARE FOUND IN ELECTRONIC WASTE(E-WASTE)

- ✓ Lead
- ✓ Mercury
- ✓ Cadmium
- ✓ Beryllium

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- ✓ Brominated flame retardants (BFRs)
- ✓ Polyvinyl chloride (PVC)

Suggest the possible solutions to reducing the environmental harm from improper disposal of e-waste

- ✓ Less pollutant manufacture: This involves the manufacturing of hardware that are less hazardous and easy to recycle.
- ✓ Recycling of old computer hardware into a second use situation
- ✓ By re-using of components from PCs that are beyond repair
- ✓ Upgrading of existing systems

CHAPTER TWO

Computer Management, Lab Care and Maintenance

COMPUTER MANAGEMENT

Bootting: This refers to the process of turning (powering) the computer and loading the operating system into memory.

OR

Bootting is the process of switching on the computer.

WAYS OF BOOTING A COMPUTER

- Cold booting(hard rest)
- Warm booting(soft rest)

Describe the process of cold booting a computer.

- ✓ The system unit is switched on from the power button
- ✓ The computer goes through Power On Self Test(POST) with the help of BIOS (Basic Input/ Output System) and the operating system is then loaded.

Distinguish between cold booting and a warm booting of a computer.

- ✓ Cold booting is the switching on the computer for the first time when it has been completely off.

While

- ✓ Warm booting is the restarting of a computer which has been previously on.

WAYS OF PERFORMING A WARM BOOT

- ✓ By choosing start> shut down> restart
- ✓ By pressing CTRL + ALT + DEL
- ✓ By pressing the computer's restart button

State two ways of booting a computer loaded with DOS or Windows OS

- ✓ Booting from a diskette or disk i.e. flash disk, compact disk, floppy disk.
- ✓ Booting from the Hard Disk.

DANGERS OF SHUTTING DOWN A COMPUTER IMPROPERLY.

- ✓ Loss of data
- ✓ Crashing of hard disk
- ✓ Corruption of files

THE COMPUTER BOOTING PROCESS

Describe the computer booting process

- ✓ The system unit is switched on from the power button thus allowing the BIOS to start up.
- ✓ The computer then goes through Power On Self Test(POST) to check whether the computer hardware is connected and working properly.
- ✓ The BIOS looks for boot program contained on either hard disk or compact/flash disk to load the operating system.
- ✓ The boot program is then loaded into memory and executed.
- ✓ BIOS then searches for the system configuration information for the OS and the rest of OS is loaded into RAM, and the desktop and Icons display on the screen.

In the process of booting, state the function of each of the following.

1. RAM

This provides temporal storage of boot program when loaded in it(memory) to enable the computer to boot.

2. CMOS

This is used to store configuration information about the computer including amounts of memory, type of disk drives, hardware, current data and time. Thus during the booting process this information is retrieved.

Mention instances when booting of a computer may fail

- ✓ In case of hard disk crash
- ✓ When there is no operating system installed on the hard disk
- ✓ When the bootable device is unavailable such as hard disk

WAYS OF KEEPING FILES IN A SYSTEM SECURE

- ✓ Regularly perform a routine maintenance on files and storage devices.
- ✓ Install an updated antivirus software on your computer
- ✓ Remove corrupt files in the system
- ✓ Perform regular scanning for suspicious malware in your computer system.
- ✓ Avoid keeping so many files in the same folder/directory.

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- ✓ Use utilities like scan disk, defragmenters to remove in fragments in directories.

COMPUTER SOFTWARE/ PROGRAM(s)

Define the term computer program.

- This is a sequence of instructions written to perform a specific task with a computer.

OR

- Is a set of instructions designed to perform a specific function with the computer.

EXAMPLES OF COMPUTER APPLICATION PROGRAMS

- i) Spreadsheet program like MS Excel, Lotus 1-2-3, Visi Calc, etc
- ii) Database like MS Access, My SQL, Dbase I, Dbase II, etc
- iii) Word processor like MS Word, Word Perfect, Apple pages, etc
- iv) Presentation Software Like MS Power Point, Harvard graphics, etc
- v) Web browser like Mozilla Fire Fox, Opera, Internet Explorer, Net scape.
- vi) Publishing programs like MS publisher, MS Front Page, HTML

STEPS OF STARTING A PROGRAM

- Click Start button
- Point to All programs/ programs
- Open programs folder by clicking on it
- Then click on the program you want to open.

OR

- Double click on the shortcut of the program on the desktop

PROGRAMS THAT ALWAYS RUN IN BACKGROUND WHEN THE COMPUTER IS STARTED

- Time and date
- Antivirus
- Network connection
- Volume control
- Battery status (in case of laptops or notebook computers)

REASONS NECESSARY TO RESTART A COMPUTER

- The computer has hanged (frozen)
- The computer had failed to boot fully
- After installing a new program.
- The speed of the computer had slowed down.

FOLDERS

Define the term a folder.

- A folder (directory) is a storage location for related files and sub-folders.

FILES

Define the term a file.

- A file is a collection of related data.

A computer file comprises of a file name and a file extension separated by a dot. Explain the function of the file extension.

- It identifies the type of file, e.g. files with extension.exe are executable.
- It helps in categorizing the files for easy retrieval. E.g. during search of files the Operating System can first isolate file of a particular extension then search among them and this makes the search process faster.

EXAMPLES OF FILES

- Alliance.doc
- Fees structure.xls
- Dog.jpg
- Letter.txt
- CMD.exe
- Sembera.mpg/.mp4
- I can't stop loving you.mp3
- Students Database.accdb/.acc/.db

FILE EXTENSION

What is meant by file extension?

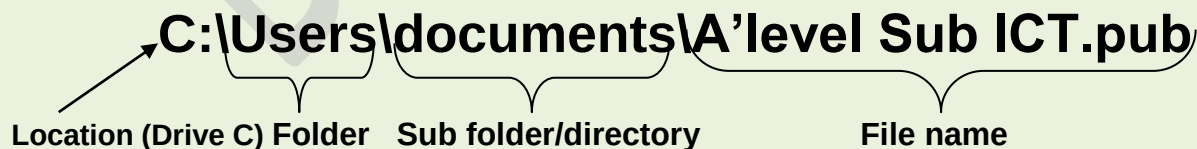
- This is a part (suffix) at the end of a filename which identifies the type of file it is.

EXAMPLES OF FILE EXTENSION AND THE FILE TYPES THEY REPRESENT

File extension	File type	Example
.Doc	MS Word	Alliance.doc
.xls	Spreadsheet	Mywork.xls
.ppt	Presentation	AIDS.ppt
.db	Database	Finance depart.db
.pub	Publication	Cards.pub
.rtf	Rich text format	Invitation.rtf
.txt	Unformatted text document	Mukeere.txt
.pdf	Portable document format	A level sub ICT.pdf
.exe	Executable file	Luzige.exe
.bat	Batch file	Autofile.bat
.gif	Graphics Interchange format	Flower.gif
.jpeg or .jpg	Joint Photographic Experts group	Davita.jpg
.png	Portable network Graphics	Ken.png
.mpeg or mpg	Moving picture Experts group	Indecent proposal.mpeg

Describe the following file path: C:\Users\documents\A'level Sub ICT.pub

- The file **A'level Sub ICT.pub** is in documents sub folder which is in Users folder found on hard disk C



Give any two details about a file that is stored by an Operating System.

- File name
- File size
- File type

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- Date and time of creation
- Storage location or path
- Attributes

FEATURE OF A FILE NAME

- A file name is made up of two parts: the file name which relates to the contents and the file extension which identifies the file type.
- The file name should be associated with the content
- Special characters such as “/”, “&” must be avoided in file names

FILE ATTRIBUTES

Define the term file attribute as used in operating system.

- A file attribute is a feature of a file that defines how it can be accessed and the operation which can be performed on it.

Explain the three file attributes

- i) **Read only:** This protects the file from any unauthorized change or deletion of data
- ii) **Hidden:** Files with this attribute are concealed and cannot be viewed in windows explorer and therefore cannot be opened or deleted unless one knows their names.
- iii) **Archive:** This attribute shows that the file has already been modified and needs to be backed up or archived. Thus when such a file gets deleted accidentally, it can be recovered using restore or undelete command.

Describe how you could search for a file in the hard disk using the operating system.

- Click start then select search or find from the start menu
- Type the name or extension of the file to be searched
- Select C: as drive to search from.
- Then click search

OR

- From the windows explorer type the name of the file in the search libraries section
- Then click on the search icon

Define an Icon in desktop environment?

- This is the smallest graphical or pictorial representation of several commands, an application, or a shortcut

State the major icons on the desktop

- My computer
- My documents
- Recycle bin
- Network places(Network)

Outline any five ways a file is kept secure in a computer

- By setting passwords to the files
- By installing an updated antivirus software
- By limiting physical access or usage of computers by un authorized users
- BY encrypting the files
- By avoiding usage of old storage devices
- By using the audit log mechanism to monitor each activity by user

List the elements that can be displayed on the desktop screen of a normal working computer.

- Icons
- The start button
- The task bar
- Notification area
- Windows desktop background

MAJOR DESKTOP ICONS AND THERE FUNCTIONS

- **My computer:**
Its a system folder that enables you to explain the contents of drives as well as manage files.
- **My documents:**
Its a personal folder for storage of personal files & graphics.
- **Recycle bin:**
It's a temporal storage location of deleted files and folders.
- **Network places:**
This displays shortcuts to shared computers, printers and other resources on network.

Explain the functions of the following parts of the desktop.

1. Start button:

When clicked, it opens the start menu which is the primary access to the programs, utilities and settings that are available in windows.

2. The task bar:

This is located at the bottom of the screen and contains the active tasks, which are icons and titles of programs that are running on the computer or folders that are open.

3. The Notification area:

This area contains the icons of special programs as well as the time of day.

State any two elements housed by the task bar.

- Start button
- Notification
- Icons of running programs

COMPUTER LAB CARE & MAINTENANCE

Computer literacy

What is meant by Computer literacy

- This refers to the comfort level someone has with using computers and programs associated with computers.

Servicing and maintenance of computers is an activity which should be done regularly. Give three reasons why computers should be regularly serviced.

- To prolong the life time of hardware
- To pre-empt problems that may render the proper performance of the computer system.
- To update outdated software programs such as Antivirus, Operating systems, application softwares, etc

COMPUTER LABORATORY

Define a computer Laboratory.

- This is a room/building equipped with computer facilities to enable the teaching and learning process to take place.

Areas of Laboratory security

Briefly explain any four areas of Laboratory security.

- **Physical security:**

This involves ensuring that all computer lab equipments such as computers and other assets are locked down through cabling or other means. So the computer lab should be monitored either in person by a lab technician, remotely through security cameras, or combination of the two.

- **Electrical Power and wiring:**

All aspects of electrical wiring as used to provide in buildings and structures must be clearly known. Ensure that all wires are properly insulated to avoid electric shocks and short circuits in computer lab usage.

- **Security Cameras(CCTV):**

The computer lab requires surveillance in areas that may need monitoring such as the inside of the lab and its outside, this is mostly done by use of CCTV and sensor devices.

- **First Aid Kit:**

It is necessary to have a first aid kit for emergencies such as accidents in the lab, sickness.

- **Fire extinguisher:**

This is used to extinguish or control small fires, often in emergency situations.

- **Air conditioning:**

Used to alter the properties of air (primarily temperature and humidity) to more favorable conditions. More generally, air conditioning can refer to any technological cooling, heating, ventilation, or disinfection that modifies the condition of air

Distinguish between safety precautions and safety practices in the Computer Laboratory.

- Safety precautions are measures that are put in place to prevent damages of computers & equipments in a computer laboratory

While

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- Safety practices are the activities that are continuously done to prevent damages in the computer laboratory.

WAYS TO ENSURE SAFETY OF COMPUTERS IN THE LABORATORY.

- Dusting and dumb proofing the computer room.
- Ensuring proper electrical wiring to avoid accidents from electric shocks and short circuits.
- Setting up a spacious lab room to enable easy movement of users within.
- Installing fire prevention and control equipments.
- Providing stable power supply
- Burglar proofing lab room
- Putting up lab rules and regulations, and enforcing these rules.

Define an Uninterruptible Power Supply(UPS)

- This is a power backup device that provides alternative source of power in case of power blackout. It also regulates the amount of power use by the computer.

Distinguish between a surge protector and a UPS.

- A surge protector only protects a computer from high voltage.

While

- A UPS protects the computer from sudden power loss.

State three reasons why a user will be required to restart a computer.

- When the computer slows down
- After installing new software programs
- After installing new hardware devices
- When a computer freezes
- After uninstalling a software program
- In case one wants to recover from errors

State three possible dangers of shutting down a computer improperly.

- Crashing of hard disk
- Loss of data
- Corruption of files

FUNCTIONS OF THE FOLLOWING ITEMS IN A COMPUTER LABORATORY

1. UPS

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- This acts temporary power storage unit for the computers in case of power blackouts.
- It regulates the amount of power into the computer, hence eliminating power surge and partial blackouts.
- It alerts the user in case of power blackout by producing a beeping sound.
- 2. Air Conditioner**
 - To modify the condition of air in the laboratory such as cooling and heating ventilation.
 - To regulate dust accumulation in the computer lab.
 - To dispel insects that could cause harm to the computer hardware
- 3. Wool carpet**
 - **Improved Safety:** Because it affords more traction, carpet helps prevent slips and falls. When falls do happen, chances of injury are greatly minimal.
 - **Carpet** provides a non-glare surface that reduces reflection and eye strain.
- 4. Internet gateway**

An internet gateway is your modem or router or any other peripheral which allows you to access the internet.
- 5. Burglar proofing**

It ensures the physical security of the laboratory against Burglars.

CAUSES OF DATA LOSS IN A COMPUTER

- Computer virus infections
- Power fluctuations/blackouts
- Improper shutting down of computers
- Accidental deletion of files
- Crashing of hard disk

PRECAUTIONS TO ENSURE PROTECTION AGAINST COMPUTER DATA LOSS

- By installing an Up-to-date antivirus software.
- Use of UPS or surge protector to protect computers from power loss or high voltage
- Shutting down the computers properly using correct procedures.
- Handling of storage media correctly.

BEHAVIOURS THAT MUST BE AVOIDED WHILE IN THE COMPUTER LABORATORY

SUB ICT SIMPLIFIED

- Avoid eating and drinking while in the computer lab, since food particles and spilled drinks can attract insects and flies, also causing rusting and short circuits.
- Avoid shouting, loud talking or cell phone use in the lab since may destruct the attention of users of the lab.
- Avoid installing, modifying or deleting of any software on the lab computers without authorization.
- Do not remove, modify, relocate, or disassemble lab equipments without permission of the lab coordinator.
- Sending/ posting harassing messages or repeatedly sending/ posting unwanted messages(electronic or paper) to others should be avoided.

FACTORS TO CONSIDER WHEN SETTING UP A COMPUTER LAB

- Laboratory physical security
- Adequate space for computers and users
- Availability of fire fighting equipments
- The room should be well ventilated
- Availability of power source
- The room should be away from dust roads
- Lab should easily be accessed by users.

What is meant by Trouble shooting?

- This is a form of problem solving, often applied to repair failed products or processes on the computer. It is a logical, systematic search for the source of a problem so that it can be solved, and so the product or process can be made operational again.

Outline the steps of trouble shooting process.

- Identification of the malfunctions or symptoms within a system.
- Generating possible causes of the symptoms and eliminating causes of the problem.
- Confirmation that the solution restores the product or process to its working state.

Explain the importance of the following items in the laboratory.

1. **Computer covers**
 - To protect dust from reaching the hardware components of the computer system.
 - To protect the computers from moisture and dust.
2. **Anti glare screen**

SUB ICT SIMPLIFIED

- To reduce electromagnetic rays from CRT monitors
- To help prevent eye strain and body fatigue caused by over bright monitors

- 3. Blower**
- To remove dust accumulation from hardware components and system units.

- 4. Surge protector**
- To regulate and protect computers from high voltage

- 5. Gaseous fire extinguisher**
- It enables to put off fire in case of an outbreak.

Explain the term formatting disks as commonly used in computer lab.

- This refers to the process taken to erasing the contents on the disk to make it empty.

State the circumstance which usually requires one to format a disk.

- In case of a virus infection
- When the disk space is full
- When the Operating system becomes corrupted
- During fresh installation

Mention three items which must be installed before computers can become usable.

- The Operating System
- Application programs such as Microsoft Office Suit
- Device drivers
- Antivirus software.

During installation, a system administrator usually has to configure certain hardware devices such as those on computer network. Explain the meaning of the term “configuration”

- This refers to the process of setting specifications required for a computer or communication device to properly be useable.

WAYS OF CARING FOR MICROCOMPUTERS IN A LABORATORY

- Computers should regularly be serviced, including peripheral devices.
- Switch off the computer in proper order i.e. by shutting down from the star button, then the UPS and finally socket.
- Avoid foods or drinks near computer hardware as these may attract files and dirt.
- Avoid switching off the computer abruptly as this may lead to corrupting of files and sometimes crushing of hard disk
- Do not service computers or open system casing when power cables are still connected as this may lead to electric shock.
- Use computer covers to protect computers from effects of dust especially when computers are not being used.
- Avoid exposure of computers to direct sunlight and heat.
- Servicing should be done by technicians to avoid mistakes.
- Use stabilizers and UPS to protect computers from power surge.

CHAPTER THREE

Computer Hardware

COMPUTER HARDWARE

- These are physical components/parts of a computer that can be seen, touched and felt.

CLASSIFICATIONS OF COMPUTER HARDWARE

- Input devices
- Output devices
- Storage devices
- Central processing unit.

FACTORS TO CONSIDER WHEN CHOOSING TO PURCHASE COMPUTER HARDWARE

- Multimedia capability
- User needs
- Portability
- Upgradeability and compatibility
- Software requirements
- Warranty given and its scope
- Multimedia capability

Define the term input device.

- This is a device used to enter or feed data and instructions into the computer for processing.

State four ways of inputting data into the computer system, giving an example for each.

- Using the keying-in device such as keyboard.
- Using a pointing device such as mouse, joystick, light pen, and track ball.
- Using data capture devices such as scanners, digital cameras, and barcode readers.
- Using Voice/speech recognition equipment such as microphones.

FACTORS TO CONSIDER WHEN BUYING A COMPUTER SET

- Hard disk capacity
- Processor speed and type
- Amount of RAM(Random Access Memory)
- State of the computer, whether new or clone
- Type of the computer, whether it is a desktop or laptop
- Technology used to make the monitor and its size.

Distinguish between soft copy output and hard copy output.

- A softcopy is the output received on the computer screen in form of a document, video or audio form. This kind of output is intangible and cannot be touched or felt.

While

- Hardcopy is a tangible output received on physical medium like a paper and it can be touched and felt by a person.

Give three situations where a hard copy is preferred to a soft copy.

- For physical data storage
- When one wants to sign a document physically
- When one wants to scan, photocopy or fax a document
- When one is unable to use display devices due to fatigue of eye strain
- When one wants to read the work away from the computer.

EXAMPLES OF ICT HARDWARE DEVICES THAT CAN AID A BUSINESS MAN

- Mobile phone for communication
- Barcode reader for reading serial numbers of packaged goods
- Sensors for monitoring security within business premises
- CCTV cameras for monitoring security within the building
- Electronic Point Of Sale(EPOS) to record purchases of consumers.

DEVICES THAT CAN BE USED AS BOTH INPUT AND OUTPUT

- Multifunctional device- is a single piece of equipment that compounds the functionalities of a printer, scanner, copy-machine and facsimile machine.
- Digital camera.
- Smart board
- Touch screens

LED (Light Emitting Diode)

Mention any two practical uses of LED(Light Emitting Diode) on a printer.

- To show whether the device is ready
- To show whether the device is idle

SUB ICT SIMPLIFIED

- To show whether the device has a fault/ problem e.g. a paper jam
- To show whether the trays of the printer are out of paper
- To show whether the device is working/ is on.

MONITORS

They are two types of monitors namely:

- LCD monitors (Liquid Crystal Display)
- CRT monitors (Cathode Ray Tube)

The LCD technology is replacing the CRT technology in display devices. State the advantages of LCDs over CRTs.

- LCDs require less desk space
- LCDs are more energy saving
- The radiation emitted by LCDs is negligible
- LCDs have a good resolutions
- LCDs are more portable

Mention the disadvantages of LCD technology as compared to CRT.

- LCDs are more expensive
- The LCD color output is usually slow and poor
- LCD monitors can only be viewed from a very narrow angle.

Define the following terms as used in display devices

1. Resolution

This refers to the number of pixels per inch on the screen which determines the clarity of images displayed and is measured in Dots Per Inch (DPI)

2. Refresh rate

This refers to how many times per second the screen is redrawn to avoid flickering.

3. Video adapter (Video card)

This is a card slotted on the motherboard that converts digital output from the computer into analogue video signals that is sent through a cable to the display device.

FACTORS TO CONSIDER WHEN CHOOSING A MONITOR

- Size of the screen needed
- Display type(technology used) either CRT or LCD
- Monitor controls such as brightness, contrast, displayed area, etc
- Resolution

Distinguish between LCD projectors and DLP projectors.

SUB ICT SIMPLIFIED

- Liquid Crystal Display (LCD) projectors are projectors that use liquid crystal display technology to produce lower quality images only.

While

- Digital Light Processing (DLP) projectors are projectors that use tiny mirrors to reflect light.

PRINTERS

- A printer is an output device that converts softcopy into hard copy.

TYPES OF PRINTERS

They are two major types of printers, and these include:

1. Impact printers
2. Non impact printers

IMPACT PRINTERS

- Impact printers are printers where the printing mechanism physically touches the printing medium e.g. a paper while printing. So they make some noise while printing.

EXAMPLES OF IMPACT PRINTERS

- Dot matrix printers
- Daisy wheel printers
- Thermal printers
- Epson

NON IMPACT PRINTERS

- Non impact printers are printers where the printing mechanism does not physically touch the printing medium e.g. a paper while printing.

EXAMPLES OF NON IMPACT PRINTERS

- Ink jet printers
- Laser printers

ADVANTAGES OF NON IMPACT PRINTERS OVER IMPACT PRINTERS

- They produce high quality output
- They are fast in printing
- They hardly make any noise when printing

DISADVANTAGES OF IMPACT PRINTERS OVER NON IMPACT PRINTERS

- They are quite slow at printing
- They are noisy while printing
- They produce poor quality output

ADVANTAGES OF DOT MATRIX PRINTERS OVER INK JET PRINTERS

- They are cheap to maintain
- They can withstand dusty environments and high temperatures
- Multi copy production is possible as long as carbonated paper is used
- Are readily available on the market

FACTORS TO CONSIDER WHEN BUYING A PRINTER

- The initial and maintenance cost
- The range of capabilities of a printer
- The availability of cartridge on the market (in some printers cartridge is costly)
- The amount of work(information) to be printed
- Speed of the printer
- Technology used in printing, either impact or non impact
- Range of capabilities of printers
- Compatibility with existing hardware technology.

AUDIO OUTPUT DEVICES

State two audio output devices

- Speakers
- Headsets (headphones)

ADVANTAGES OF AUDIO OUTPUT DEVICES

- No permanent copy of audio is produced
- A computer can only repeat the same message exactly the same way, thus if the message is not understood the first time, even when repeated may not be understood.

Define the term multi media as used in computing.

- It is a feature that combines audio, video, text and images to enable interactive and effective way of communication.

List any four requirements of a multimedia system.

- Sound card
- Audio output device (speakers or headsets)
- Multimedia software (such as windows media player)
- A CD or DVD drive
- A super video graphic adapter monitor.

Mention five ICT hardware devices that can be used in the bank.

- Ink jet printer

SUB ICT SIMPLIFIED

- PC camera
- Digital cameras
- Flat bed scanner
- Electronic Fund Transfer Point Of Sale (EFTPOS)
- CCTV cameras for monitoring security within the building
- Sensors for monitoring security within business premises
- Automatic Teller machine (ATM)
- Magnetic Ink Character Recognition Reader (MICR)

What is meant by “dead tree edition” as used in output devices?

- This refers to the printed paper version of written work

State one specialized application of each of these hardware devices

1. Light pen

They are used to design work and also by health care professions such as dentists to record patients' information easily.

2. Stylus and graphic tablet

They are used for computer aided designs and drafting by Architects, mapmakers, artists and designers.

3. Barcode reader

They are used mostly in supermarkets, departmental stores, sales outlets to read barcodes(serial numbers) on packaged products.

4. Optical Character Recognition (OCR) Reader

They are often used to read characters on turn around documents such as water bills, electricity bills, etc.

Complete each sentence below (a)-(e) using one item from the list.

A USB

A register

A cache memory

A serial port

An Arithmetic Logic Unit

- a) **Cache memory** is a high speed memory built into the CPU that temporarily stores data during processing.
- b) **Control unit** is a part of the CPU that coordinates its operations
- c) **Register** is a fast memory chip located on or near the CPU to help speed up processing.

SUB ICT SIMPLIFIED

- d) **A USB** is used to connect up to 127 peripheral devices to a computer without additional expansion.
- e) **An Arithmetic Logic Unit** is part of the CPU that performs logical operations.

PERIPHERAL DEVICES

Explain what is meant by the term “Peripheral device”.

- This is an external device attached to the system unit.

List down any four Peripheral devices

- Key board
- Monitor
- Mouse
- External CD drives
- Printer
- Speakers

Write short notes on the following terminologies in computer hardware

1. BUS

This is a path on the mother board of the CPU used to send data from or to peripherals, RAM and CPU.

2. Serial ports

These are sockets that can be used to connect devices that use a serial interface such as modem, scanner, mouse, etc. A serial port transmits data bits one after the other (serially) over a single line.

3. Parallel ports

These are sockets on the computer that are used to connect newer versions of printers or other peripheral devices such as a portable hard disk, tape backup, scanners, CD-ROM that need a faster data transfer than a serial port.

4. PS/2 ports

These are ports used to connect the PC to its keyboard and mouse.

5. USB port

This is an external port that can allow the user to connect up to 127 external PC peripheral devices such as USB keyboards, mice, printers, modems, scanners and external disk drives.

What is meant by Peripheral device interface (ports)?

- These are sockets or slots that peripheral devices connect to on the motherboard.

Distinguish between a USB port and SCSI port.

- Universal Serial Bus (USB) port uses a serial method of data transfer (one bit of data at a time in a single wire)

While

- Small Computer System Interface (SCSI) port uses the parallel method of data transfer (eight bits of data at a time, each in its own wire).

USB is an emerging technology which is becoming common today. State three advantages that make USB more superior to other technologies.

- They are quite faster in data transfer compared to PS/2 ports
- They can be used to connect up to 127 external PC peripheral devices since they are universal
- At least eight other peripheral devices can be chained and connected to the same port.

What is meant by a storage device?

- This is a hardware component that writes and reads data to and from storage media.

Distinguish between Primary memory and secondary storage. Give an example of each.

- **Primary memory/ RAM (Random Access Memory)**, is where both data and instructions are temporarily held for immediate access and use by the computer's microprocessor e.g. RAM, Cache memory and Buffer.

While

- **Secondary storage** is any storage device designed to retain data and instructions (programs) in a relatively permanent form e.g. hard disks, CD-ROM, flash memory.

Distinguish between writing data and reading data as used in storage devices.

- Writing data means recording data from memory to a storage medium.

While

- Reading (retrieving) data means transferring data from storage medium into the computer's memory for use by the operating system or application program.

State one use of each of the following memory

1. Cache memory

This stores data temporarily during processing

2. Firmware

These are ROM chips that contain data, instructions and information which is recorded permanently by the manufacturer of hardware.

3. Buffer

This is a region of memory that is used to temporarily hold data while it is being moved from place to place.

4. CMOS

CMOS (Complementary Metal Oxide Semi-conductor) is memory used to store configuration information about the computer which includes amount of memory, types of drives current date, time, etc.

5. Virtual memory

This is a kind of memory where the operating system allocates a portion of storage (usually on the hard disk) to function as additional RAM or in the absence of RAM.

In the space provided below, fill in the correct answer of hardware device

1. Enters instructions in the command line user interface

Keyboard

2. It is used to input text and graphics from hard copies into electronic format

Scanner

3. Used to ease in the processing of cheques by reading magnetized ink

MICR (Magnetic Ink Character Recognition reader)

4. Is used to move a pointer in GUI

Mouse

5. Used to make multiple choice questions in an examination
OCR (Optical Character Recognition Reader)

State the difference between Data Bus and Address Bus

- Data (memory) Bus is used to transfer data and instructions from memory to the CPU for execution.

While

- Address Bus consists of all signals necessary to define any possible memory address locations within the CPU.

Explain the meaning of the following features in computer interfaces

1. Plug and play

This is an interface that allows devices to be installed, configured and immediately removed from the computer while it is still working e.g. USB

2. Hot swappable

This is a peripheral device feature that allows devices that can be plugged and removed from computers while the computer is running e.g. USB devices.

State one function of a ROM chip

- To store permanent data (firmware) required by the computer.

COMPONENTS OF THE SYSTEM UNIT

- Mother board
- Power supply
- Memory chips
- CPU/Microprocessor
- Expansion slots
- Data buses
- Ports and connectors.

MICROPROCESSOR

- This is an integrated circuit (chip) that does all the processing of a full scale computer.

MAJOR PARTS OF A MICROPROCESSOR

1. Control Unit
2. Arithmetic and Logic Unit (ALU)

3. Registers

State the function of the following parts of the microprocessor

1. Control Unit

- It coordinates all operations within the CPU(microprocessor)
- Directs data that require mathematical operations from memory to ALU
- Sends processed data to either output device or storage device
- Erases information from memory when instructed

2. Arithmetic and Logical Unit (ALU)

- It performs arithmetic, comparison and logical operations

3. Registers

- It stores data within the CPU during processing
- It holds the work area, programs and data input.

Distinguish between clock rate and word size

- Clock rate/speed is defined as the frequency at which the processor executes instructions or processed data.

While

- Word size is the number of bits that the CPU can process at a time

State the function of the following in a computer

1. System clock

This is an internal clock that generates a signal that is used to synchronize the operation of the CPU and data movement around the other components of the computer.

2. PCI slot

This is an interface that allows additional hardware such as network cards to be connected.

3. Motherboard

This is a surface used for mounting computer components such as CPU, memory cards, expansion slots.

What is meant by machine cycle?

- This is a series of operations needed to execute a single instruction.

Briefly describe each of the operations of the machine cycle.

1. Fetching

The process of obtaining data items from memory.

2. Decoding

This is the process of translating instructions into commands that the computer understands.

3. Executing

This is the process of carrying out commands.

4. Storing

This is the process of writing the results into memory.

Explain the following terms as used with computers

1. Parallel processing

This is the use of multiple processors to execute a program at the same time

2. Multi-processing

This is the execution of more than one program by the same processor.

3. Multi-programming

This is the processing of two or more programs by the same processor at the same time.

VERSIONS OF MICROPROCESSORS ON THE MARKET TODAY

- Pentium duo-core
- Pentium i-3 core
- AMD Duron
- AMD Sempron
- AMD Duo Core

CHAPTER FOUR

COMPUTER SOFTWARE

Computer Software:

- This is a collection of computer programs and related data that provides instructions to the computer hardware.
- OR**
- Computer software is an interface between hardware and its users.

WAYS THROUGH WHICH COMPUTER SOFTWARE CAN BE ACQUIRED

- By buying standard software from vendors or licensed software companies.
- From user developed (in-house/ tailor made) software.

Explain why is it important to consider user needs when purchasing computer software.

- It is because the analysis of user needs determines the types of software the user will acquire.

ITEMS THAT SHOULD COME ALONG WITH AUTHENTIC SOFTWARE

- User manual
- License
- Installation guide
- Refresh guide
- Provision for trial version

Distinguish between open source and proprietary software

- **Open source software** is one whose source code is freely made available to users so that they can modify, use and distribute the modified product.
- While**
- **Proprietary software** is one whose source code is hidden from the users and therefore modification is for only the copyright owner/ software developer.

HARDWARE DETAILS THAT SHOULD BE CONSIDERED BEFORE INSTALLING A SOFTWARE.

- Hard disk capacity
- Processor speed and type
- Amount of RAM

TERMS USED REGARDING TO SOFTWARE

1. General purpose software

Is software purchased off-the-shelf by the user and is used for wide variety of purposes.

2. In-house developed software

These are customized programs written by a programmer or software developer under contract.

3. Software release

Is the public or private distribution of a new or upgraded version of software product.

4. Software version

Software version also known as software upgrade is the major upgrade in a current software leading to even changes in software interface and some procedures.

5. Software suit

This is a collection of individual application software packages sold as a single package. E.g. Database software, spreadsheet, presentation, word processor, publisher software are all sold under Microsoft Office suit.

ADVANTAGES OF GENERAL PURPOSE SOFTWARE

- It is relatively cheap and readily available
- They are user friendly i.e. they are easy to learn and use
- They are provided with supportive documentation in both soft and hard copy format.
- They can be used to accomplish a number of tasks e.g. word processor generates a variety of documents (letters, notices, memos, speeches)

TYPES OF COMPUTER SOFTWARE

They are two major types of computer software namely:

1. System software
2. Application software.

SYSTEM SOFTWARE

- **System software** is set of programs that manage and control the operations of a computer and other types of software that run on it.

CLASSIFICATIONS OF SYSTEM SOFTWARE

There are three classification of system software namely:

- **Operating systems**
This is a set of programs containing instructions that coordinate all the activities among computer hardware devices. E.g. Windows, MS DOS, Linux.
- **Utility programs**
This is software designed to help analyze, configure or maintain a computer system. E.g. Antivirus, back up, screen saver, debugger, file compression, etc.
- **Programming languages**
These are artificial languages designed to communicate instructions to a machine, particularly a computer. E.g. BASIC, COBOL, FORTRAN, PASCAL, etc.

OPERATING SYSTEM

What is meant by Operating system?

- This is a set of programs containing instructions that coordinate all the activities among computer hardware devices.

FUNCTIONS OF AN OPERATING SYSTEM

- It makes hardware to start operating
- Supervises and coordinates hardware operation
- It helps in memory management
- It monitors system performance
- It helps in storage management
- Helps in configuring devices. Operating system supports plug and play and can configure devices automatically.

SUB ICT SIMPLIFIED

- It administers security. Operating system enables system administrators do system audits on system users.
- It acts as an interface between the user and hardware.

EXAMPLES OF OPERATING SYSTEM

- Windows such as Windows XP, Windows 7 and Windows 8
- Netware
- LINUX
- Windows Vista
- MS DOS
- UNIX
- Palm OS for PDAs
- Solaris for e-commerce applications
- Pocket PC 2002

FEATURES OF A USER FRIENDLY OPERATING SYSTEM

- It works at faster speeds thus enabling application programs to run more frequently
- Uses GUI in its commands.
- It is multitasking. It enables the user to carry out many tasks at ago.
- It allows a user to recover from errors using the Undo and Redo actions.

WAYS OF IDENTIFYING AN ILLEGAL COPY OF OPERATING SYSTEM

- Frequent interruptions of warning messages about genuinity of operating system being used (fatal error)
- Prices of software are extremely cheap
- There is no documentation material accompanying it.
- There is no license agreement or certificate of authenticity.

WAYS OF ACQUIRING AN OPERATING SYSTEM

- By downloading open source copies of the software from internet.
- By purchasing a licensed copy from authorized software vendors.

List two devices controlled by an operating system

SUB ICT SIMPLIFIED

- Secondary storage devices
- RAM
- Processor
- Input devices such as keyboard, digital camera, mouse, etc
- Output devices such as printer, monitor, speakers, etc
- Communication interfaces and ports

FACTORS TO CONSIDER IN ORDER YOU TO UPGRADE AN OPERATING SYSTEM

- Capability features with existing hardware of the computer.
- User friendliness of the operating system.
- The cost
- Security features related to the operating system
- Reliability of the operating system
- Hardware configuration of the computer

A University student on internship has just bought a new laptop for his computing work, suggest any three software programs he will need installed on his computer before using it.

- Operating system such as Windows 7, Windows 8
- Antivirus software
- Software suit such as MS office
- Device driver software.

TERMINOLOGIES USED IN OPERATING SYSTEM

1. Patch

This is a program code that rectifies a bug (error) in an operating system or seals the weak points of an operating system.

2. Interrupt

This is an act of suspending the normal sequential processing of instructions in a computer.

3. Troubleshoot

This is to trace, identify and correct any hardware and software related problems.

UTILITY PROGRAMS

- Is software designed to help, analyse, configure or maintain a computer system

FUNCTIONS OF THE FOLLOWING UTILITY PROGRAMS

1. **Disk defragmenters**
This moves fragments of files to one location.
2. **File compression**
This adjusts or reduces the size of a file with programs such as winZip to improve system performance.
3. **Diagnostic utility**
This compiles technical information showing identified problems in the system.
4. **Network utility**
These analyse the computer network connectivity, configures network settings, checks data transfer or log events
5. **Antivirus**
This is a utility that scans a computer system for the possibility of malware and viruses.
6. **Backup software**
These are utilities that make copies of all information on a disk and restores either the entire disk (e.g. in the event of disk failure) or accidental deletion of information.
7. **File viewer**
This is a program that enables the viewing of the location and access to contents of a file.
8. **Cryptography utility**
This enables data to be transmitted over a communication channel without it being hacked.
9. **Memory testers**
Checks for failures in the storage of the system.
10. **Disk cleaners**
This enables the computer user to decide on what to delete in case the hard disk is full.
11. **Screen savers**
Prevents phosphor burn out in most CRTs and LCDs

PROGRAMING LANGUAGES

What are programming languages?

- These are artificial languages designed to communicate instructions to a computer system.

CLASSIFICATIONS OF PROGRAMMING LANGUAGES

There are four major classifications of programming languages.

- Machine languages
- Assembler languages
- High level languages
- Fourth generation languages

Distinguish between machine language and high level language

- **Machine languages** are languages written in binary form in the computer and are extremely hard for humans to read and write.
- While**
- **High level languages** are languages consisting of statements or sequences of text including words, numbers and punctuations much written like natural languages.

EXAMPLE OF HIGH LEVEL LANGUAGES

- BASIC (Beginners All Purpose Symbolic Instruction Code)
- COBOL (Common Business Oriented Language)
- FORTRAN (Formula Transition)
- Pascal

ADVANTAGES OF HIGH LEVEL LANGUAGES

- They are machine independent
- They are user friendly.
- They are problem oriented
- They are easier to learn, write, correct and revise than assembler and machine languages.

Distinguish between Command Line Interface and Graphical User Interface

- **Command Line Interface** is an action that prompts a computer user to type in commands or instructions e.g. Log in details such as user name and password.
- While**
- **Graphical User Interface** is an action where a user clicks on any command especially graphics e.g. copy and paste or moving an item.

ADVANTAGES OF USING GUI

Give two advantages of using a GUI in operating systems

- It is easy to learn and use unlike command line interface
- It is user friendly since it offers guidelines on how to navigate programs
- It is easy to transfer data from one window to another by the drag and drop method
- One does not need to master the commands in order to use it unlike the command line interface

Mention one similarity between GUI and menu driven interfaces

- Both interfaces have a list of options to choose from

DIFFERENCES BETWEEN GUI AND MENU DRIVEN INTERFACES

- The menu driven interface has no icons unlike GUI that has icons
- The menu driven interface does not allow the use of pointing devices unlike GUI

APPLICATION SOFTWARE

- **Application software** are programs designed to help a user perform a specific task.

Examples of application software

- ✓ Word processor e.g. MS Office Word, K-Word, Apple pages, King soft Word, etc
- ✓ Database software e.g. MS Office Access, Dbase, Paradox, Oracle
- ✓ Spreadsheet e.g. MS Office Excel, apple numbers, MS Visi Calc, etc
- ✓ Presentation software e.g. MS Office Power Point, Harvard Graphics, Corel presentation, etc
- ✓ Browsing software e.g. Mozilla Fire Fox, Internet Explorer, Opera, Netscape, etc

SUB ICT SIMPLIFIED

- ✓ Multimedia software e.g. Windows media player, VLC player, Real player for playing audio and video files.
- ✓ Educational software such as Encarta, Matlab, Google earth, etc
- ✓ Web design software such as Java, HTML, MS Front Page, etc
- ✓ Desktop publishing software such as MS publisher, Adobe photo shop, etc

CATEGORIES OF APPLICATION SOFTWARE

➤ **Licensed**

This is where a user purchases only the license of the software and not the software itself. The vendor allows for the installation of this software on a specific number of machines.

➤ **Commercial/ sold**

This is where the user purchases a legitimate copy of the software itself from the vendor and is free to resale it.

➤ **Freeware**

This is the application software that can be downloaded, used and copied without any restrictions. E.g. internet software such as web browsers, file converters, etc.

➤ **Shareware**

This is an application software that can be downloaded but you need to pay some amount of money for using them.

➤ **Open source**

This is software made available with their source code that a user is free to modify its code and use it

➤ **Needs to be installed**

This is software that has to be installed before you use your computer, such as device driver software, MS office suit

➤ **Run online**

This is software that runs online and you do not necessarily need to download it e.g. Antivirus, Internet games

Distinguish between Custom made software and off the shelf software.

- **Custom made (special purpose/ tailor made software)** is software written to meet the needs of an organization that cannot be usually satisfied by other sources of software.

While

- **Off-the-shelf (general purpose/ packaged software)** is software which may be used for a wide variety of purposes, e.g. Word processors, spreadsheets, databases, presentations, etc

Distinguish between single user license and site license.

- **Single user license** is a license agreement that gives a software buyer the right to install and use software on a single computer at a time.
While
- **Site license** is a license that gives a software buyer the right to use and install the software on multiple computers at a single site.

WORD PROCESSORS

What is meant be Word processing?

- This refers to the process of entering and manipulating a document using a computer and computer programs.

What is aword processing application?

- Word processing applications are computer programs that are used for of creating, saving, editing, formatting and printing text documents.

EXAMPLES OF WORD PROCESSORS/ WORD PROCESSING APPLICATIONS

- Microsoft Office Word,
- Corel WordPerfect,
- Lotus WordPro,
- Apple Pages,
- OpenOffice.org Writer,
- LibreOffice Writer,
- KWord
- Celframe Write
- Kingsoft Writer etc.

COMMON FEATURES OF WORD PROCESSING APPLICATIONS

- **Word Wrap:** this feature automatically sends a word that does not fit within the margin settings onto the next line, without the user pressing Enter key.
- **Find:** allows the user to locate all occurrences of a particular character, word or phrase.
- **Replace:** allows the user to substitute existing characters, words or phrases with the new ones.
- **Spell checker:** allows the user to check spellings of the whole document at one time or to check and even correct the spelling of individual words as they are typed (Autocorrect)
- **Grammar checker:** this reports grammatical errors, usually by a wavy green line, and suggests ways to correct them.

SUB ICT SIMPLIFIED

- **Thesaurus:** suggests alternative words with the same meaning (synonyms) for use in the document.
- **Mail Merge:** This is feature used to create similar letters to be sent to several people. The names and addresses of each person can be merged with one single main document.
- **Automatic page numbering:** numbers the pages automatically in a document
- **Tables:** allow users to organize information into rows and columns.
- **Multi-columns:** arranges text into two or more columns that look similar to newspaper or magazine
- **Drop cap** – Formats the first letter in paragraph to be dropped across two or more lines.
- **Clipart:**, refers to pre-made images about various subjects used to illustrate concepts in documents.
- **Templates:** establish the initial document layouts and formats for various document types.
- **Printing:** allows a user to obtain a hard copy of a document from the printer.
- **Word Count:** Establishes the number of words, characters, paragraphs, etc. in a document.
- **Headers and Footers:** Used to insert text in the top and bottom margin through the document.
- **Footnotes and Endnotes** are used as references that provide additional information about a word or phrase within a document.
- **Insert and Delete** allows a user to add and remove portions of text while editing document.
- **Bold, Underline, Italic** – These and other formatting features that allow a user to improve on the appearance of text.
- **Cut, Copy, Paste, Insert, Delete** – These and other editing features that enable a user to make changes to the text in a document.

Distinguish between copy and cut command in word processing

- **Copy command** allows the duplication of text or data to get two or more copies leaving the original behind.
While
- **Cut command** allows the complete transfer of data or text from one position to another without leaving the original behind.

Distinguish between Save and Save As in word processing

- **Save command** is used when the file is being saved for the first time on the computer or any other storage location
While
- **Save As command** is used when the user wants to change the name of the file. **OR** When the user wants to change the storage location of a file.

TERMS USED IN WORD PROCESSORS

1. **Thesaurus**

SUB ICT SIMPLIFIED

This is a word feature that allows you to search for meanings or synonyms without leaving the word processor.

2. Drop cap

This is an artistic (enlarged) beginning letter of a sentence or paragraph

3. Macros

This is a character of words that represents a series of key strokes such as text or commands

4. Footnote

This is a note at the bottom of a page of a book or manuscript that comments on or cites a reference for a designated part of the text.

5. Mail merging

This is the combination of two files into a single file when creating mass productions.

SPREADSHEET SOFTWARE

- Is an application software used to organize data in rows and columns in order to perform calculations on it.

EXAMPLES OF SPREADSHEET PROGRAMS

- MS Excel
- VisiCalc
- Lotus 1-2-3
- Symphony
- Apple numbers
- Jazz
- Open Office. Org Calc
- VP planner

TERMS USED IN SPREAD SHEETS

1. A workbook

This is a collection of worksheets in a spreadsheet. It is the main file of a spread sheet program

2. A worksheet

This is a single work space in a worksheet. It contains rows and columns (cells) where you enter data.

3. Active cell

This is the cell where the cursor is currently stationed

4. Data range

This refers to a group of cells that usually acts as a basis for calculations in worksheets including creating charts.

5. Automatic recalculation

SUB ICT SIMPLIFIED

This is a feature in spread sheet that automatically adjusts the results of a formula if there is any adjustment in data.

EXAMPLES OF DATA TYPES STORED IN A SPREADSHEET

- Labels
- Numbers
- Formulas
- Text
- Functions

What is meant by cell reference?

- This is the location / address of a cell in a worksheet e.g. B5, F9, DG43

Distinguish between relative cell referencing and absolute cell referencing

- **Relative cell reference** is a reference to cells relative to the position of the formula.
- While**
- **Absolute (fixed) cell reference** is a reference that always refers to cells in a specified location

State the function of the following as used in spread sheets.

1. **Auto fill/ copy hand tool**
 - This is a feature that automatically fills in cells, create row of data and even copy formatting from one cell to the next
2. **Freezing and unfreezing panes**
 - This is a feature that enables headings of a large worksheet to be viewed as you scroll through a work sheet. In this case these headings do not disappear as you scroll through.

SPREADSHEET CELL ERROR MESSAGES

1. **#VALUE**
An argument or operand used is of wrong type
2. **#NAME**
It means that Microsoft Excel does not recognize the name used in the formula
3. **#####**
The cell width needs to be adjusted such that the values are all seen.

ERROR VALUES IN MS EXCEL

1. **#DIV/0!**
The formula is trying to divide by 0
2. **#N/A!**
No value is available
3. **#NULL!**
You specified an intersection of two areas that do not intersect
4. **#NUM**
There is a problem with a number used in the formula
5. **#REF**
The formula refers to a cell that is not valid e.g. 6E instead of E6.

Distinguish between a formula and a function as used in spread sheets

- **A formula** is a sequence of instructions for performing calculations and displaying result in a cell.
- While**
- **A function** is a predefined formula that provides shortcuts for commonly used calculations.

FUNCTIONS USED IN SPREAD SHEETS

- Sum
- Average
- Maximum
- Minimum
- Count
- Mode
- If
- Median
- Rank

Explain the meaning of these terms as used in charts & graphs in spreadsheets

1. **Data series**
These are related data points or data markers that are plotted in a chart usually distinguished by a unique colour or pattern.
2. **Legends**
These are small boxes that identify the pattern or colour assigned to data series in a chart.
3. **Embedded Chart**
This is a chart which is created within the same worksheet having the table.

4. Chart sheet

This is a chart displayed on a separate worksheet from the real sheet onto which a table is formed.

Mention one data management tool used in spread sheet.

- Sorting
- Filtering
- Creating charts/ graphs

State the use of the functions in spread sheets

1. RANK

This enables one to create positions of data in either ascending or descending order

2. What IF

This is used to forecast the outcome and test different parameters with the help of the automatic recalculation feature.

3. COUNT

This is used to count and return the number of items in a given range.

Distinguish between data filtering and sorting

- **Data filtering** involves the display of data rows that meet certain criteria.
- While**
- **Sorting** is the arranging of data alphabetically or numerically either in ascending or descending order.

The worksheet below shows a school's expenses (in dollars) paid out for the first three months of the year.

	A	B	C	D	E	F	G	H
1	Item	Jan	Feb	Mar	Total	Average		
2								
3	Food	72	72	106				
4	Electricity	24	37	66				
5	Telephone	96	90	114				
6	Stationary	48	38	64				
7								
8	Total							

a) Write down a formula to calculate the following

i) **Total expenses for electricity**

=SUM(B4:D4) or =(B4+C4+D4)

ii) **Average expenses for stationary**

=AVERAGE(B6:D6)

iii) **The highest expense for telephone**

=MAX(B5:D5)

- b) Write down a formula to count all the items with expense greater than 70 dollars

=COUNTIF (B3:D6,">70") or +COUNTIF (B3:D6,">70")

The spread sheet table below was created using MS excel and shows part of sub ICT results for some students of S.6 in a certain school in an end of term exam.

	A	B	C	D	E	F	G	H
1								
2	NAME	S850/1	S850/2	S850/1 OUT OF 40%	S850/2 OUT OF 60%	TOTAL	GRADE	POSITION
3								
4	Ann	55	80	22	48	70	D1	2
5	Fred	57	59					
6	James	43	50					
7	Kezia	73	80					
8	David	59	59					
9	Mariam	66	70					
10	Suzan	49	40					
11	Oliver	50	57					

- a) Suggest two other spread sheet programs that could be used to create the records besides MS Excel.

- Lotus 1-2-3
- Visi Calc
- Open Office. Org Calc
- Jazz
- Symphony
- VP planner

- b) Which function is suitable for grading students in column G
- The IF function

- c) Write down a formula for creating position for students in column H basing on total.

- =RANK (F4, F\$4:F\$11,0)

- d) State the tool you would employ to complete the positioning for all the students.

- The Auto fill/ copy handle tool

ELECTRONIC PUBLISHING

Define the term electronic publishing?

- This is the production, distribution and user interaction of computer based production of text, graphics and other interactive media.

OR

- It is the type of publishing that is used to create and disseminate information by electronic means including email and via web.

EXAMPLES OF ELECTRONIC PUBLICATIONS

- E-mail publishing
- E-books
- Web publishing
- Print on demand
- Digital content
- E-journals
- Electronic ink
- Online portfolios
- Digital scrap books and online photo galleries.

TERMS USED IN PUBLISHING SOFTWARE

- 1. Frame**
These are areas within a publication which may include text areas and picture areas
- 2. Handles**
These are small circles which appear around the edge of a frame when you click on the frame
- 3. Template**
These are tools having pre-chosen design styles that can be used in publisher to help easily create basic publications.

EXAMPLES OF PRINT ON DEMAND/ DESKTOP PUBLICATIONS

- Books
- Journals
- Magazines
- Flyers
- Posters

SUB ICT SIMPLIFIED

- Calendars
- Brochures
- Certificates
- Banners
- Business cards
- News letters
- Greeting cards

FEATURES OF ELECTRONIC PUBLISHING SOFTWARE

- Has inbuilt templates which users can adopt in publications
- Text formatting features
- Has inbuilt fonts
- Has ability to import text and graphics from other programs
- Has standard shapes such as lines used for drawing
- Has page formatting features such as margins, footers

EXAMPLES OF ELECTRONIC PUBLISHING SOFTWARE

- Microsoft Publisher
- Adobe InDesign
- Adobe page maker
- iStudio publisher
- Corel Draw
- Frame maker

DATABASE MANAGEMENT SYSTEM

What is meant by Data base management system (DBMS)?

- This is a collection of programs that enables users create and maintain a database.

EXAMPLES OF DATABASE MANAGEMENT SYSTEMS

- Microsoft Access
- Oracle
- My SQL
- Visual basic
- Dbase
- Paradox

ADVANTAGES OF DBMS OVER FILE BASED SYSTEM

- It controls data redundancy. Data of similar type is stored in single location hence easing independence updates.

SUB ICT SIMPLIFIED

- It helps restrict unauthorized access. Many DBMS use security mechanism to protect data.
- DBMS provide facilities for recovery from hardware and software failures. (Provides backup and recovery)
- Improves data accessibility and responsiveness. It is possible to access data via an internet connection.
- It improves maintenance through data independence.
- It enforces standards in databases.
- It avoids duplication of data.

The table below shows part of a database for employees of a certain company

EMPLOYEE ID	FIRST NAME	LAST NAME	SEX	BASIC SALARY (\$)	DEPARTMENT
K001/04	Glenn	Tucker	M	25000	Security
K007/008	James	Bond	M	30000	Security
K013/098	Jessie	Elvis	M	20000	Messenger
K009/003	Mary	Heitinga	F	40000	Council
K156/145	Leon	Melanine	F	58000	Council
K345/001	Jack	Freddie	M	10000	Cleaner
K728/024	Hwan	Sung	M	33000	Technical

a) Write down the data type for the following fields

i) SEX

Text

ii) EMPLOYEE ID

Text

iii) BASIC SALARY (\$)

Currency

b) How many records are there in the table

- 7 records (number of rows used for data entry)

c) What is the suitable primary key?

EMPLOYEE ID

d) Write down a query for employees in the security and council department.

- "Security" or "Council"

The table below shows part of a database. Use it to answer the questions that follow.

StudentID	Name	Math	Eng	Sci	SST	Total marks
014/KSS/005	Senind	80	60	78	46	
014/KSS/102	Opondo	90	81	72	60	
014/KSS/009	Kutosi	56	92	66	46	

(a) What is the most appropriate primary key?

Student ID

(b) (i) Write down the criteria for students whose name ends with letter O and have a mark in math greater than 70

Name: Like"*O", math>70

(ii) Write down a criteria for students whose names begin with letter K to S and have a mark in science less than 68

Name: Like"[K-S]*", Sci<68

(c) The formula entered in design view for total mark is

Total:[Math]+[Eng]+[Sci]+[SST]

(d) Write down an input mask for Student ID to contain KKK and incremental by one "KKK1"

Distinguish between the following terms as used in database management system.

(i) Validation rule and validation text

Validation rule gives an expression that must be true whenever you add or change a value in a field e.g. "F" or "M"

While

Validation text is a text that appears when a value violates the validation rule e.g. required to enter F or M for this field.

(ii) Data validation and data verification

Data validation involves checking data input against / predetermined data limits coded in the computer

While

Data verification involves checking data input manually by comparing input data with source data

State the importance of an input mask as a field property in databases

An input mask controls the value of a record and sets it in a specified format e.g. a telephone number field can be formatted with an input mask to accept ten digits that are automatically formatted as "(555) 123-4567. And the blank field will look like () ____ _"

CHAPTER FIVE

INTERNET & WWW

- ❖ Introduction to Internet
- ❖ Internet Services
- ❖ Electronic Mail (E-mail)
- ❖ World Wide Web (WWW)

INTRODUCTION TO INTERNET

Internet is the global connection of computers linked together sharing information and other resources.

OR

This is a worldwide network with more than 100 million users that are linked for the exchange of data, news, conversation and commerce.

OR

It is an arrangement of connected computers which allows the computer users all over the globe to exchange information.

SUB ICT SIMPLIFIED

It is based on the networking technologies of the Internet Protocol Suite. It is the successor of the Advanced Research Projects Agency Network (ARPANET) developed by DARPA of the U.S Department of Defense. The internet is also the communications backbone underlying the World Wide Web (WWW).

COMMUNICATION PROTOCOLS

The term protocol or internet protocol refers to the set of rules that govern transmission of data across all computers on a network

OR

These are rules that govern how communication over the internet is carried out.

EXAMPLES OF COMMUNICATION/ INTERNET PROTOCOLS

(i) IP

Internet Protocol (IP) is an address that uniquely identifies each computer on the internet.

(a) TCP/IP

Transmission Control Protocol/ Internet Protocol (TCP/IP) is a protocol suite that determines how computers connect, send and receive information on the internet.

(b) HTML

Hyper Text Markup Language (HTML) refers to the Hypertext authority system which transmit text, data, images, sound and multimedia file types across a network.

(c) HTTP

Hyper Text Transfer Protocol (HTTP) is a set of rules by which an HTML document is transferred over the web.

(d) FTP

File Transfer Protocol (FTP) enables file sharing between hosts.

OR

FTP is an internet standard that allows users to upload and download files with computers

Key functions of FTP are:

- To promote sharing of files (computer programs and or data)

SUB ICT SIMPLIFIED

- To encourage indirect or implicit (via programs) use of remote computers.
- To shield a user from variations in file storage systems among hosts.
- To transfer data reliably and efficiently.

(e) IMAP

Internet Message Access Protocol (IMAP) is a method of accessing electronic mail bulletin board messages that are kept on a mail server.

(f) POP

The Post Office Protocol (POP) is designed to allow a workstation (PC) to dynamically access a mail drop on a server host.

(g) SMTP

Simple Mail Transfer Protocol (SMTP) is a protocol designed to transfer electronic mail reliably and efficiently. SMTP is a mail service modeled on the FTP file transfer server. SMTP transfers mail messages between systems and provides notification regarding incoming mail.

(h) TCP/IP

Transmission Control Protocol/ Internet Protocol (TCP/IP) is a protocol also used in data transfer on a network

(i) TELNET

Is an Internet protocol that enables a computer to function as a terminal working from a remote computer.

EFFECTS OF INTERNET TO SOCIETY

Today internet is becoming part of the human life. Describe the positive and negative effects of the internet to society.

POSITIVE EFFECTS

- **Information resources:** The internet offers research information to students and users with the help of numerous websites and search engines.
- **Faster communication:** It offers a speedy communication when using services such as Email, Chat, commerce, etc
- **Entertainment:** Internet improves access to entertainment such as online games, musical downloads, etc
- **E-commerce:** Online shopping is possible using numerous websites such as eBay, olx, etc
- **Social Networking:** Using numerous sites such as Facebook, twitter, MySpace, etc people can find both new and lost long friends, find jobs, business opportunities, etc

SUB ICT SIMPLIFIED

- **Online Services:** One can access online services such as hotel reservations, pay for bills and taxes, etc
- **Formation of communities:** The internet enables many forums involving debates and discussions.

NEGATIVE EFFECTS

- **Theft of personal Information:** When using an unsecure connection, there is a high possibility that your personal details such credit card number, bank account information; etc may be accessed by Con artists or hackers.
- **Virus threat:** Computers on the internet are very prone to virus attacks, which may lead to crushing of hard disk or data loss.
- **Spamming:** Unwanted/ Unsolicited E-mails in bulk are very frustrating as they make the internet connection very slow and unreliable.
- **Easy access to pornography:** Un restricted access to porn may have a bad influence on the mental health especially in Teens/ Children. It may also play a havoc in marital and social lives of adults.
- **Social disconnect:** When many people are engulfed in a virtual world of socialization, this may lead to drifting apart of friends and family members.

INTERNET SERVICES

REQUIREMENTS NEEDED FOR AN INTERNET CONNECTION

- Internet Service Provider
- Internet software such as Lotus Notes, Microsoft Office Outlook, Internet browsers, etc
- Transmission medium such as wireless (satellites, microwaves, Wi-Fi, etc)
- Data Terminal Equipment.

USES/SERVICES PROVIDED BY THE INTERNET

- World wide Web (WWW)
- E- mail
- Mailing Lists
- Chat Rooms
- Instant Messaging (IM)
- News Groups
- Internet Forums / Message boards
- Provides Voice Over Internet Protocol (VoIP)
- E-commerce

SUB ICT SIMPLIFIED

- E-banking
- E-Learning
- E-publishing

EXAMPLES OF INTERNET CONNECTION SERVICES

(i) Email

Is the process of sending and receiving messages across the internet or other computer networks.

(ii) Chat rooms

This is any technology ranging from real-time online chat over instant messaging and online forums to fully immersive graphical social environment.

(iii) Mailing lists

This is a group of subscribers to an organization, thus the organization uses this list to send material to multiple recipients.

(iv) News groups

These are discussion groups/ forums within the Usenet system for messages posted from different locations.

(v) Internet forums/ message boards

These are online discussion sites where people hold conversations in the forms of posted messages. In this case messages are temporarily archived.

(vi) Instant Messaging (IM)

This is a set of communication technology used for text-based communication between two or more participants over the internet or other types of networks.

(vii) VoIP

VoIP (Voice Over Internet Protocol) refers to the communication protocols, technologies, methodologies and transmission techniques involved in the delivery of voice communications and multimedia sessions over the Internet Protocol network (IP)

OR

SUB ICT SIMPLIFIED

VoIP (Voice Over Internet Protocol) is a method for taking analog audio signals and converts them into digital data that can be transmitted over the internet.

TERMS USED IN INTERNET

1. URL

This is an acronym for Uniform Resource Locator. It is the global address of a website and other resources on the World Wide Web (WWW).

It is a string of characters that specify the address of a web page.

2. Uploading

This is the process of transferring files (Text, Audio, Video, Graphics and Sound) from one computer system to another or to a storage device or the internet.

3. Downloading

This is the process of retrieving or bringing copies of files (Text, Audio, Video, Graphics and Sound) over to your computer or storage device from another location such as the internet.

4. Surfing

This is the process of accessing Data from the internet.

5. Online meetings

This is where people who are far away can chat virtually using services such as video conferences, Skyping, etc

FACTORS THAT LEAD TO A SLOW INTERNET SPEED

- **Quality of ISP:** Some ISPs provide slow connections because of outdated technology.
- **Age of the computer set used:** Computers with outdated technology (old processors) slow down the connection speed of internet.
- **Type of communication medium used:** Physical transmission medium is slow compared to Wireless
- **Distance of broadcasting services:** If far away from the computers.
- **Type of browsers:** Some browsers may be using outdated technology.
- **Weather:** Prevailing bad weather conditions may affect internet speed.

SUB ICT SIMPLIFIED

PRECAUTIONS THAT SHOULD BE TAKEN WHEN DOWNLOADING CONTENT FROM THE INTERNET

- Do not download content from unknown or an trusted sites.
- Be sure that your internet connection is stable and does not break.
- Ensure that there exists enough storage space for the content to be downloaded.

INTERNET SERVICE PROVIDER (ISP)

This is a company that supplies connections to the internet, including offering services related to the internet.

EXAMPLES OF ISPs IN UGANDA TODAY

- Uganda Telecom Limited
- Orange Telecom
- Mobile Telecommunications Network (MTN)
- Airtel Uganda
- Iway Broadband
- Inconnect
- Smile Telecom

SERVICES PROVIDED BY INTERNET SERVICE PROVIDERS (ISPs)

- Internet access
- Domain name registration
- Web hosting
- Providing customized e-mail addresses
- Access to software tools
- Provide fire walls
- Manage internet traffic e.g. By use of dedicated lines
- Provide backup storage of data.

FACTORS TO CONSIDER WHEN CHOOSING A SUITABLE ISP

- Initial connection cost
- The subscription fees for the services
- The available band width

SUB ICT SIMPLIFIED

- Authenticity
- Additional value services such as web hosting, firewalls, backup storage of data
- Experience in the internet field

E-MAIL (Electronic mail)

Electronic mail (e-mail) is the process of sending and receiving messages over the internet.

OR

This is a method of exchanging digital messages from one or more recipients across the internet or other computer networks.

EXAMPLES OF EMAIL SOFTWARE

- Gmail
- Yahoo mail
- Hotmail
- Ovimap

FEATURES OF THE ELECTRONIC MAIL

- Enable people to send many copies of the same message to many people
- The mail is sent instantly
- Enables attachments
- Indication for read and unread messages
- Sent box for storing sent mails
- Outbox for storing outgoing mails
- Inbox for storing incoming mails

TERMS USED IN ELECTRONIC MAIL

- 1) **Compose**
This is the process of creating a new mail
- 2) **Inbox**
This is the storage location of incoming mails
- 3) **Sign out**
This is the process of exiting an email account
- 4) **Forward**
This feature allows one to pass on a received mail to other recipients
- 5) **Check mail**
This feature enables a user to view all the details of a received mail.

FUNCTIONS OF THE COMMON FIELDS ASUSED IN EMAILS

- 1) **To:**
This specifies to whom the message was sent. This is a field where you type in the recipients email address.
- 2) **Subject:**
This provides a hint as to what the message is about.
- 3) **Carbon Copy (CC):**
This tells us that the message was “carbon copied” to another user. In other words this field is often used when one wants to send a copy of the email to other recipients.
- 4) **Blind Carbon Copy (BCC):**
This field is often used if additional copies of the same email message are to be sent out to other recipients but without each of them knowing about it.
- 5) **Attachment:**
This is a textual, graphical, audio or video file that accompanies an email. It can be downloaded by the recipient for use

An email compose window consists three main elements: header, message and signature. Mention three items contained in the header.

- Recipient's address
- The subject
- Attachment

What do you understand by mail etiquette as applied in sending mails

- These are the moral rules (Social code) to be observed when using email services.

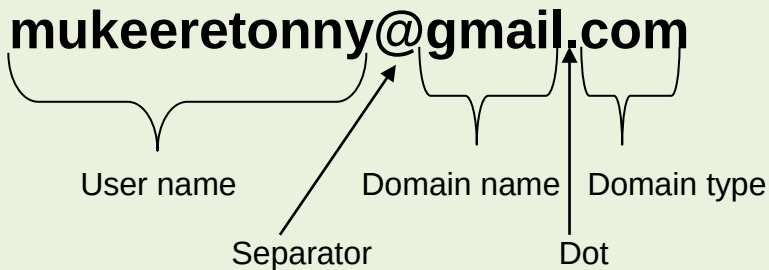
State one mail etiquette

- Usenet

SPAM is known to be unsolicited mail inbox. State any three potential harm that such mail can bring.

- It leads to congestion of mail box
- They sometimes carry malware / viruses
- They can be an avenue for fraud
- They sometimes contain porn content which is undesirable.

PARTS OF AN EMAIL ADDRESS



- **User name:** This specifies the identity of the recipient of the email
- **@:** This at symbol acts as a separator between the user name and the domain name
- **Domain name:** This identifies the domain that hosts the email account
- **Dot(.):** This separates the domain name and domain type
- **Domain type:** This specifies the category of the domain

Distinguish between sign in and sign up as used in emails

- **Sign in** is used to access content of an already existing email account by use of a user name and password.
- **While**
- **Sign up** is used to create a new email account.

Hope's company email address is hopem@mutocorporation.com

Describe the parts of his email address

- **hopem:** is the user name
- **@:** the separator between the user name and domain name
- **mutocorporation:** is the domain name
- **Dot(.):** this separates the domain name from the domain type
- **Com:** this identifies the category of the domain in this case commercial

ADVANTAGES OF E-MAILS IN BUSINESS

- It offers a speedy communication to business clients
- It is reliable compared to other forms of communication

SUB ICT SIMPLIFIED

- One can communicate to multiple people at the same time using the mail merge feature.

ADVANTAGES OF USING EMAIL SOFTWARE OVER THE POSTAL SYSTEM.

- The message sent is usually instant
- Email is more secure since there may be no intermediate persons
- It is possible to send the message content to multiple people. E.g. using the mail merge feature.
- It is convenient to as one can access it from anywhere at any time.
- A person can access a variety of information cheaply such as audio, text, graphics and video as attachments.
- It takes less time to compose and send the message.

DISADVANTAGES ASSOCIATED WITH THE USE OF EMAILS AS A COMMUNICATION METHOD

- The Email system is not possible without an internet connection
- The Email sent with attachments is susceptible to virus and worm infection
- The Email system does not favor illiterates
- It only accommodates Intangible data(soft copy)
- It is expensive to setup an internet connection for sending emails.

ADVANTAGES OF USING MOBILE PHONES FOR INTERNET ACCESS

- They are portable, thus can be used from anywhere
- They are readily available
- They are cheaper to users
- Immediate contact is available
- They offer faster connections
- They save time which would be wasted in crowded internet cafes.

DISADVANTAGES OF USING MOBILE PHONES FOR INTERNET ACCESS

- They can easily be lost
- They have limited display of content
- Large volumes of data may not be downloaded
- Some phones cannot browse certain websites
- Better internet connections may be limited to advanced phones such as smart phones

POPULAR TYPES OF E-COMMERCE CARRIED OVER THE INTERNET

i. Online shopping:

Customers can view, select, order and pay for their merchandize online. Customers can use Electronic fund transfer method or E-money to pay goods and services.

ii. Online advertisement:

SUB ICT SIMPLIFIED

Advertisers pay for the search service and customers can be targeted for specific types of Ads on the basics of their search results.

iii. **Online banking:**

This involves the Electronic transfer of funds using credit card applications

SOME MOBILE NETWORK ACRONYMS

i. **GSM**

Global System for Mobiles

ii. **GPRS**

General Packet Radio Service

SOME INPUT/OUTPUT DEVICES NEEDED FOR VIDEO CONFERENCING

- Webcam
- Microphone
- Speaker

DOMAIN NAME

This is a unique name that denotes a company, an organization, society or group on the internet.

TYPES OF DOMAIN NAMES

- i. **Com**
Commercial business e.g. www.microsoft.com or www.mountofolivescollege.sc.ug
- ii. **Org**
Nonprofit Organization e.g. www.ngo.org
- iii. **Edu**
Education e.g. www.ncsu.edu or www.duke.edu
- iv. **Net**
Networking organization e.g. www.earthlink.net
- v. **Gov**
Government agency e.g. www.epa.gov or www.whitehouse.gov
- vi. **Mil**
Military e.g. www.army.mil or www.navy.mil

EXAMPLES OF TOP LEVEL DOMAIN NAMES

- .com
- .org
- .net
- .gov
- .edu

➤ .mil

WORLD WIDE WEB (WWW)

This is an internationally complex system of servers or web servers that communicate with each other over the internet with client machines running browser software.

TERMS RELATED TO WORLD WIDE WEB

i. **Web browser**

This is an application software or program designed to enable users access, retrieve, view documents and other resources on the internet. E.g. Internet Explorer, Mozilla fire fox, Google chrome, Opera, Netscape, etc

ii. **Search engines**

These are programs that enable users to search for documents and other resources on the World Wide Web. E.g. Google, Yahoo search, Dog pile, Bing, etc

iii. **Web server**

This is a computer that satisfies users requests for web pages.

iv. **Web master**

This is a person who creates, maintains and manages a website often for a business, organization or company.

v. **Home page**

This is the starting page or table of contents for a website. OR **Home page** is the first page of a web site.

WEB BROWSERS

A **web browser** is a application software or program that enable users to access, retrieve, view documents and other resources on the internet.

COMMON EXAMPLES OF WEB BROWSERS

- Google chrome
- Mozilla fire fox
- Internet explorer
- Opera
- Netscape
- Mosaic
- Lobo
- Kmeleon
- Safari
- Sea Monkey
- Abaco
- Camino
- Epiphany

SUB ICT SIMPLIFIED



Flock

FEATURES OF A WEB BROWSER



The display area for displaying a webpage



Navigation tools for moving back and forward in a site



The refresh button for reloading a webpage



A combo box for entering the URL address



A button for terminating the process of loading a web page



A button for taking a surfer to the default homepage for the browser

WEBSITES AND WEB PAGES

A website is a collection of related web pages

While

A web page is a file (document) that can be accessed and read over the World Wide Web.

TYPES OF WEBSITES



Online Social Networking websites

These are websites that build online communities who want to interact with other people socially or meet people to share their interests e.g. My space, Google+, twitter, facebook, whatsapp, etc.



Information websites

These are websites that enable information or resource sharing over the web e.g. Wikipedia.org



E-commerce websites

These are dynamic websites that carry out business transactions electronically over the internet. E.g. business.com, google business, Alibaba.com



Internet blog

These are sites generally used to post online diaries, journals which may also include discussion forums. Eg. Xanga, blogger, Usually a blog owner will logon daily or whenever necessary post anything.



A wiki

This is a website that allows its users to add, edit its contents via a web browser using a richtext editor or markup language.

MAIN ACTIVITIES THAT CAN BE CARRIED OUT ON AN ACTIVE WEB PAGE



Select an item

SUB ICT SIMPLIFIED

- Navigate through the web page
- View the content on the web page
- Print the page
- Follow links to other pages
- Download items that are downloadable on the page
- Copy items from the webpage
- Refresh the page

TERMS USED IN THE WORL WIDE WEB

❖ **Surfer**

This is a person who spends time exploring the world wide web.

❖ **Web portal**

This is a website that provides specialized services such as email, searching, news updates financial updates, weather and links to selected websites.

❖ **Browsing**

This is the process of navigating through web pages and websites over the internet.

❖ **Newsgroups**

These are organized groups of internet users sharing ideas and interests.

❖ **VoIP**

VoIP (Voice Over Internet Protocol) refers to the communication protocols, technologies, methodologies and transmission techniques involved in the delivery of voice communications and multimedia sessions over the internet protocol network (IP)

Or

VoIP (Voice Over Internet Protocol) is a method for taking analog audio signals and turning them into digital data that can be transmitted over the internet.

URL (Uniform Resource Locator)

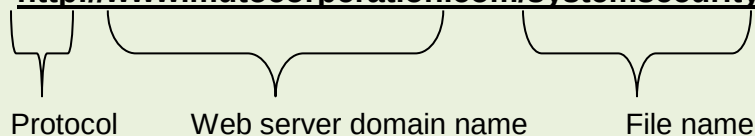
This refers to a string of characters that specify the address of a web page.

MAIN PARTS OF THE URL

There are three main parts of the URL and these include;

- Protocol
- Web server's domain name
- File name

Example. <http://www.mutocorperation.com/systemsecurity.pdf>



Distinguish between a Hyperlink and a Hypertext

SUB ICT SIMPLIFIED

- **A hyperlink** is a string of clickable text or graphics that points to another Webpage or document
- While**
- **A hypertext** is a web page that has hyperlinks to other web pages

Study the URL below and answer the questions that follow:

<http://www.muto.org/countries/uganda>

- (a) **Explain the various parts of the URL**
- **http** (hyper text transfer protocol) defines the protocol or scheme by which to access the web page
- www.mock.org identifies the domain name of the computer (server) where the webpage resides. In this case the server is located at MOCK
-
- **Countries** represents the path of the URL
- **Uganda** is the document name that is currently opened
- (b) **What is the top level domain in this URL**
- Org (Organization)

MULTIMEDIA SITES

These are websites for entertainment and news mainly containing videos, music, radio broadcasts, Live TV

EXAMPLES OF MULTIMEDIA SITES

- You Tube
- Tube mate
- Fan fridge
- Free Book search
- FindSounds
- Meta Café
- Musgle

Write these acronyms in full

- **PDA**s
Portable Digital Assistants
- **HTML**
Hyper Text Markup Language
- **IBM**
International Business Machine
- **Wi-Fi**
Wireless Fidelity
- **ATA**

SUB ICT SIMPLIFIED

Analog Telephone Adapters

INTERNET TELEPHONY

This refers to a broad range of services over the internet such as transmitting voice, video and even fax that were traditionally performed over the telephone line.

OR

Internet telephony also known as VoIP (Voice Over Internet Protocol) refers to the communication protocols, technologies, methodologies and transmission techniques involved in the delivery of voice communications and multimedia sessions over the internet protocol network (IP).

WAYS HOW INTERNET TELEPHONY CAN BE USED

- **PC to PC**
Users can connect to anyone else using the same program for real time video chatting over an existing internet connection.
- **PC to Phone**
Some programs such as Skype allow a computer user to dial an actual phone number, so that one user is using their computer microphone and the other user is using telephone.
- **PC to fax**
Internet telephony can be used to send fax transmissions to a physical fax machine.
- **Phone to Phone services**
This is where a user needs to purchase a device (such as Casio PAP2T) and plug it into an internet router, thus allowing any phone to be plugged into it.
- **Mobile phone to PC or Phone**
Skype provides free applications for smart phones to enable Skype users to make free phone calls or at a small fee.

The emerging usage of the ICTs has seen the Electronic form of payment (e-payment) used by most service sectors and organizations. Mention any three companies/ government entities in Uganda using e-payment system.

- UMEME
- Uganda Revenue Authority (URA)-e-tax
- Kampala Capital City Authority (KCCA)
- National Water and Sewerage Cooperation

State any four advantages of using the e-payment method rather than the traditional way of money payment.

- It is fast and more reliable
- Money paid is in safe custody
- It eliminates issues of fraud since there is no cash payment in the organization

SUB ICT SIMPLIFIED

- It also eliminates issues of forgery and bribery.

TECHNOLOGICAL REQUIREMENTS FOR PEOPLE WHO USE INTERNET TELEPHONY SUCH AS SKYPE.

- Pentium – use a class PC with standard sound card having microphone input jack and headphone/ speaker output jack.
- An internet connection of high speed.
- Internet telephone software such as Microsoft Net Meetings
- A microphone to pick up your voice
- A speaker for hearing the person you are calling
- PC headset
- PC camera / Webcam

PROGRAMS THAT SUPPORT INTERNET TELEPHONY

- Skype
- Yahoo messenger
- AOL instant messenger
- Whatsapp
- Palm Chat. etc

SERVICES OFFERED BY INTERNET TELEPHONY

- Voice (VoIP)
- Video e.g. Video phone calls
- Transmitting fax

ADVANTAGES OF USING INTERNET TELEPHONY OVER THE TRADITIONAL CONVENTIONAL WAY

- **Low cost**
Some internet telephony companies allow free PC to PC calls for its software users. PC to phone connections are cheaper than traditional phone calls.
- **Support conference calling**
Some Internet telephony companies offer a conference calling features as part of the service package for no additional fee
- **Portability**
Internet telephony services work anywhere thus the user finds available broadband access.
- **Support video**
Internet telephony features has the ability to enable users to transmit and watch video, thus one with a laptop and internet connection can use internet telephony from anywhere.
- **Area Code**
A subscriber to VoIP/ internet telephony service is able to assign any area code to his phone number for incoming calls.

SUB ICT SIMPLIFIED

- Internet telephony provides free features such as caller ID, call waiting, call forwarding unlike with traditional phone lines.

DISADVANTAGES OF USING INTERNET TELEPHONY OVER THE TRADITIONAL CONVENTIONAL WAY

- Poor broadband connection makes internet telephony unreliable.
- Internet telephony may suffer from problems like dropped calls and audio distortion due to network problems.
- Older computers with slower processors and limited RAM may not support internet telephony calls.
- It has limited emergency call support. Emergency services cannot trace internet telephony calls since the calls do not originate at a physical location.
- Internet telephony may give poor sound quality compared with analog phone lines.
- Electricity: If power goes out or if the laptop battery runs low, it may take with an internet connection, thus no internet telephony.
- Compatibility issues: Some VoIP may have compatibility issues with hardware devices such as conventional modems that may need to dial out an outgoing number. Some of these devices may be set up for voice and not data.

TELECOMMUTING

This is a work arrangement in which an employee works at a location that is remote from the actual business facility at which he is employed.

TECHNOLOGICAL REQUIREMENTS SUPPORTING TELECOMMUTING.

- Teleconferencing
- Video conferencing
- Internet telephony
- Mobile telecommunication technology

Telecommuting is a work arrangement in which an employee works at a location that is remote from the actual business facility at which he is employed.

Mention three employees concerns about telecommuting.

- **It can negatively affect a person's career**
There is a decrease in productivity (poor performance) due to a lack of supervision. Also the employee is less likely to get promoted since he hasn't consistently seen and appraised.
- **Worry of technological issues involved**
Such worries such as hardware failures, poor internet connection and failure to trouble shoot.
- **Social isolation**
An employee may miss out on the physical interaction with co-workers due to work at remote area.
- **Missing out on important issues at actual company**

SUB ICT SIMPLIFIED

Telecommuters may miss out on information sharing and new corporate strategy and at high projects at their companies.

➤ **Co-worker jealousy**

Other employees miss out on telecommuting may imagine that one who telecommutes receives more privileges thus jealousy may come up.

BENEFITS OF TELECOMMUTING TO A COMPANY

- Telecommuting insures continuity of operation in the event of a disaster.
- It is more environment friendly (greener approach to work).
There is a reduction to office energy consumption since few hardware devices will be using energy.
- It is cost efficient/ saving
Companies may save on costs of transport, fuel, food on their employees.
- Increase productivity of employees
Telecommuting removes office distractions and allows employees to concentrate on their duties.
- Telecommuting equalizes personalities and reduces potential for discrimination.
This ensures that people are judged by what they do and not their looks.
- It increases collaboration
Once telework technology is in place, employees and contractors can work together without regard to logistics. This increases collaboration options.
- Telecommuting reduces staffing redundancy and offers scale up and scale down options.
The need to over staff just in case is greatly reduced.
- Enables companies to retain key employees.
Losing a valuable employee can cost an employer millions of shillings
- Saving space
Telecommuting ensures that the company saves a lot an office space and utility (heating, air conditioning, water, and sewer)

CHAPTER SIX

DATA COMMUNICATION & NETWORKING

- ❖ Introduction to the data communication
- ❖ Elements of data communication
- ❖ Transmission media
- ❖ Networking

INTRODUCTION TO DATA COMMUNICATION

This refers to the high speed data exchange between computers and or other electronic devices via network media such as cables or wireless.

ELEMENTS OF DATA COMMUNICATION

1. **Sender:**
This refers to a device that initiates an instruction to transmit data and instructions e.g. a source computer.
2. **Receiver:**
This is a device that accepts the data signals from the source device.
3. **Transmission medium:**
This is a communication channel or path over which the data signals are sent e.g. Cables or wireless.
4. **Protocol:**

SUB ICT SIMPLIFIED

This is set of rules and procedures for exchanging information among computer devices on a network. E.g. Ethernet, TCP/IP.

5. Messages:

The data/content/information to be relayed or transmitted over a transmission medium.

6. Communication device/ signal converter:

Is a device that converts data or instructions from the sending device into signals that can be carried out by a transmission medium to a receiver.

DATA COMMUNICATION TOOLS

These are tools that enable you to emulate and test networks.

FEATURES THAT DIFFERENTIATE MANUAL FROM ELECTRONIC DATA COMMUNICATION

Manual data communication	Electronic data communication
Does not require subscription or payment for usage in communication	Requires some form of subscription or payment for the service
They are cheap to maintain	Are costly to maintain especially when servicing of hardware.
The range of communication is limited to a gathering especially in ceremonial settings such as rituals	The range of communication is quite wide, globally using tools such as Emails, mobile phones, IMs
Usually involve messengers or couriers e.g. Postal services, Office messengers	Usually involve use of internet facility such as email, IMs

SERVICES OFFERED BY DATA COMMUNICATION

1. Email

This is a method of exchanging digital messages from the author to one or more recipients across a computer network or the internet.

2. Skype

This is a service that allows users to communicate with peers by voice using a microphone or by video using a webcam and instant messaging using internet.

3. Instant messaging (IM)

This is a set of communication technologies used for text based messaging between two or more participants over the internet or other types of networks.

4. News groups

These are forums or discussion groups usually within the Usenet system for messages posted from many users in different locations.

TERMS USED IN DATA COMMUNICATION

SUB ICT SIMPLIFIED

1. **Bandwidth**

This is the maximum amount of data that can be transmitted by a communication medium in a fixed amount of time.

2. **Network topology**

This is the physical or logical arrangement / organization of computers, communication media and other peripherals to facilitate data flow in a network.

3. **Duplex**

This refers to a transmission of data signals in both directions simultaneously.

4. **Simplex**

This is transmission of signals in only one direction.

5. **Half duplex**

This is where transmission of signals is possible in both directions but one direction at a time.

TRANSMISSION MEDIA

There are two types of transmission media namely;

➤ **Physical transmission media:**

Physical transmission media provides a physical path usually restricted along which signals are relayed e.g. cables

➤ **Wireless transmission media:**

Wireless transmission media transmits data signals from one point to another without using physical connections.

FEATURES TO CONSIDER WHEN CHOOSING A COMMUNICATION/ TRANSMISSION MEDIA

- The speed of data transmission
- The cost of media
- Installation costs
- Data security
- Country/ company policy
- Availability in the market
- Size of the network

TRANSMISSION MEDIA IN DATA TRANSMISSION

- Twisted pair cables
- Coaxial cables
- Optical fibers

ADVANTAGES OF OPTICAL FIBRE IN DATA TRANSMISSION

- It has greater capacity in terms of bandwidth. Thus hundreds of GBps over tens of kilometers have been demonstrated.
- Are highly secure, as they are resistant to tapping, eaves dropping.
- Enable faster data transmission since they use light to transmit data,
- They are immune to electromagnetic and electrical interference which bring about noise.

SUB ICT SIMPLIFIED

- It is cheap when transmitting for longer distances.
- Has lower attenuation which is constant over a wide range.
- They are smaller in size and lighter in weight hence offering an easier way to use along public ways and reducing structural support requirements.

Distinguish between Digital and analog data

- **Digital / Discrete data** is data that is encoded / represented as distinct values usually in form of zero's and one's
- While**
- **Analog / Continuous data** is data encoded according to a continuous physical parameter such as height, amplitude of a waveform.

CAUSES OF DEGRADATION OF SIGNALS DURING DATA TRANSMISSION

- Presence of noise (unwanted signals)
- Interference from other signals
- Poor weather / environmental conditions which leads to signals to travel lesser distance
- Faulty or poor quality hardware equipments
- Distortion due to change of shape of signal which may be caused by attenuation.

WIRELESS TECHNOLOGY

Define the term wireless technology?

Wireless technology is a kind of data signal transmission from one point to another without using physical connections.

EXAMPLES OF WIRELESS TECHNOLOGIES USED IN DATA TRANSMISSION

- Microwaves
- Infrared
- Wi-Fi (Wireless Fidelity)
- Blue tooth
- Satellite (broadcasting) communication

ADVANTAGES OF USING A WIRELESS TECHNOLOGY COMPARED TO OTHER TRADITIONAL TECHNOLOGIES.

- It is a cheaper deployment of a Local Area Networks (LANs)
- It is faster to setup as it does not have many connections to be made.

SUB ICT SIMPLIFIED

- Enables expandability. Adding new devices/ computers to the network is easy.
- Mobility. It can be used in remote locations provided there is network coverage.

LIMITATIONS WITH THE USE OF WIRELESS FIDELITY (Wi-Fi) AS A COMMUNICATION MEDIUM,

- Wi-Fi networks have a limited range making it difficult to communicate with distant devices.
- Wi-Fi has a high power consumption compared to other standards such as Bluetooth. This makes the battery life in Wi-Fi mobile devices a concern.
- Mobile use of Wi-Fi over wide ranges is limited, for instance in automobile moving from one hotspot to another.
- Wi-Fi is less secure than wired connections (Ethernet) since the intruder does not need physical connection. Unencrypted internet access can easily be detected by intruders.
- Wi-Fi possesses a high data security risk especially for open wireless access to a LAN (unencrypted Wi-Fi networks)
- There is always interference. Wi-Fi connections can be disrupted or the internet speed lowered by having other devices in the same area. There is Wi-Fi pollution due to the excessive number of access points in the same area which can lead to interference/ overlapping of signals.

Distinguish between Baseband and Broadband transmission of signals in data communication.

- Baseband transmission is where only one signal can be sent at a time since it utilizes the full capacity of the transmission medium.
While
- Broadband transmission is where several signals can be sent simultaneously over the same transmission medium but at different frequencies.
- Baseband is digital signal **while** Broadband is analog signals.

DATA TRANSMISSION TECHNIQUES

1. **Blue tooth**
This is a technology standard of exchanging data over short distances using short wavelength radio transmission over mobile devices.
2. **Satellite**
This is a broadcasting transmission in effect from a microwave relay station where two or more ground based microwave transmitters/ receivers are linked together
3. **Wi-Fi**

SUB ICT SIMPLIFIED

This is a popular technology that allows an electronic device to exchange data wirelessly (using radio waves) over a computer network, including high speed internet connection.

4. **Infrared**

This is a technology standard of electromagnetic radiation used for data transfer in industrial, scientific and medical applications.

5. **Microwaves**

This is a technology of radio waves which is extensively used for point to point telecommunication

ADVANTAGES AND DISADVANTAGES OF USING MOBILE PHONES

The advantages out way disadvantages of using Mobile phones as a communication medium compared to the traditional way of communication. Mention three advantages and two disadvantages of using mobile phones.

Advantages

- The message can be received instantly
- Mobile phones are faster in communication
- They offer at most privacy during communication
- They favor both literates and illiterates
- They can be used from anywhere as long as there is reception of network signals
- Most mobile phones provide a variety of services e.g. Internet connectivity, games, radio, TV and teleconferencing, etc

Disadvantages

- They are expensive in terms of usage and maintenance
- Can easily be stolen
- When they develop mechanical faults, they ruin communication
- They cannot be used in areas where there is no network coverage
- Sometimes they are susceptible to tapping or eaves dropping by security agencies of the government hence depriving privacy of an individual.
- Over exposure to radiation from these phones leads to health problems.

APPLICATIONS OF DATA TRANSMISSION TECHNIQUES

1. **Twisted pair cables**

- It is used within buildings for LANs supporting personal computers.
- It is the most commonly used medium for digital signaling e.g. for connections to a digital switch.
- It is the most commonly used medium in wired telecommunication networks and is the workhorse for communication within buildings

2. **Satellite**

- They are used in television distribution e.g. for broad cast feeds to and from television networks and local affiliate stations; in video conferencing; in distance learning by universities and schools.
- In long distance telephone transmission.
- In private business network e.g. when transmitting a program from a central location to the satellite, to local stations then to individuals.
- For radio audio services in instances where listeners roam a continent listening to the same audio programming anywhere.
- For Internet access in remote areas that cannot access broadband connection.
- Communication satellites are used for military communication applications, such as Global Command and Control systems

3. Blue tooth

- For wireless control of and communication between a mobile phone and a hands free headset.
- Wireless networking between PCs in a confined space and where little band width is required.
- Personal security application on mobile phones for prevention of theft or loss of items
- For wireless control of and communication between a mobile phone and a blue tooth compatibility car radio stereo system.
- For wireless communication with PC input and output devices such as keyboard, mouse, printer, etc.
- For sending small advertisements from Bluetooth enabled advertising hoardings to other discoverable Bluetooth devices.
- For dial up internet access on personal computers or PDAs using a data capable mobile phone as a wireless modem.
- Allowing a DECT phone to ring and answer calls on behalf of nearby mobile phone.

4. Wi-Fi

- Used in personal computers to connect a network resources such as the internet via a wireless network access point.
- Used in Video game consoles to connect a network resource.
- Used in digital audio player to connect a network resource.

5. Microwaves

- They are used in broadcasting and telecommunication transmissions such as televisions broadcasts, in Metropolitan Area Network (MAN) such as WiMAX.
- In Radars(found in air traffic control, whether forecasting, navigation of ships and speed limit enforcement) when detecting the range, speed and other characteristics of remote objects.
- They are used in Radio astronomy e.g. when determining the distance to the moon or mapping the invisible surface of Venus through cloud cover,
- They are used in navigation such as the Global Position System(GPS)
- They are used in micro ovens during the heating process in industrial process (drying and curing of products)

SUB ICT SIMPLIFIED

- They are used to transmit power over long distances e.g. in Solar power satellite with systems with large solar arrays that would beam power down to the earth's surface via microwaves.

COMPUTER NETWORK

This is a collection of computers and other hardware devices linked together using a communication media for purpose of communication and resource sharing.

IMPORTANCES OF A COMPUTER NETWORK

Mount of Olives College Kakiri is considering to set up a computer network in its premises. They have contracted an IT professional for advices on this task.

Briefly mention any five purposes of computer networking

- To enable resource sharing like printers, storage devices, modems, application programs, data, etc
- To enable remote communication between devices using services such as Emails, chat rooms, instant messaging, teleconferencing, etc
- To enable distributed processing facilities e.g. data can be processed from many terminals and then stored on a common server.
- To enable data security and management. A network administrator can easily monitor and manage a company's sensitive data with high security.
- To ensure cost effectiveness and reliability in an organization.
- Reduces chances of data loss in an organization.

FACTORS TO CONSIDER WHEN SETTING A COMPUTER NETWORK

State five factors an IT professional considers before setting a computer network for an organization.

- Cost of installation. An organization has to choose a network basing on cost effectiveness and within their budget.
- Number of computers and other hardware devices to be connected.
- The communication medium to be used such as cabling system or wireless technology.
- The architecture of the building to be used.
- The number of users to access the network. A bigger number of users require a large network.

SUB ICT SIMPLIFIED

- The bandwidth to be used on the network. This will determine the speed of the network.
- Expansion of the network. The future expansion of a network will depend on the topology used i.e. a star topology easily enables future expansion
- The technicalities involved in setting the network.
- The safety provisions of the network.

MAJOR REQUIREMENTS OF SETTING UP A COMPUTER NETWORK

- Clients/ workstations
- Server
- Transmission media (either physical or wireless)
- Network interface card (NIC)
- Router/ bridges/ gateways
- Protocols
- Network software like Networking Operating System.

LIMITATIONS OF COMPUTER NETWORKING

- Costly and Complicated: It is expensive to install cables and other network devices, and also to maintain the network.
- Unreliability/ Downtime: A fault in the network server, switches, routers can prevent the whole network from working.
- Security concerns: Networks are prone to hackers and crackers.
- Harmful content: The internet is full of harmful/ inappropriate content such as porn. The internet may also expose one to scams.
- Virus issue: Computer networks are prone to virus attacks which may eventually spread and corrupt the data.

USES OF LAN TO A SCHOOL ENVIRONMENT

- It acts as an Access point to the internet with installation of a gateway.
- For monitoring of computer users
- For communication among users
- To enable resource sharing like printers, storage devices, modems application programs, data, etc among others.

NETWORK DEVICES

1. Terminal

This is the work station/ client/ computer from where data is sent or received

2. Server

- This is the main computer that facilitates data communication
- It is the computer running software that enables it to serve specific requests from other computers/ clients

3. Transmission media

This is a communication channel or path over which data signals are sent from source to destination. This can either be physical or wireless

4. Network Interface Card (NIC)/ Network Adapter

It is a piece of computer hardware designed to allow computers to physically access a networking medium.

5. A switch

- This is a multipurpose device which gets input signals in one port and sends the output signals to many of its other ports.
- It is a device that forwards and filters chunks of data communication between ports based on MAC addresses in the packets.

Distinguish between Peer to Peer and Client server models in computer networks.

A Peer to Peer network model allows any user to share files and other resources with any other user and does not require a central dedicated server computer. In this model each computer acts as a server and a client.

While

In a server client network model environment, files are stored on a centralized high speed file server PC that is made available to client PCs. All other networking devices such as printers and internet connections are routed through the server.

Mention five advantages of a client server model over Peer to Peer client model network.

- Centralization: Resources and data security are controlled through the server.
- Scalability: Any or all other elements can be replaced individually as the needs increase.
- Flexibility: New technologies can easily be integrated into the system.
- Interoperability: All components (client, servers & network) work together.
- Accessibility: Servers can be accessed remotely and across multiple platforms.

TYPES OF COMPUTER NETWORKS

1. Local Area Network (LAN)

This is a network used to interconnect many computers within a given local area, usually in premises of a single organization building such as a school. A very high speed data transmission can be attained within a limited geographical area.

2. Metropolitan Area Network (MAN)

This refers to the interconnection within geographical limits of a city or town. MAN may be owned by a single company that has many other branches across a metropolitan area each having its own LAN. The use of WIMAX.

3. Wide Area Network (WAN)

This is a network used to interconnect a number of widely dispersed computers in various cities of a country or different countries. These networks usually have telephone lines, underground

SUB ICT SIMPLIFIED

coaxial cables, and satellite and microwave communication. WAN usually consist of many LANs and MANs connected together to form a large network.

4. **Wireless Local Area Network (WLAN)**

This is a kind of local area network that uses non-physical network connection to communicate to nodes. Usually this does not involve use of cabling medium for data transmission.

5. **Wireless Fidelity (Wi-Fi)**

This is a popular technology that allows an electronic device to exchange data wirelessly (using radio waves) over a computer network, including high speed internet connection.

Give four reasons why WLAN would be preferred to a cabled LAN.

- It is a cheaper development of a Local Area Network (LANs)
- It is faster to setup as it does not have many connections to be made.
- Enables Expandability: Adding new devices/ computers to the network is easy.
- Mobility: It can be used in remote locations provided there is network coverage.

Give three reasons why the school should not use blue tooth technology to create a network.

- Blue tooth technology is limited to only a few devices.
- It cannot support higher band width applications.
- It has a limited range / area of reach.

NETWORK TOPOLOGIES

A network topology is the layout in which computers are arranged for communication with other devices.

MAJOR TYPES OF NETWORK TOPOLOGIES

1. **STAR TOPOLOGY**

In this topology all devices/ nodes are connected to a central hub. Nodes communicate across the network by passing data through the hub. The hub does the broadcasting to other nodes.

ADVANTAGES OF STAR TOPOLOGY

- Easy to install and wire.
- No disruption to the network when connecting and removing devices.
- Easy to detect faults in the network and to remove parts.

DISADVANTAGES OF STAR TOPOLOGY

- Requires more cable length than in bus topology.
- If the hub, switch, or connector fails, nodes or computer devices attached are disabled.
- More expensive than bus topology because of the high costs of a hub, switch etc

2. BUS/ LINEAR TOPOLOGY

All devices are connected to a central cable called the bus (backbone). In this topology, only one device can send data at a time.

ADVANTAGES OF BUS TOPOLOGY

- Easy to connect computers and peripheral devices to a linear bus
- Requires less cable length than a star topology.

DISADVANTAGES OF BUS TOPOLOGY

- The entire network shuts down if there is a break in the main cable.
- Terminators are required at both ends of the backbone cable.
- Difficult to identify the problem if the entire network shuts down.
- It is not meant to be used as a stand-alone solution in a large building.

3. RING TOPOLOGY

In this topology, all devices and computers are connected to one another in a circle. Each station/ node is responsible for regenerating and retransmitting signals across the network to its neighbours.

ADVANTAGES OF RING TOPOLOGY

- Transmission of data is fairly simple as it only travels in one direction.
- No data collision.
- Easy to expand the ring topology network.

DISADVANTAGES OF RING TOPOLOGY

- The network fails to work if a single computer device is disconnected.
- If the cable connecting devices breaks, the network fails to work.
- Data must pass through a computer until it reaches its destination. This can make slower than other networks.
- If there is any problem in the network, there is difficult to identify the cause.

4. TREE TOPOLOGY

This topology consists characteristics of **bus** and **star** topology. It consists of star-configured work stations connected to a linear bus backbone cable. It allows the expansion of an existing network.

ADVANTAGES OF TREE TOPOLOGY

- Enables point-to-point wiring for individual segments.
- Its supported by several hardware and software venders.

DISADVANTAGES OF TREE TOPOLOGY

SUB ICT SIMPLIFIED

- Overall length of each segment is limited by the type of cabling used.
- If the backbone line breaks, the entire segment goes down.
- More difficult to configure and wire than other topologies.

INFRARED

This is a technology standard of electromagnetic radiation with longer wavelengths than those of visible spectrum used for data transfer in industrial, scientific and medical applications.

APPLICATIONS OF INFRARED

- They are used in night vision, using infrared illuminations to observe people and animations without being detected.
- Are used in Astronomy by using sensor equipped telescopes to penetrate dusty regions of space; to detect objects such as planets.
- Infrared imaging cameras are used to detect heat loss in insulated systems, observing changing blood flow in the skin, etc.
- Infrared imaging is used extensively for military and civilian purposes such as target acquisition, surveillance, night vision, environment monitoring, etc
- Whether forecasting

Describe these classifications of Network software

1. Network Operating system

This operating system is specifically designed to optimize the networked computers ability to respond to service requests.

2. Network Protocols

This is a set of rules and procedures for exchanging information among computer devices on a network. They define how the transmission channel is established, how information is transmitted and how errors are detected & corrected.

Distinguish between Logical and Physical Topology, give an example of each.

Logical topology deals with the way data flows on one device to another on a network e.g. Ethernet, token Ring.

While

SUB ICT SIMPLIFIED

Physical topology refers to the physical layout or arrangement of computers on a network e.g. Star, bus, ring, tree and mesh topology.

NETWORKING OPERATING SYSTEM

This is the operating system specifically designed to optimize the networked computers ability to respond to service requests.

EXAMPLES OF NETWORK OPERATING SYSTEMS

- UNIX/ Linux
- Macintosh Operating System (Mac OS)
- Novell Netware
- Cisco Internetwork Operating System
- Microsoft windows server

FUNCTIONS OF NETWORKING OPERATING SYSTEMS

- It responds to requests from application programs running on the network.
- It helps to implement network security features
- It enables various processes on the network to communicate with each other.
- It provides access to network resources such as printers, files and modems.
- Supports network services like network card drivers and protocols.

State true or false for each of these networking concepts

WAN is faster than LAN in data communication	False
Infrared can be used in military lasers	True
Twisted pair cables are more expensive than optical fiber cables	False
WIMAX technology is used in MANs	True
Ethernet and token ring are examples of physical topology	False

Mention one function of the following networking devices

1. Router

This is a device that connects multiple computers to a single Digital Subscriber Line (DSL) for internet access.

2. Bridge

SUB ICT SIMPLIFIED

This reduces the amount of traffic on a LAN by dividing it into segments and address filtering.

3. Gateway

This is network device that allows access from one network to another e.g. from one LAN to another WAN.

4. Repeater

This is a device that receives a data signal from one point of a network, then amplifies it and then retransmits it to other parts of the network.

5. Network software

This is software that is designed to help set up, manage and monitor computer networks.

Coaxial cables are the most commonly used networking cable medium sometimes in the absence of twisted cables.

ADVANTAGES OF COAXIAL CABLES AS A TRANSMISSION MEDIUM

- They are more suitable mediums for faster data transfer
- They are more stable under high network traffic
- They are immune to electromagnetic interference (EMI)
- It is more efficient at carrying many analog signals at high frequencies

DISADVANTAGES OF COAXIAL CABLES AS A TRANSMISSION MEDIUM

- They are very expensive for long distance data transmission.
- They are not suitable for long transmission due to high attenuation
- They are susceptible to tapping and eaves dropping.

Distinguish between a Hub and switch.

A Hub broadcasts the packets to all computers on the network

While

A switch forwards data packets directly to the address node using the point to point method.

CHAPTER SEVEN

COMPUTER SECURITY, ICT ETHICAL ISSUES & EMMERGING TECHNOLOGY

- ❖ Computer Security
- ❖ Privacy and ethical issues
- ❖ Emerging Technologies
- ❖ ICT industry

COMPUTER SECURITY

Computer security refers to all processes and mechanisms by which Computer-based equipment, information and services are protected from unintended or unauthorized access, change or destruction.

FORMS OF COMPUTER SECURITY

There are two basic forms of computer security and these namely;

1. **Data security**
This is a means of protecting a database from destructive forces or unauthorized users. Data security can be in form of backup, data masking, disk encryption.
2. **Physical security**
This is the protection of personnel, hardware, programs, networks and data from physical circumstances and events leading to serious losses or damage to an organization. This security

SUB ICT SIMPLIFIED

can be in form of use of Fire extinguishers, use of sensors and CCTV to guard against intrusion by burglars.

Describe the meaning of “Hardware Security Threats”

This is a situation where a person can maliciously, forcefully gain access to hardware with intent to steal or destroy data saved on them.

SOME POSSIBLE THREATS TO DATA SECURITY

- Computer viruses
- Hacking
- Cracking
- Phishing
- Fraud
- Eaves dropping
- Sabotage
- Surveillance

TERMS USED IN COMPUTER SECURITY

1. Hacking

This is the act of breaking into a computer system to gain unauthorized access. This may also involve defeating security capabilities, use of unauthorized revelation of passwords.

2. Denial of service Attack

This is an attempt to make a machine or network resources unavailable to its intended users. It may also involve efforts of one or more people to temporarily or indefinitely interrupt or suspend services of a host connected to the internet such as those of banks, credit card payments gateways.

3. Spam

This is the unsolicited sending of bulk email to users of internet such as email accounts

4. Data masking

This is the process of obscuring or masking of specific data within a data base table or cell to ensure that data security is maintained and sensitive information is not exposed to unauthorized personnel.

5. Data erasure

This is a method of software based overwriting that completely destroys all electronic data residing on a hard drive or other digital media to ensure that no sensitive data is leaked when an asset is retired or reused.

Natural disasters can cause loss of data

WAYS OF ENSURING THAT DATA AND PROGRAMS ARE NOT LOST IN CASE OF A DISASTER.

- Make a backup; this is the making a duplicate (copy) of the data
- Store data/ information in a network
- Store data to areas which are not prone to natural disasters.
- Build information systems with structures that can withstand natural disasters

WAYS OF RECOVERING DATA LOSS THROUGH ACCIDENTIAL ERASURE

- Make use of backups

SUB ICT SIMPLIFIED

- Use the data recovery utility
- Recover the lost data through the recycle bin, restore it
- Use the Undo command to reverse the deleted action

SOME OF THE INTERNET/ NETWORK ATTACKS

- **Hardware security mechanisms:** Is a situation where a person can maliciously, forcefully gain access to hardware with intent to steal or destroy data saved on them.
- **Denial of service attack:** This is a mechanisms mostly by hackers meant to disrupt data flow and communication between machines, making system resources unavailable.
- **Sniffing:** This is the act of intercepting TCP/IP packets while getting transferred on a network.
- **Virus and worms:** These attacks have the ability to destroy data and also replicate themselves in programs and files.
- **Trojan horses:** These are useful tools by hackers to try and break into a private network. Hackers usually attach these Trojan horses to files which triggers viruses.

WAYS OF PROTECTING AGAINST EACH OF THE ABOVE NETWORK/ INTERNET ATTACKS

- We can protect against Hardware security threats by use of a locking mechanism for desktop or use of chain mechanisms for laptops
- We can protect against Denial of service attacks by use of fire walls
- Sniffing can be protected against by use of fire walls
- Virus and worms can be protected against by use of Genuine Antivirus software.

COMPUTER VIRUSES

What is a computer virus?

This is a piece of malicious software designed to infect a computer system.

OR

This is a mini program that duplicates itself and attaches its replica to other programs and documents.

TYPES OF COMPUTER VIRUSES

1. Trojan horse:

These are malicious programs that enter a system or network under the disguise of another program.

2. Stealth virus:

This is a virus type that disguises itself by redirecting commands around itself to avoid detection.

3. Computer worm:

This is a malicious programs that auto replicates or is capable of multiplying itself there by using up all the available memory space.

4. Polymorphic virus:

This is a kind of virus which changes form/ hides from the antivirus software or it will encrypt itself to avoid detection.

5. Retro virus:

SUB ICT SIMPLIFIED

This kind of virus directly attacks or bypasses the antivirus software installed on the computer.

6. Time bomb:

This is a time of a virus that lies dormant until a certain date or time or for a period of time and cause harm to the computer system when triggered.

OR

This is a malicious program that is programmed to detonate at a specific time and release a virus into the computer system or network.

7. Logic Bomb:

This is a malicious software that is triggered with time.

WAYS THROUGH WHICH COMPUTER VIRUSES ARE SPRAED

- By use of a removable storage device such as flash disk, memory card which Hs been contaminated.
- Through an email attachment.
- Through downloading content on the internet from un trusted sites/ sources.
- As part of another program
- From pirated software (Demo/ trial version software i.e. software without a product key/ license key)
- Through infected proprietary software

SYMPTOMS OF VIRUS INFECTION IN A COMPUTER SYSTEM

- Programs on your computer system start to load slower.
- They computer system mysteriously shuts itself down or starts itself up and there is a great deal of unanticipated disk activity.
- You mysteriously loose access to a disk drive or other system resources because the virus has changed the setting on the device to make it un-useable.
- Your system suddenly does not reboot or give you unexpected error messages during start up.
- Program sizes change from the installed versions
- Unusual files appear on your hard drive or files start to disappear from your system.
- Your web browser, word processing application or other software begins to exhibit unusual operating characteristics. Screens or menus may change.

In any country, security agencies usually employee the mechanisms of Trapping and Eaves dropping in case of tracking down crime.

Distinguish between the terms Eaves dropping and Tapping.

Eavesdropping is the act of secretly listening to the private conversation of others without their consent.

While

Tapping involves acquiring data by redirecting a communication between two entities.

CRIMES ASSOCIATED WITH CYBER TERRORISM

- Online black mail
- Denial of service attacks
- Identity theft

BACKUP

Explain the term Backup as regards to Computer Security

Backup refers to the making of copies of data so that additional copies may be used to restore the original after a data loss event.

INSTANCES WHERE BACKUP IS REQUIRED

- In case of natural disasters such as fire outbreaks, earth quakes.
- In the event of hackers or cracking of the original data
- In case of data loss due to virus infection.
- In the event hard disk crashing.

WAYS OF ENSURING EFFICIENT BACKUP OF DATA

- Make regular backups
- Backup on removable storage such as secondary storage
- Keep backup media off site
- Set up automatic backup intervals

Explain the following terms as used in computer security.,

1. Spoofing:

Is a situation in which a person or program successfully masquerades as another by falsifying data and there by gaining an illegitimate advantage.

2. Fraud:

Is the use of a computer system to create fake documents, hide information or cheat unsuspecting public with intention of gaining monetary benefits.

3. Logic Bomb:

This is a malicious software that is triggered with time.

4. Denial of service:

SUB ICT SIMPLIFIED

Is a attempt to make a machine or network resources unavailable to its intended users.

5. Sniffing:

Is the act of intercepting TCP/IP packets while they are transferred on a network

HACKING

This is the act of seeking and exploiting weaknesses in a computer system or computer network. A hacker may be motivated by multitude of reasons such as profit, protest or challenge.

FORMS OF HACKING

- **Vulnerability scanner:** Is a tool used to check computers on a network for known weaknesses.
- **Password cracking:** Is the process of recovering passwords illegally from data that has been stored or transmitted by a computer.
- **Packet sniffer:** Is an application that captures data packets which can be used to capture passwords and other data in transmission over a network.
- **Spoofing attack (Phishing):** Is a situation in which a person or program successfully masquerades as another by falsifying data and there by gaining an illegitimate advantage.
- **Root kit:** a set of programs which work to subvert control of the operating system from its legitimate operators.

CONTROL MEASURES TO GUARD AGAINST THE ILLIGAL ACTIVITIES OF HACKERS

- Use data encryption mechanisms
- Using fire walls at the network gateway
- Use of burglar proof doors and security lock devices
- Use of security mechanisms such as sensors and CCTV against.
- Ensuring password usage by authenticate users of data system.

DATA CONTROL

This refers to the measures and policies taken to ensure data security.

Jenifer the school secretary has lost all her data in the computer. She suspects that either someone hacked into her computer system or it was due to a hardware failure.

Mention four ways she should have to protect her data.

- Making regular backups of your data.

SUB ICT SIMPLIFIED

- By use of hardware based mechanisms for protecting data such as use of security tokens to physically access a hardware resource.
- Data masking
- Use of Guinean antivirus software to scan against malicious software.
- Perform regular maintenance of computers
- Password-protect documents
- Use disk encryption mechanisms
- Install a Firewall at the Network gateway to guard against intrusion of hackers.

Distinguish between Data encryption and pass wording documents

Data encryption involves the transformation of data information in to non-readable forms to ensure data security when being transmitted over a network.

While

Pass wording of documents is a mechanism used to protect documents from unauthorized access.

WAYS TO GUARD YOUR SYSTEM AGAINST COMPUTER VIRUSES

- Installing an Updated Antivirus software
- Using Guinean Antivirus software
- Scanning all external storage devices for viruses before using them with the computer.
- Avoid downloading content from the internet without being sure of their safety.
- Sensitivity computer users on the damage caused by viruses.
- Installing a fire wall at the network gateway.

COMPUTER CRIME

What is meant by the term Computer Crime?

This refers to any criminal activities which involve the use of ICTs to gain illegal or unauthorized access to a computer system with intent of damaging, deleting or altering computer data.

FORMS OF COMPUTER CRIME

- **Computer viruses:**
These are malicious programs that can replicate themselves and harm the computer system on a network without the knowledge of the system users.
- **Spam:**
This is the unsolicited sending of bulk email mostly to gain unauthorized access.
- **Hacking:**
Is the illegal act of getting access to computer resources or data without prior knowledge from the owner.
OR
This is the activity of breaking into a computer system to gain unauthorized access.
- **Fraud:**

SUB ICT SIMPLIFIED

This is the use a computer system to create fake documents, hide information or cheat unsuspecting public with intention of monetary benefits.

- **Obscene or Offensive Content:**
Is the access and use of harmful or offensive content such as pornography
- **Trespass:**
This is the illegal access to data sent over a network.
- **Phishing:**
This is the act of attempting to acquire sensitive information like passwords, usernames and credit card details by disguising as a trust worth source.
- **Cyber stalking:**
This is the use of ICTs mainly the internet to torture other individuals through false accusation, transmission of threats, damage to data and equipment.
- **Identity theft:**
This is the act of pretending to be someone else by using someone else's identity as one's own.
- **Drug trafficking:**
This involves the use of ICTs such as encrypted emails, restricted chat rooms to sell illegal substances.
- **Cyber terrorism:**
This is the act of terrorism committed through the use of cyberspace or computer resources, e.g. the use of propaganda on the internet such as "there is a bomb planted on a shopping mall"
- **Tapping:**
Is the use of intelligent programs (spyware) to gain unauthorized access to information and data during transmission.
- **Alteration:**
This is the illegal amendment of data or information with the intension of misinforming others or gaining money.
- **Salami attack:**
This is the use of programs to alter data in one's computer system by little margin and then transferring the stolen data to another computer system.
- **Eavesdropping:**
This is the act of secretly listening to the private conversation of others without their consent.

METHODS OF PROTECTING COMPUTERS AGAINST COMPUTER CRIME

- **Log files**
These are system files installed in a computer system to maintain a record of how the system is being used.
- **Audit trail**
Is the thorough study done mostly by system administrators on an information system to identify any loopholes that can be used to illegally access the system resources
- **Biometric Analysis**

SUB ICT SIMPLIFIED

This is the form of access control measure that takes the user's attributes such as finger prints and facial recognition.

➤ **Data encryption**

This is the use of unreadable data (cipher) during data transmission which is later decrypted by the recipient. In most cases it is only the sender and the receiver who is able to understand the encrypted data.

➤ **Security patch**

This is a piece of code that is added to software so as to enhance its security feature.

➤ **Firewalls**

This is a security mechanism installed at the network gateway to block any unauthorized access to the system resources.

➤ Use of Guinean and updated antivirus software.

DATA SECURITY PRACTICES THAT CAN BE CARRIED OUT BY A SYSTEM ADMINISTRATOR OF AN ORGANIZATION

- Making regular backup of data
- Keeping a copy of backup off side for eventualities of disasters such as fire outbreaks
- Performing regular maintenance on a system
- By use of hardware based mechanism for protecting data such as use of security tokens to physically access a hardware resource
- Data masking
- Use of Guinean antivirus software to scan against malicious software on the system
- Use of log files: System files installed in a computer system to maintain a record of how the system is being used
- Audit trail: This is the thorough study done mostly by system administrators on an information system to identify any loopholes that can be used to illegal access the system resources.
- Educate users of the system on the importance of data security. This is to ensure that users are able to detect and avoid actions that may pose insecurity to system data.
- Use of licensed software on the system. This is to reduce on the risks of virus infections.
- Put up file level and share level security. This is where permission is set through user accounts and passwords for system users.
- Use of secure wireless transmission to guard against hackers. The configuration of the network should be secured.

COMPUTER FRAUD

What is meant by Computer fraud?

This is the use of computer systems to create fake documents, hide information or cheat unsuspecting public with the intention of monetary benefits.

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SUB ICT SIMPLIFIED

FORMS OF COMPUTER FRAUD

- Hacking
- Online identity theft
- Salami attack

Define the following terms

1. Data security

This involves protecting data against unauthorized access, alteration and destruction.

2. Privacy

This involves securing personal data so that only the owner can access it.

Mention four precautions a financial institution such as a bank can take to prevent illegal data alteration.

- Use of authentication method where the bank staff will be required to log into system whenever they are to gain access.
- Use of Audit trails
- Setup log files in the system
- Limit the level of access to data

Distinguish between Financial Identity theft and Identity Cloning

Financial Identity theft involves use of false identity to obtain goods and services.

While

Identity Cloning is the use of another user's information to pose as a false user.

FORMS OF IDENTITY THEFT

- Illegal migration
- Black mail
- Terrorism

CYBER STALKING

This is the use of ICTs mainly the internet to torture other individuals through false accusation, transmission of threats, damage to data and equipment.

FORMS OF CYBER STALKING

- False accusations
- Obscene emails

SUB ICT SIMPLIFIED

- Abusive phone calls.

TERMS USED IN COMPUTER SECURITY

1. Spyware

This is an intelligent program that is able to secretly track and collect information about users and their computers e.g. cookies

2. Password

This is a code that must be keyed in so as to gain access to a file or system resource.

3. Authentication

This is the process of verifying the log on credentials of a user requesting access to a computer system resource.

4. Log files

These are system files installed in a computer system to maintain a record of how the system is being used

5. Fire wall

This is a security mechanism installed at the network gateway to block any unauthorized access to the system resource.

Distinguish between a hacker and a cracker

A hacker is a person who illegally accesses computer resources without prior knowledge from the owner with an aim of archiving wrong endings

While

A cracker is a malicious intruder who breaks into a computer system for illegal purpose. This person usually knows the codes and passwords of the system.

SOME DESTRUCTIVE ACTIVITIES THAT HACKERS DO ON A COMPUTER SYSTEM

- Copying of information
- Altering the passwords and user names on the system.
- Deleting the files
- Modifying information in the system
- Infecting systems with computer viruses

What is meant by firewall?

This is a security mechanism installed at the network gateway to block any unauthorized access to the system resources.

FUNCTIONS OF A FIREWALL

- Filtering of network traffic or data packets on the computer.
- Concealing or hiding the true network address during data transmission.

SUB ICT SIMPLIFIED

Distinguish between Cyber terrorism and Cyber extortion.

A Cyber terrorist is someone who intimidates or coerces a government or organization to advance his or her political or social objectives by launching computer based attacks against Computers, Network and Information stored on them.

While

Cyber extortion is a form of cyber terrorism in which a website, email server or computer system is subjected to repeated denial of service or other malicious attacks by demanding money in return for promising to stop the attacks.

FORMS OF CYBER TERRORISM

- Online blackmail
- Denial of service
- Identity theft

ICT ETHICAL ISSUES

The industry is faced with several ICT ethical issues that affect the moral values and principles.

Briefly explain the following ICT ethical issues

1. Plagiarism

This is the wrongful appropriation, purloining and publication of another author's language, thoughts, ideas or expression and the representation of them as one's own original work.

2. Software Piracy

This refers to the act of duplicating and installing of a copy of software into your computer without authorization.

OR

It refers to the un authorized use or distribution of software

3. Copyright Infringement

This refers to the copying of intellectual property without written permission from the copyright holder, which is typically a publisher or other business representing or assigned by the work's creator.

4. Misuse

This refers to the use of harmful content such as porn, spam, stolen ID, hacking, etc

5. Issue of Libel

This is the publication of false statement that injures one's business or personal reputation.

OTHER TERMS USED IN COMPUTER ETHICS

1. Intellectual Property

This means that a company or person owns the rights to some kind of technology or innovation.

2. Information Privacy

SUB ICT SIMPLIFIED

This is about understanding how information technologies impact the privacy of individuals and developing new privacy preserving and secure technologies to protect them.

FORMS OF INTELLECTUAL PROPERTY

- Copyright
- Trademarks
- Patent rights

State the importance of the following symbols as used on ICT products.

1. TM

This is a symbol which signifies an unregistered trade mark. Is a recognition, sign, design or expression which identifies products or services of a particular source from those of others

2. ®

This is a symbol which signifies a registered trade mark.

3. ISBN



International Standard Book Number (ISBN) is a unique numeric commercial book identifier based upon the 9 digit Standard Book Numbering (SBN) code.

State the importance of a serial number on a product

- To enable electronic distribution and allow extensive evaluation of a product before purchase.
- To distinguish a product from one another

Distinguish between a patent and a trade mark.

A patent is a legal instrument (intellectual property law) declaring that a company or a person has ownership over an idea/ technology.

While

A trade mark is a recognizable sign, design or expression which identifies products or services of a particular source from those of others.

AREAS WHERE COPYRIGHT IS USED

- In publication works such as e-publishing, desktop publishing
- Digital broadcasters such as webcasting, digital films and televisions online
- Protection of data bases
- Responsibilities of online providers
- Peer to peer file sharing systems such as Napster

WAYS IN WHICH SOFTWARE PIRACY CAN BE REDUCED

- Educate users about the dangers of using pirated software
- Reduce the cost of software
- Use of holograms, a component which comes with the original copies and cannot be duplicated
- Putting up and enforcing legal rules on software piracy
- Encourage the use of open source software
- Use of flexible licensing of software.

What is meant by Brand Piracy?

This refers to the unauthorized usage of trade marks by producing and trading counterfeit consumer goods.

MEASURES TO MINIMISE BRAND PIRACY

- Owner of trade mark may pursue legal action against trade mark infringement
- A company formally register their trademarks with the government

Briefly describe any five emerging technologies

1. 4G cellular communication

This is a system that provides mobile ultra-broadband internet access. For example use of laptops with USB wireless modems to smart phones, and to other mobile devices.

2. Artificial intelligence

This refers to the intelligence of machines, robots and the branch of computer science used to create it.

3. 4G optical disks, hologram data storage

Any form of optical data storage in which information can be recorded or read with 3dimensional resolution one that has the ability to enable DVD sized disks to store data to a level of petabyte

4. General Purpose computing on graphics processing units(GPGPU or GPU)

This is the utilization of graphics processing unit to typically handle computation for only computer graphics.

5. Radio Frequency Identification(RFDI)

The use of a wireless noncontact system that uses radio frequency electromagnetic fields to transfer data from a tag attached to an object, for purpose of automatic tracking and identification.

TERMINOLOGIES USED IN EMERGING TECHNOLOGIES

1. Artificial Intelligence

This refers to the simulation or creating of intelligence similar to humans

2. Digital Forensics

This is a branch of science encompassing the recovery and investigation of material found in digital devices, often related to computer crime.

SUB ICT SIMPLIFIED

3. Simulation

This is a science of representing behavior of real life situations using a computerized model

4. Virtual reality

This is a condition in which a person becomes psychologically submerged in an artificial setting generated by a computer system

TYPICAL EXAMPLES OF VIRTUAL REALITY

- In computer aided learning
- Computer games
- Interactive mapping such as Google maps
- 3D movies
- Online tourism
- Online advertisement

APPLICATION AREAS OF SIMULATIONS

- Traffic control, using traffic lights
- Training the use of missiles by the army
- Training pilots on how to charter planes
- Training on use of fire fighting equipment

EXAMPLES OF ARTIFICIAL INTELLIGENCE

- Robotics (motion and simulations)
- Natural language processing. This is the ability of machines to have the ability to read and understand the languages humans speak.
- Perception. This is the ability to use input from sensors to deduce aspects of the world
- Deduction, reasoning & problem solving. This is the use of problem solving algorithms that imitate humans
- Expert systems. Software is designed to make a computer function like a human expert in specific narrow areas of specialization
- Social intelligence

ROBOTICS

Define the term Robot

This is a computer controlled device that mimics a human being in carrying out routine tasks.

APPLICATION AREAS OF ROBOTICS

State any five application areas of robotic technology

- Car assembling
- Bomb detonation

SUB ICT SIMPLIFIED

- Inside dangerous chemical areas
- In military operations which may seem to be risky
- Fire fighting
- In repetitive and monotonous tasks in which human performance might depreciate with time such as in packaging of goods.

CAREERS RELATED TO THE ICT INDUSTRY

1. ICT trainer/ Computer teacher/ Instructor:

This one is responsible for training people on how to use ICTs and various application programs.

2. Computer programmer/ Software developer:

Designs computer applications by the use of programming languages.

This is a profession for the development or continued development and maintenance of software programs.

3. Computer analyst:

Checks for correctness of data and programs on the computer

4. System analyst:

This one is able to test out all features of a product for any problems or use ability issues

5. Computer technician / Hardware engineer:

Is responsible for the maintenance and repairs of the computer equipment

This is a profession responsible for assembling, testing, maintaining and repairing of electronic equipment.

6. Computer operator:

Controls the operations of the computer from the input of data and instructions to the output

7. Data entrant:

This is a professional responsible for taking data from a hard copy or other source and enter it into electronic format.

8. Database administrator:

This is a job that requires creating, testing and maintaining on or more databases

This one also designs and maintains the databases of the organization, prepares and enforces standards for the use and security of information in databases.

9. Webmaster:

This is a job which requires one to create / completely design a website and maintain it.

10. Network/ System administrator:

This is a professional responsible for designing, setting up and maintaining a network, including monitoring of network resources.

11. Graphical designer:

This is a professional who designs and creates either graphics or #D animations for software, games, movies, web pages, etc.

12. Information Systems manager:

He/ she plans and directs the activities of the entire information services organizations.

13. Secretary:

Uses computers to keep all the necessary information instead of keeping paper files

14. Systems development manager:

He / she direct the activities of systems Development Personnel and projects.

And many others

REVISION PAST PAPERS

Name _____ **MARKING GUIDE** _____ Stream _____

FUNCTIONAL COMPUTING
Subsidiary Subject ICT

Paper 1

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

Write your name and stream in the spaces above.

*This paper consists of **three** sections, **A, B** and **C**.*

*Section **A** contains **15 compulsory** objective-type questions. The correct alternative **A, B, C** or **D** must be written in the box provided on the right hand side of each question.*

SUB ICT SIMPLIFIED

Section **B** contains **two** general question crossword puzzles. Answer only **one**. Answers to section **B** must be written in the spaces provided in the question paper.

Section **C** contains **three** essay type questions. Answer only **one**. Answers to section **C** must be written in the answer booklet provided.

FOR EXAMINER'S USE ONLY						
SECTION A	SECTION B		SECTION C			TOTAL
MCQ 1-15	Q16	Q17	Q18	Q19	Q20	
30	30	30	20	20	20	100

SECTION A:(30 MARKS)

Answer **all** questions in this section. Each question carries 2 marks.

1. Which of these is a pointing device used to manipulate on-screen objects?
A. Compact Disk (CD)
B. Keyboard.
C. Monitor.
D. Mouse. D
2. Musa has copied a calculation result from a calculator application to the Clipboard. Which option can he use to get it from the clipboard into a text document?
A. Cut.
B. Put.
C. Paste.
D. Save. C
3. Using a pointing device to move an on-screen item from one location to another on the screen is known as:
A. Dragging.
B. Clicking.
C. Pointing.
D. Double Clicking. A
4. The maximize button is used to:
A. Return a window to its previous size.
B. Expand a window to fill the computer screen.
C. Scroll slowly through a window.
D. Send a window to the task bar. B
5. What appears if a window contains more information than can be displayed in the window?
A. Program icon.
B. Cascading menu.
C. Scroll bars.
D. Check boxes. C
6. For most windows programs, the save command is located on which menu?
A. File.
B. Edit.
C. Help.
D. Save. A
7. Which of the following function keys is used for getting help on a computer?
A. F1.
B. F2.
C. F3.
D. F4. A

SUB ICT SIMPLIFIED

8. Which of the following is a computer classification by size?

- A. Analog.
- B. Mainframe.
- C. Digital.
- D. Special Purpose

B

9. Which of the following processor types is the fastest and strongest?

- A. 80486.
- B. Pentium II.
- C. Pentium IV.
- D. Pentium Duo Core.

D

10. _____ is a general word for all characters such as letters, numerical digits, symbols and marks that combine to form words and sentences.

A. Data.

B

B. Text.

C. Works.

D. A byte.

11. How many binary digits are required to store a file with 300 bytes?

A. 300 bits.

D

B. 800bits.

C. 1200bits.

D. 2400 bits.

12. Which of the following key combinations below is used to change case between uppercase, lower case and sentence case?

A. Ctrl+Z.

B

B. Shift+F3.

C. Ctrl+A.

D. Ctrl+U.

13. What is the correct sequence of launching Microsoft Office Word Processing Application?

For more ICT related resources:

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SUB ICT SIMPLIFIED

- A. Start>Microsoft Office>All programs>Microsoft Office Word.
- B. Microsoft Office >All programs> Start> Microsoft Office Word.
- C. Start> All programs>Microsoft Office> Microsoft Office Word.
- D. All programs> Microsoft Office Word >Start>Microsoft Office.

C

14. Tangible components of a computer are called?

- A. Software .
- B. Hardware.
- C. Peripherals.
- D. Equipment.

B

15. Zooming.

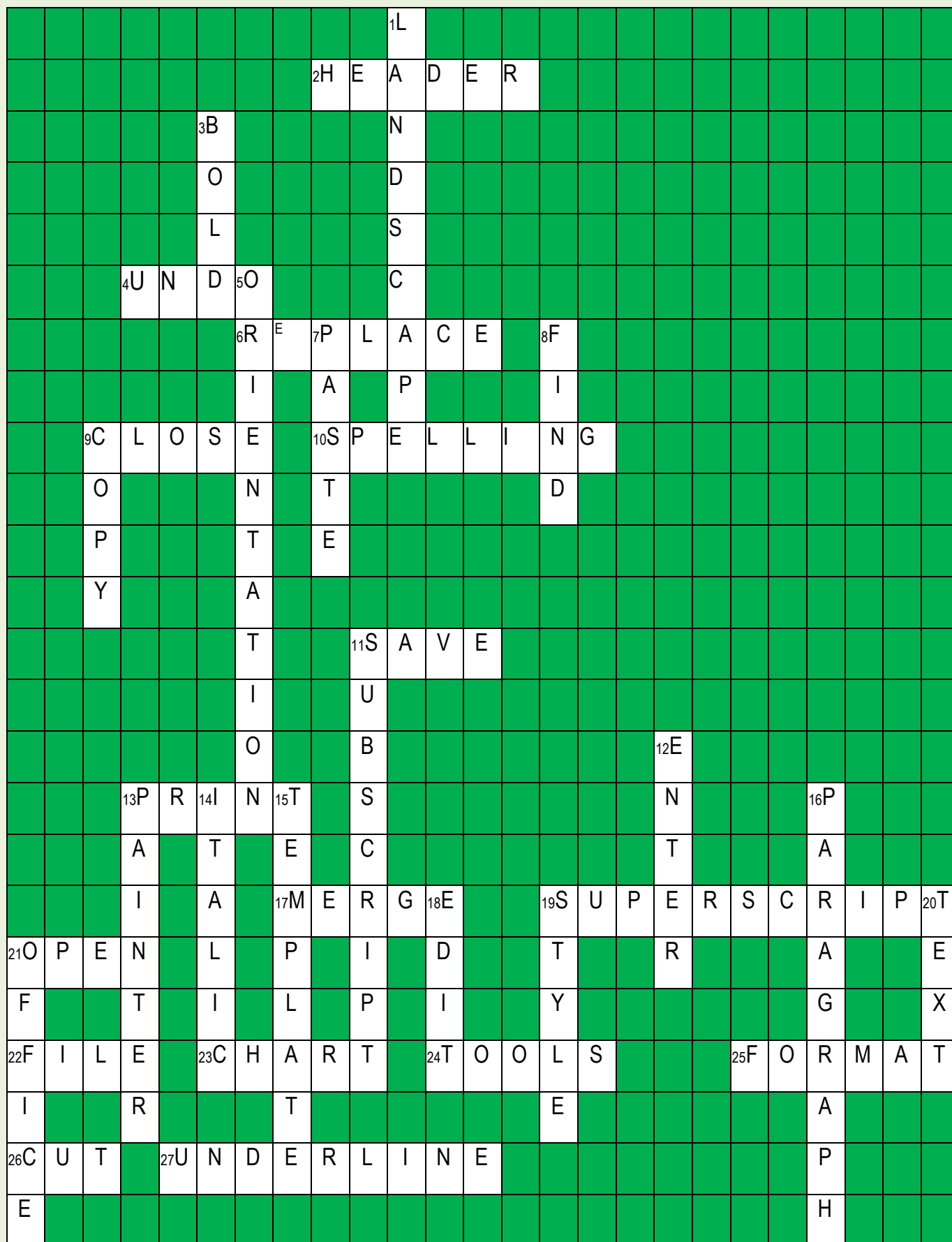
- A. Increases the font size.
- B. Increases the page size.
- C. Reduces the font size.
- D. Increases or reduces the magnification.

D

SECTION B (20 MARKS)

*This section contains two **optional** crossword-puzzle type questions. Choose One.*

16. Crossword Puzzle 1: **Microsoft Word 2003**. Use the Clues given below to solve this puzzle.



CLUES ACROSS

2. Area at the top of the page which can display information on every page. (6).
4. Ctrl+Z is equivalent to which command? (4).
6. Ctrl+H is equivalent to which command? (7).
9. Alt+F4 is equivalent to which command? (5).
10. What will pressing the F7 key check? (8).
11. Ctrl+S is equivalent to which command? (4).
- 13 Ctrl+P is equivalent to which command? (5).
17. The process of combining letters with addresses to produce mass mailing letters is referred to as mail: (5).
19. Formatting used to display a character above the line. (11).
21. Ctrl+O is equivalent to which command? (4).
22. Under which drop down menu will you find page setup options? (4).
23. An alternative name for a graph. (5).
24. Under which drop down menu will you find the word count option? (5).
25. Under which drop down menu will you find paragraph formatting options? (6).
26. Ctrl+X is equivalent to which command? (3).
27. Ctrl+U is equivalent to which command? (9).

CLUES DOWN:

1. Pages can be formatted as portrait or: (9).
3. Ctrl+B is equivalent to which command? (4).
5. Portrait and Landscape are types of page: (11).
7. Ctrl+V is equivalent to which command? (5).
8. Ctrl+F is equivalent to which command? (4).
9. Ctrl+C is equivalent to which command? (4).
11. Formatting used to display a character below the line. (9).
12. To insert a manual page break, press the Ctrl key and which other key? (5).
13. The tool used to copy formatting from one item of text to another is called the Format: (7).
14. Ctrl+I is equivalent to which command? (6).
15. A special type of Word document, which can be recycled and used to create new documents. (8).
16. When typing, pressing ENTER key inserts a break (9)
18. Under which drop down menu will you find the Cut, Copy and Paste commands? (4).
19. A set combination of formats that has a name and can be applied to text with a single click. (5).
20. What does 'T' stand for in RTF? (4).
21. Microsoft Word is often used within a suite of programs, called Microsoft: (6).

SUB ICT SIMPLIFIED

17. Crossword Puzzle 2: **Basic concepts of IT.** Use the Clues given below to solve this puzzle.

	1D	I	2R	E	C	T	O	R	Y										3D		
	I		A															4M	I		
	G		N					5S						6O				7U	P	S	
8W	I	N	D	O	W	9S		10C	P	U				C					L		C
	T		O			P		A			11M	I	C	R	O	S	O	F	T		A
	A		M			E		N											12I	N	T
	L				13M	A	I	N	F	R	A	14M	E						T		T
						K		E			I			15S	O	F	T	W	A	R	E
						E		R			C			H					S		
		16L	A	S	E	R				17H	A	R	D	W	A	R	E			K	
		A									O			R					I		
		18P	R	19I	N	T	E	R		20T		21P	O	W	E	R	P	O	I	N	T
		T		N						O		H			W					G	
		O		T					22I	N	F	O	R	M	A	T	I	O	N		
		P		23R	E	A	D		N	E		N			R						
				A					K	R		24E	X	25C	E	L		26V	I	R	27U
				N					J					O			28E			N	
		29F	R	E	E	30W	A	R	E					P			M			I	
				T		O			T			31K	E	Y	B	O	A	R	32D		V
						34R	A	35M						R			I		O		E
						D		O		36E		37U	N	I	X		L		38S	C	R

SUB ICT SIMPLIFIED

								U		I				G							S			T
			39	A	C	C	E	S	S	40	G	R	A	P	H	I	C	A	L		A			R
									E		H				T						L			A
											T													L

CLUES ACROSS

1. Alternative name for a folder. (9).
7. A device that protects your computer in the event of a loss of power. (3)
8. Operating system found on most personal computers. (7)
10. The brain of the computer. (3).
11. The company that manufactures Windows Operating Systems. (9).
12. A well-known CPU manufacturer. (5).
13. A very large, powerful computer used by large organizations. (9).
15. Electronic Instructions that tell a computer how to perform tasks. (8).
16. A common, high speed nonimpact printer, used in offices. (5).
17. Tangible parts of the computer system such as the Monitor. (8).
18. An output device that produces a hard copy output on a physical material like paper. (7).
21. Application program from Microsoft used for creating presentations. (10).
22. What is the 'I' in IT short for? (4).
23. What does the 'R' in ROM stand for? (4).
24. An application program form Microsoft for producing spreadsheets. (5).
26. Malicious software that can infect a computer. (5).
29. A type of software that can be freely copied and distributed. (8).
31. Main input device, consisting of a set of buttons used to operate a computer (8).
34. When a program loads from disk, it is copied into: (3).
37. An operating system developed in the early 1970s by scientists at Bell Laboratories. (4).
38. A display output device that visually conveys text, graphics, and video information. (6).

39. An application program from Microsoft used for producing databases. (6).

40. What does the 'G' in GUI stand for? (9).

CLUES DOWN

1. Personal computers are not analogue, they are: (7).
2. What does the 'R', in RAM stand for? (6).
3. Another name for a floppy disk is: (8).
4. The ability to run more than one program at a time is called. (12)
5. A device that allows you to digitize a picture or printed page and import the data into your computer. (7).
6. Optical Character Recognition acronym (3).
9. An audio output device. (7).
14. An audio input device. (10).
15. Software that can be used for a limited trial period. (9).
16. Small popular portable computer. (6).
19. Small private version of Internet, used only within an organization. (8).
20. Powdered ink, which is packaged in a print cartridge. (5).
22. Common type of nonimpact printers, which use liquid ink. (6).
25. A term referring to the legal protection given to text and images that have been placed on the WWW. (9).
27. What does the 'U' in USB stand for? (9).
28. Term used to describe sending messages between computers. (5).
30. An application program from Microsoft used for word-processing. (4).
32. Operating system widely used before Microsoft Windows. (3)
33. What does the 'C' in CPU stand for? (7).
35. A pointing input device commonly used with desktop computers. (5).
36. Number of bits in byte. (5)

SECTION C (40 marks)

Answer any two questions from this section. Answers must be written in the answer booklet provided.

18. (a) Discuss any **six** areas where computers are applied and used in today's society. (12 marks)

▪ **Computers are applied in the areas of**

- 1) **Education**, - for Computer Aided Learning (C.A.L), Computer Aided Assessment (C.A.A.), for calculating mathematical arithmetic by students, for creating electronic report cards, creating student databases etc.
- 2) **Research**, - for searching literature and references through use of electronic databases on the World Wide Web, Research Data Storage, Research publishing etc.
- 3) **Business**, - for electronic commerce, business advertising, electronic money transfers, Telecommuting, etc.
- 4) **Health**, - for managing and storing health Records, printing labels, monitoring/Diagnosis of patients, etc.
- 5) **Communication**, through Video Conferencing, E-mail, social networking, faxing, etc.
- 6) **Military/security**, for monitoring security through cameras, coordinating the personnel, capturing data for Police National Computer Databases, operating dangerous weapons, etc.
- 7) **Homes** – for managing household finances, accessing internet and running Programs for Children at home, playback of Digital Media such as Video/ Music, Family Pictures at home, etc.
- 8) **Entertainment / leisure**. For Computer games, for faster instant audio and video playback in cinema halls and during functions, in the entertainment industry to produce Music, Edit Video, etc.

1 mark for mentioning area of application, 1 mark for explanation

- (b) Explain any **four** demerits associated with computer usage in today's society. (08 marks)

1. **Risk of addiction**. Many people don't care about anything else because they spend many hours in front of computer.
2. **Eye Strain**. Using computer for long can be very harmful to the eyes.
3. **Expensive**. Computers are still costly because The initial costing and maintenance cost of a computer are very high.
4. **Need for Extra Training**. Use of computers require additional training before use of computers and their programs.
5. Computers have led to **unemployment** of people such as Tellers, shop workers, postal workers; since many manual/repetitive jobs have been replaced by computers like ATMs
6. **Electronic fraud**: Stealing money electronically through practices like Credit card cloning
7. **Impact on Environment**: Computer manufacturing processes and computer waste are depleting natural resources and polluting the environment.
8. **Hacking**: Unauthorized access into computers possibly to access information, compromising privacy. E.g. Wikileaks
9. **Software Piracy**: Stealing software, not paying for licenses through cracking.

10. **Presence of Viruses:** You may lose all your programs, files, documents folders and other data on your computer because of virus infection on the computer.
11. **Moral Decay:** The internet has websites with content such as pornography, which have a bad impact on the users especially the young children.
12. Computers are **delicate** and they must be given a great care.
13. **Over dependence.** Our creativity, skills and reasoning can decrease when we are too dependent to the computer. For instance, with email replacing the hand-written letter, Onscreen Art designing instead of Hand drawing Art etc.
14. **There is dehumanizing of people.** Excessive use of computer for communication with others threatens to reduce the intensity of personal bond that often develops between people. For instance, the telephone calls

Any four x 2 marks = 8 marks.

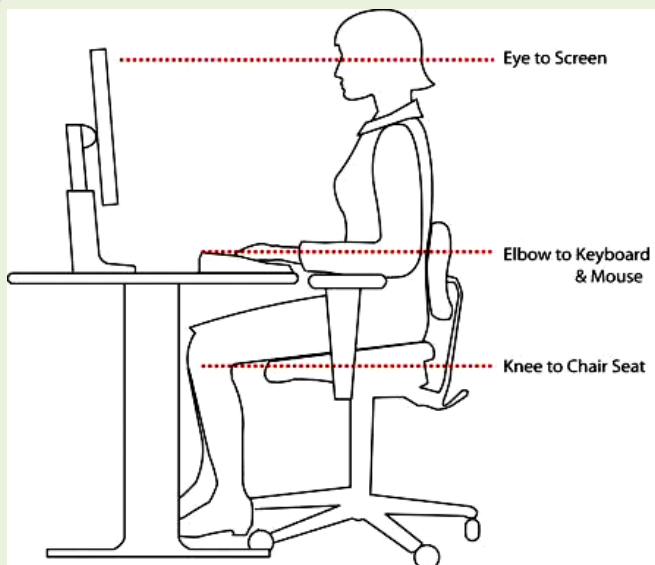
19. (a) What is a **Computer Laboratory**?

(02 marks)

- A computer laboratory is a building in which computers are setup in an environment conducive for safe use.

(b) Mention any **four** good habits that you should exhibit while using computers in a computer laboratory. (08 marks)

1. Never use your computer during a **storm**. The computer is connected to electricity and that means that lightning could be conducted to the computer.
2. **Proper shut down** of computers should be followed to avoid disk and system failure (avoid abrupt switching off)
3. Be careful when **using the internet**. Do not accept downloads from Internet sites that you don't know and trust. And never open an email attachment unless you know and trust the person who sent it.
4. Handle **delicate storage devices with care**. Don't touch the inner surface of Compact disks and Floppy disks. Safely remove Flash disks from the system.
5. Always **Sit upright** to avoid muscle pains and back aches caused by poor sitting posture. (see figure)
6. Always use **surge protectors**, Uninterruptible Power supply (UPS) or voltage stabilizers to ensure steady power supply to safeguard their system.
4. **Proper cable installation** and placement. Cables should be preferably along walls, avoiding danger of exposing the use to electric shock



6. Always install **lightening conductors** to the computer laboratory to protect the machines and the users of the computers.
8. **Ventilation should be good.** Good aeration enables the computer to cool and hence avoids overheating
9. **Minimize Electrical noise** / interferences in the computer environment. ELECTRICAL NOISE refers to externally radiated signals or electrical that cause undesirable additions to the current voltage. Electrical noise is commonly generated by devices like Fluorescent lights of high frequency, Motors, Battery Chargers, Inverters, Radios, television, and Cell phones.
12. A computer room should have enough **light** avoid eyestrain, and headaches, and Radiation filter screens can be fitted to reduce the light that reaches the eye.
13. **Standard furniture:** The table on which a computer is placed must be strong and wide enough to bear the weight and accommodate all the peripheral devices
14. **Regular servicing** should be done for hardware and software updates to ensure proper working conditions of the computers
15. Always use and regularly updated **antivirus** software. Viruses and worms are horrible computer-unfriendly programs that can crash your system and cause damage.
16. Carry out **Disk Defragmentation** when necessary. Disk Defragmentation is the process in which scattered pieces of individual files and free space are reorganized and stored in an adjacent manner (next to each other) on the disk.

Any four x 2 marks = 8 marks

(c) State any **five** activities and/or items that are prohibited in your school's computer laboratory. (10 marks)

1. Avoid exposing computers to dust. Computer components should be kept **dust-free**.
2. **Never try to remove** the cover on your computer or touch inside the system unit. There are many sensitive components. Instead, take it to a qualified technician.
3. Keep all **liquids and food** items away from your computer. Liquids and food crumbs can cause rusting and corrosion and damage electronic circuits. Also, mixing liquids and electronic components can cause serious electrical shock!
4. Physically, be careful, **avoid knocking and dropping** any hardware to the ground as this could cause any of the delicate components to break or be damaged and stop working.
5. Avoid making **hardware connections to the motherboard** when the computer is on. Eg keyboard, monitor and mouse connections.
6. Don't bring **magnetic devices** to the lab. The computer has magnetic disks which can be spoilt if they come near other magnetic fields.
7. Avoid excessively **bright and flickering** computer monitors. The brightness of the computer monitors should be adjusted to avoid eye strain.
8. **Avoid direct sunlight** and high Temperatures that may damage hardware components.
9. Do not access **pornographic** websites from the internet. Observe computer ethics and standards as a good computer user.
10. **Overcrowding** of either machines or people should be avoided.

11. **Bulky items such as bags** should not be brought into the computer laboratory to avoid possible theft of hardware components.

Any five x 2 marks = 10 marks

20. (a) What is a **word processing application**? (02 marks)

- Word processing applications are computer programs that are used for creating, saving, editing, formatting and printing text documents.

(b) Outline any **four** examples of word processing applications you know. (08 marks)

Examples of word processors include:

- Microsoft Office Word,
- Corel WordPerfect,
- Lotus WordPro,
- Apple Pages,
- OpenOffice.org Writer,
- LibreOffice Writer,
- KWord
- Celframe Write
- Kingsoft Writer etc.

(c) State any **seven** features common to most word processing applications. (14 marks)

COMMON FEATURES OF WORD PROCESSING APPLICATIONS

1. **Word Wrap:** this feature automatically sends a word that does not fit within the margin settings onto the next line, without the user pressing Enter key.
2. **Find:** allows the user to locate all occurrences of a particular character, word or phrase.
3. **Replace:** allows the user to substitute existing characters, words or phrases with the new ones.
4. **Spell checker:** allows the user to check spellings of the whole document at one time or to check and even correct the spelling of individual words as they are typed (Autocorrect)
5. **Grammar checker:** this reports grammatical errors, usually by a wavy green line, and suggests ways to correct them.
6. **Thesaurus:** suggests alternative words with the same meaning (synonyms) for use in the document.
7. **Mail Merge:** This is feature used to create similar letters to be sent to several people. The names and addresses of each person can be merged with one single main document.
8. **Automatic page numbering:** numbers the pages automatically in a document
9. **Tables:** allow users to organize information into rows and columns.
10. **Multi-columns:** arranges text into two or more columns that look similar to newspaper or magazine
11. **Drop cap –** Formats the first letter in paragraph to be dropped across two or more lines.
12. **Clipart:** , refers to pre-made images about various subjects used to illustrate concepts in documents.
13. **Templates:** establish the initial document layouts and formats for various document types.

SUB ICT SIMPLIFIED

14. **Printing:** allows a user to obtain a hard copy of a document from the printer.
15. **Word Count:** Establishes the number of words, characters, paragraphs, etc. in a document.
16. **Headers and Footers:** Used to insert text in the top and bottom margin through the document.
17. **Footnotes** and **Endnotes** are used as references that provide additional information about a word or phrase within a document.
18. **Insert** and **Delete** allows a user to add and remove portions of text while editing document.
19. **Bold, Underline, Italic** – These and other formatting features that allow a user to improve on the appearance of text.
20. **Cut, Copy, Paste, Insert, Delete** – These and other editing features that enable a user to make changes to the text in a document.

Any seven x 2 marks = 14 marks

Name _____ Stream _____

S850/1**SUBSIDIARY ICT****Paper 1****Aug/Sep,20XX**

Senior Five Beginning of Term Three Examinations
INFORMATION AND COMMUNICATION TECHNOLOGY
(Subsidiary Subject)

Paper 1**1 hour 45 minutes****INSTRUCTIONS TO CANDIDATES:**

Write your name and stream in the spaces above.

*This paper consists of **three** sections, **A, B** and **C**.*

*Section **A** contains **20 compulsory** objective-type questions. The correct alternative **A, B, C** or **D** must be written in the box provided on the right hand side of each question.*

*Section **B** contains **one** general question crossword puzzle. Answers to section **B** must be written in the spaces provided in the question paper.*

*Section **C** contains **four** essay type questions. Answer any **two** questions. Answers to section **C** must be written in the answer booklet provided.*

FOR EXAMINER'S USE ONLY						
MCQ	Q21	Q22	Q23	Q24	Q25	TOTAL

SECTION A:(20 MARKS)

Answer all questions in this section.

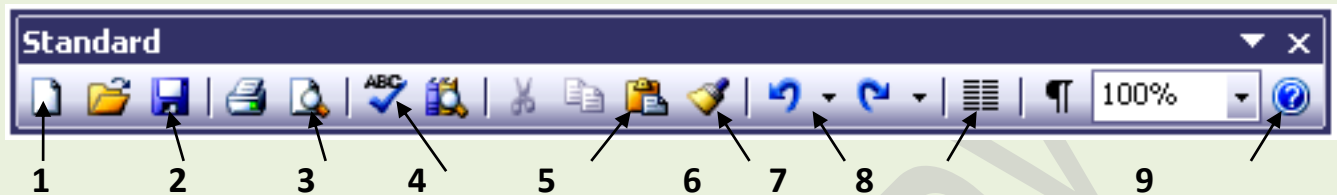
SUB ICT SIMPLIFIED

1. Which of the following is not a computer peripheral?
 - A. Monitor.
 - B. Printer.
 - C. RAM.
 - D. Scanner.☐
2. The process of restarting a computer is referred to as
 - A. Warm booting.
 - B. Logging off.
 - C. Cold booting.
 - D. Hibernating.☐
3. Selected Text can be copied by
 - A. Right clicking and choosing copy command from the pop up menu.
 - B. Pressing Ctrl + C on the key board.
 - C. Clicking on the copy button on the standard tool bar.
 - D. Any of the above methods.☐
4. In a spreadsheet, how many cells are in the range B10:D13?
 - A. 9 cells.
 - B. 12 cells.
 - C. 10 cells.
 - D. 13 cells.☐
5. Which one of the following is not a special purpose computer?
 - A. Laptop.
 - B. Automatic teller machine (ATM)Storage.
 - C. Traffic light control computer system.
 - D. Thermometer.☐
6. Which of the following does not appear on the start menu?
 - A. Help.
 - B. Programs.
 - C. Accessories.
 - D. Recycle bin.☐
7. What appears if a window contains more information than can be displayed in the window?
 - A. Program icon.
 - B. Scroll bars.☐

SUB ICT SIMPLIFIED

- C. Cascading menu.
- D. Check boxes.

The screen shot below shows some of the buttons on a standard tool bar. Use it to answer question 8.



8. Which of the buttons represents the format painter?
- A. Button 2.
 - B. Button 3.
 - C. Button 5.
 - D. Button 6.
9. All these devices are imaging devices except
- A. Digital Camera.
 - B. Camcorder.
 - C. Barcode Reader.
 - D. Webcam.
10. The points of connection between a computer (motherboard) and its peripherals are.....
- A. Connectors.
 - B. Cables.
 - C. Expansion Slots.
 - D. Ports.
11. Formatting in Word Processing refers to:
- A. Preparing a new floppy diskette for use.
 - B. Changing the appearance of text in a document.
 - C. Cleaning diskettes for storage.
 - D. Erasing all data on a diskette.
12. Which of the following are arranged in ascending order according to size?
- A. Byte → GB → MB → Bit → KB → TB.
 - B. Bit → Byte → KB → MB → GB → TB.

C. GB → MB → KB → TB → bit → KB.

D. TB → GB → MB → KB → Byte → bit.

13. Some Students want to draw a sketch map of Uganda on a computer. Which of the following devices would you recommend to them?

A. Stylus pen and Digitizing tablet.

B. Touch Pad.

C. Mouse.

D. Joystick.

☐

14. A problem that is likely to occur due to future advancement of computers is:

A. Computer illiteracy.

B. Un employment.

C. Eye problems / impairment.

D. Insecurity.

☐

15. As Tekpe was typing his Class Notes using Microsoft Office Word, the application noticed wrong grammar of a phrase in page two of his Document. How did the computer alert him to rephrase correctly?

A. A wavy red line appeared below the erroneous phrase.

B. The erroneous phrase was automatically underlined.

C. A wavy green line appeared below the erroneous phrase.

D. A screen tip was displayed above the erroneous phrase.

☐

16. _____ is an input device that is specially suited to portable computers.

A. Mouse.

B. Joystick.

C. Webcam.

D. Keyboard.

☐

17. Blackberry is an example of _____ computer.

A. PC.

B. Palmtop.

C. Notebook.

D. Desktop.

☐

18. A factor which would influence a person to adopt a computer is its

A. Accuracy

- B. Reliability
- C. Speed
- D. All of above

☐

19. All the following are proper sitting positions while using the computer except one.

Which one is it?

- A. High enough seat allowing eyes to be on level with top of the screen
- B. Face being very close to the monitor so that you have a clear view.
- C. Feet firmly straight on the floor.
- D. Placing keyboard at elbow level and arms relaxed at your sides.

☐

20. High humidity in the computer laboratory affects computers by causing:

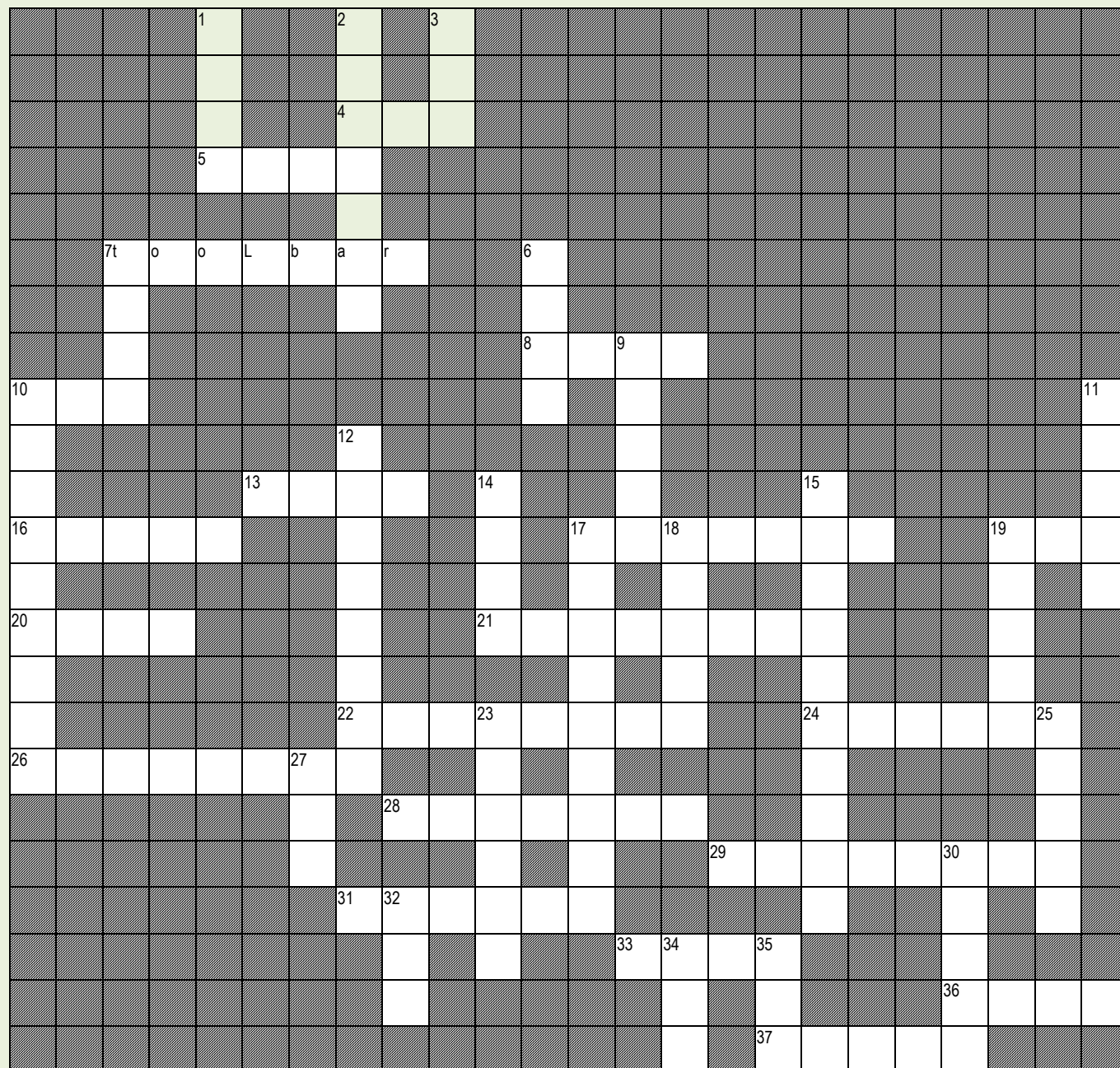
- A. The metallic parts to rust.
- B. Computer components to crack and break
- C. Computer color to fade
- D. Computers to be slow in performance.

☐

SECTION B (40 MARKS)

This section contains **one** crossword-puzzle type question. Use the Clues given to solve the puzzle.

21. Crossword Puzzle on the Basic skills of Using Windows OS and Managing files.



CLUES DOWN:

1. Software errors. (4)
2. Text editing application included with Windows. (7)
3. Common file name extension used by compressed files (3).
6. Files with a .DOC or .DOCX file extensions are created using Microsoft(4)
7. General word for all characters that combine to form words, sentences, paragraphs, pages etc. (4)
9. Cross shaped icon at the top right corner of a window (5)
10. Area of memory that you can copy or cut items to. (9)
11. A device which demodulates an incoming telephone analog signal into a digital one which a computer can process. (5)
12. Name of the button used to make window fill the screen (8)
14. What is normally displayed when you press F1 key? (4)
15. Microsoft is an example of a presentation program (10).
17. An alternative word for the word 'folder' (9)
18. Name of the Desktop button, normally displayed at the bottom left of the screen. (5)
19. Files with a .XLS or .XLSX file extensions are created using Microsoft..... (5)
23. What does 'R' stand for in RAM? (6)
25. You can prevent images from being 'burnt' onto your screen by using a screen:....(5)
27. To switch between open applications, press Alt+.....(3)
30. The pictorial representation of programs on the windows desktop. (5)
32. To display the Start Menu you would press the Ctrl key and which other key? (3).

34. File name extension used by Adobe Acrobat Files. (3)
35. Acronym for Graphics Interchange Format. (3)

CLUES ACROSS:

4. Common file extension used to indicate a temporary file. (3).
5. Within the Windows Explorer you can sort files by: (4).
7. Name of a bar containing icons or buttons, displayed across the top of an application. (7).
8. What does the 'R' in RTF stand for? (4).
10. What command is the shortcut Ctrl + X for? (3).
13. What does the 'R' in ROM stand for? (4).
16. What command is the shortcut Ctrl + V for?(5).
17. The term used to describe the screen background on a Windows based computer. (7).
19. File extension used for program files. (3).
20. Command located under the file drop down menu (4).
21. What does the 'P' in PDF stand for? (8).
22. A special type of desktop icon that points to an actual program, and is indicated by a small arrow. (8).
24. Computer screen resolution is measured in: (6).
26. An alternative name for a floppy disk (8).
28. An operating system from Microsoft (7).
29. Name of button that sends a window down to the taskbar (8).
31. When you load an application program, it is copied into (6).
33. File extension for high-quality pictures with small file sizes. (4).
36. To log on to a multi-user computer, you need a user password and a user..... (4).
37. When using an application program, you store your documents on disk as (5).

SUB ICT SIMPLIFIED

SECTION C (40 MARKS)

Answer **any two** questions from this section. Answers must be written in the answer booklet provided.

22. (a) What is **Computer Hardware**? (02 marks)
 (b) Discuss the functionality of any **nine** commonly used peripherals. (18 marks)
23. (a) (i) What is an **Operating System**? (02 marks)
 (ii) Mention any **four** examples of operating systems. (04 marks)
 (iii) Mention any **four** functions of operating systems. (04 marks)
 (b) (i) What is **Application Software**? (02 marks)
 (ii) With examples, state any **four** types of popular application software. (08 marks)
24. (a) State the difference between:
 (i) the **Internet** and the **World Wide Web**. (04 marks)
 (ii) **Search Engines** and **Web Browsers**. (04 marks)
 (b) Give any **four** advantages and any **two** disadvantages of using the internet. (12 marks)
25. (a) (i) What are **Electronic Spreadsheets**? (02 marks)
 (ii) Give any **two** examples of spreadsheet application programs. (02 marks)
 (b) Give any **four** advantages of using electronic spreadsheets as compared to using manual workbooks. (04 marks)
 (c) With examples in each case, state the difference between **Relative** cell references, **Absolute** cell references and **Mixed** cell references. (06 marks)
 (d) Mention any **six** features common to most spreadsheet application programs. (06 marks)

Good Luck

END.

KKK JOINT MOCK EXAMMINATIONS 20XX**Uganda advanced certificate of education****Subsidiary ICT****Paper 1****Duration: 2hours 30minutes****INSTRUCTIONS TO CANDIDANTES**

This paper consists of 20 questions

Answer all the questions

All answers must be written in the spaces provided in this question paper

For Examiner's Use Only

QNS	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Q1 0	Q1 1	Q1 2	Q1 3	Q1 4	Q1 5	Q1 6	Q1 7	Q1 8	Q1 9	Q2 0	TOT AL
MAR KS																					

1. (a) Define the term computer? (1Marks)

.....

- (b) List down any four characteristics of a computer. (1Marks)

i.

ii.

iii.

SUB ICT SIMPLIFIED

iv.

2. (i) What is a computer laboratory? **(1Marks)**

.....

(ii) List down any four devices used for computer care and maintaince. **(4 Marks)**

i.

ii.

iii.

iv.

3. Describe the steps one would take when saving a presentation file with file name BEN in a folder created on the desktop. **(5 Marks)**

.....

4. (a) What do you understand dy the term “removable storage”? **(1Marks)**

.....

(b) Give two advantages and disadvantages of removable storage devices.

Advantages **(2Marks)**

i.

ii.

Disadvantages **(2Marks)**

SUB ICT SIMPLIFIED

- i.
- ii.

5. Briefly describe one way how ICT can be applied in each of the following areas. **(5Marks)**

- i. Education.....
.....
- ii. Healthy.....
.....
- iii. Security.....
.....
- iv. Governance.....
.....
- v. Recreation.....
.....

6. Write short notes on the following terms as applied in ICT.

(5Marks)

- i. EFT.....
.....
- ii. Wi-Fi
.....
- iii. Edutainment.....
.....
- iv. CMOS.....
.....
- v. BIOS.....
.....

SUB ICT SIMPLIFIED

7. Given the types of software in a table below and different tasks they can accomplish fill in the gaps.

Task	Type of software that can be used
.....	Ms-Word
Presenting information in a conference	
.....	Ms-Excel
Reading information on the internet	
.....	Adobe Photoshop.

8. (a) i). Write ISP in full.

(1Marks)

.....

ii). State any two services offered by the ISPs

(2Marks)

.....

(b). State two factors that leads to a slow down of the internet speed.

(2Marks)

.....

9. (a) Give three services enjoyed by schools with networked computer Laboratories over those with stand alone computers .

(3Marks)

- i.
- ii.
- iii.

(b). state the function of each of the following in the process of establishing a network. **(2 marks)**

SUB ICT SIMPLIFIED

(i). Bridge.....

(ii). Firewall.....

10. (a) Define the term computer virus? **(1 marks)**

.....

(b). state four symptoms of a computer viruses. **(4 marks)**

- i.
- ii.
- iii.
- iv.

11. (a). Differentiate between hardcopy and softcopy **(2 marks)**

.....

(b). State two advantages of softcopy over hardcopy. **(2 marks)**

- i.
- ii.

(c). state one device that can be used to convert softcopy into hardcopy. **(1 marks)**

.....

12. (a). Distinguish between worm booting and cold booting? **(2 marks)**

- i.

SUB ICT SIMPLIFIED

ii.

(b). Describe three situations which may require a computer user to perform a worm boot. **(3 marks)**

i.

ii.

iii.

13. Sserubaale wants to start up his computer from the wall socket. Describe the step-by-step process His computer will take in order to show up the desktop. **(5 marks)**

.....

.....

.....

.....

.....

.....

14. (a) Give any two examples of primary storage devices you know. **(2 marks)**

i.

ii.

(b) Briefly explain three advantages of compact disc over Flash disk. **(3 marks)**

i.

ii.

iii.

15. Describe any two circumstances under which computers can be used in a hospital. **(2 marks)**

i.

SUB ICT SIMPLIFIED

ii.

(b). Give three advantages of using computers rather than relying on nurses only in a hospital.

(3 marks)

i.

ii.

iii.

16. Describe any five measures that should be taken to ensure that computers in the computer laboratory are safe.

(5 marks)

i.

ii.

iii.

iv.

v.

17. (a) Define the following terms as applied in DBMS

(3 marks)

i. Primary key.....

.....

ii. Record.....

.....

iii. Field.....

.....

(b). Give any two reasons as to why a company which has adopted to an electronic database management system should continue keeping the manual filling system.

(2 marks)

i.

ii.

SUB ICT SIMPLIFIED

18. Define the following terms as applied in ICT.

(3 marks)

i. Hacking.....

.....

ii. Cracking.....

.....

iii. Piracy.....

.....

(b). Give any two ways of preventing software piracy in developing countries like Uganda. **(2 marks)**

i.

ii.

19. (a). Computer software can be classified into two major classes. Briefly describe them. **(2 marks)**

i.

.....

ii.

.....

(b). Briefly explain the three major categories of systems software.

(3 marks)

i.

ii.

iii.

20. List any five advantages of a monitor over a printer.

(5 marks)

SUB ICT SIMPLIFIED

- i.
- ii.
- iii.
- iv.
- v.

SUB ICT SIMPLIFIED

CANDIDATE NAME:

INDEX NO:

Uganda Advanced Certificate of Education

SUBSIDIARY ICT PAPER 1

(Application of Knowledge)

Duration: 2hrs and 30mins

INSTRUCTION TO CANDIDATES:

Attempt ALL questions in this Part

The maximum time allocated to this paper is 2 hours and 30 minutes

The paper is made up of 20 equally questions.

Write in dark blue or black pen in the space provided on the question paper. All answers should be written in the spaces provided.

No additional materials (answer sheets) are required.

The number of marks is given in brackets () at the end of each question or part question.

The maximum mark for the paper is 100%

SUB ICT SIMPLIFIED

1. Name the input devices A, B, C and D using the words from the list. (4mks)

A B

C

D



Digital camera joystick Light pen Remote control

Scanner Touch pad Tracker ball Web cam

A..... C.....

B..... D.....

2. Ring two items which are input devices. (2mks)

Flash memory card Graph plotter Graphic tablet

Laser printer Magnetic disc Touch pad

3. Tick TRUE or FALSE next to these statements (5mks)

	TRUE	FALSE
Presentation software is used to create slide shows		
Laptop computers do not have USB ports		
DTP software can be used to produce database		
Measurement software is used to create databases		
Buzzers are input devices		

4. Complete each sentence below using one item from the list. (5mks)

A bar code reader A dot matrix printer A graphics tablet

A laser printer A microphone A multimedia projector

SUB ICT SIMPLIFIED

A presence check A speaker A temperature sensor A web cam

- (a).....is used to input data about products at a POS.
- (b).....is used to produce high quality hard copy.
- (c)sends data to a microprocessor in a washing machine.
- (d).....is used to input freehand drawings to a computer.
- (e).....is used to output sounds from a computer.

5. A school teacher is creating a slide show to describe a typical week in her life. **(3mks)**

(a) Tick **three** items of hardware which she would use.

Database	
Key board	
Spreadsheet	
Animation	
Microphone	
Video camera	

6. Draw four lines on the diagram to match the use to its most appropriate output device. **(4mks)**

Use Output device

To produce very large high quality hard copy	dot matrix printer
To display data from a computer on to a large screen	plotter
To alert mobile phone users to incoming calls	multimedia projector
To output multipart forms	buzzer

7. Tick TRUE or FALSE next to each of these statements. **(5mks)**

	TRUE	FALSE
A number pad is used to enter a PIN		
Presentation software is used to write letters		
Database software is used to create queries		
A Graphical User Interface uses icons to represent applications		
Sensors are used to store data		

SUB ICT SIMPLIFIED

8. A school keeps records of every student including details of their names, addresses, phone numbers and dates of birth. Using this information, define the terms;

File.....
.....

Field.....
.....

Record.....
.....

9. A school keeps records of its students, using many different types of data. Explain what is meant by Boolean, alphanumeric and numeric data, using examples of student data.

a) Boolean

Meaning

.....
..... Example
.....

b) Alphanumeric

Meaning

.....
.....
Example

.....

c) Numeric

Meaning

.....
.....

SUB ICT SIMPLIFIED

Example

.....

10. The manager of a bookshop uses a database to store data about the books she sells. This is part of the database.

Title of book	Fiction (Y/N)	Number in stock	Next order due
Full Circle	N	15	20/06/11
Sandi ton	Y	18	08/07/11
Vanity Fair	Y	23	16/10/11
The Old Curiosity Shop	Y	10	23/06/11
Pride and Prejudice	Y	12	23/06/11
Sense and Sensibility	Y	26	02/10/11
The Moon's a Balloon	N	12	21/08/11

- a) How many fields are there in this part of the database? **(1mk)**

- b) How many records are there in this part of database? **(1mk)**

- c) The records shown are to be sorted in ascending order of number of stock. What will be the title of the book which will be the first record in the database after it has been sorted? **(1mk)**

- d) What is the title of the book which has no more than 12 copies in stock and is not Fiction? **(1mk)**

- e) Describe a validation check which could be carried on the Next order due field. **(4mks)**

.....

11. The main memory in computers is becoming very large. Give three reasons why backing storage is still needed. **(3mks)**

SUB ICT SIMPLIFIED

1.
.....
2.
.....
3.
.....

12. A large company has a LAN. The manager of the company wants to allow all the network users to access the internet and communicate with other companies.

- (a) Describe the hardware and software he would need in order to achieve this. **(4mks)**

.....

- (b) Give three disadvantages to the manager of allowing workers to access the internet. **(3mks)**

1.
.....
2.
.....
3.
.....

SUB ICT SIMPLIFIED

13. Describe two safety issues, caused by working with computers. For each safety issue give a method of prevention. Do not use the same method for both answers. **(4mks)**

Safety issue 1

.....

Method of prevention

.....

Safety issue 2

.....

Method of prevention

.....

14. A head teacher wants to send the school magazine to parents. Discuss the advantages and disadvantages of sending it by e-mail or delivering it by hand. **(6mks)**

Advantages

.....

Disadvantages

.....

SUB ICT SIMPLIFIED

15. International companies may use video conferencing instead of face-to-face meetings.

(a) Describe the use of three hardware devices in video conferencing. **(3mks)**

1.
.....
.....
2.
.....
.....
3.
.....
.....

(b) Describe two advantages to the company of such a system. **(2mks)**

1.
.....
.....
2.
.....
.....

16. Rogers is preparing a slide show presentation to aid his speech during a board meeting.

(a) Define the term **slide show**. **(1mk)**

.....
.....

(b) Give **two** reasons why a slide show presentation would be better than just printing out and distributing a newsletter. **(2mks)**

SUB ICT SIMPLIFIED

(i).....

(ii).....

(c) Mention any **two** qualities / features that Rogers should consider while designing his presentation to make it more interesting and suitable. **(1mks)**

(i)

(ii)

(d) Apart from Microsoft PowerPoint, suggest the names of any **two** other application packages that Rogers can choose to install on his PC to help him with this task. **(2mks)**

(i)

(ii)17.The

spreadsheet below has been designed to calculate the Fuel Consumption Rate for 6 cars:

	A	B	C	D
1	Car	Distance (km)	Fuel Consumed (liters)	Fuel Consumption Rate (liters/km)
2	Prado	48	4.0	
3	Datsun	160	9.0	
4	Corona	70	4.5	
5	Benz	?....	20.0	0.1
6	Rav 4	150	33.0	
7	BMW	300	15.0	
8			AVERAGE RATE:	
9			BEST RATE:	

SUB ICT SIMPLIFIED

- (a) What formula should be entered in cell D2 to calculate the Fuel Consumption Rate for the Prado? (1mk)
- (b) The Fuel Consumption Rate for a Benz is known to be 0.1 liters per kilometer. Which formula should be entered in cell B5 to calculate the number of kilometers the Benz will cover with 20.0 liters of fuel?(1mk)
-
- (c) What formula should be entered in cell D8 to calculate the average Fuel Consumption Rate for all the 6 cars? (1mk)
- (d) What formula should be entered in cell D9 to calculate the best (most economical) Fuel Consumption Rate? (1mk)
- (e) If cell B7 was changed to 200, which cells would be automatically updated?(1mk)
- (f) Fuel Consumption Rate is often given in the form; **liters per 100 km**. A column E is to be added to the spreadsheet to contain these new values. What formula needs to be in E7 to find Fuel Consumption Rate of car 6 in liters per 100 km? (1mk)
-
- (g) Apart from performing calculations, outline any **four** other features common to spreadsheet application software packages.
- (i)
- (ii)
- (iii)
- (iv)
18. a) What is a **microprocessor**?(1mk)
-
-
- b) With the help of an illustration, explain the difference between Digital and Analog computers.
- (i) Digital computers. (1mks)
-
-
-

SUB ICT SIMPLIFIED

(ii) Analog computers. **(1mks)**

.....

.....

.....

18. Define the following terminologies as used in word processing: **(5 marks)**

(i) Word Wrap.

.....

.....

(ii) Paragraph spacing.

.....

.....

(iii) Clip board.

.....

.....

(iv) Mail Merge.

.....

.....

(v) Drop cap.

.....

.....

.....

.....

.....

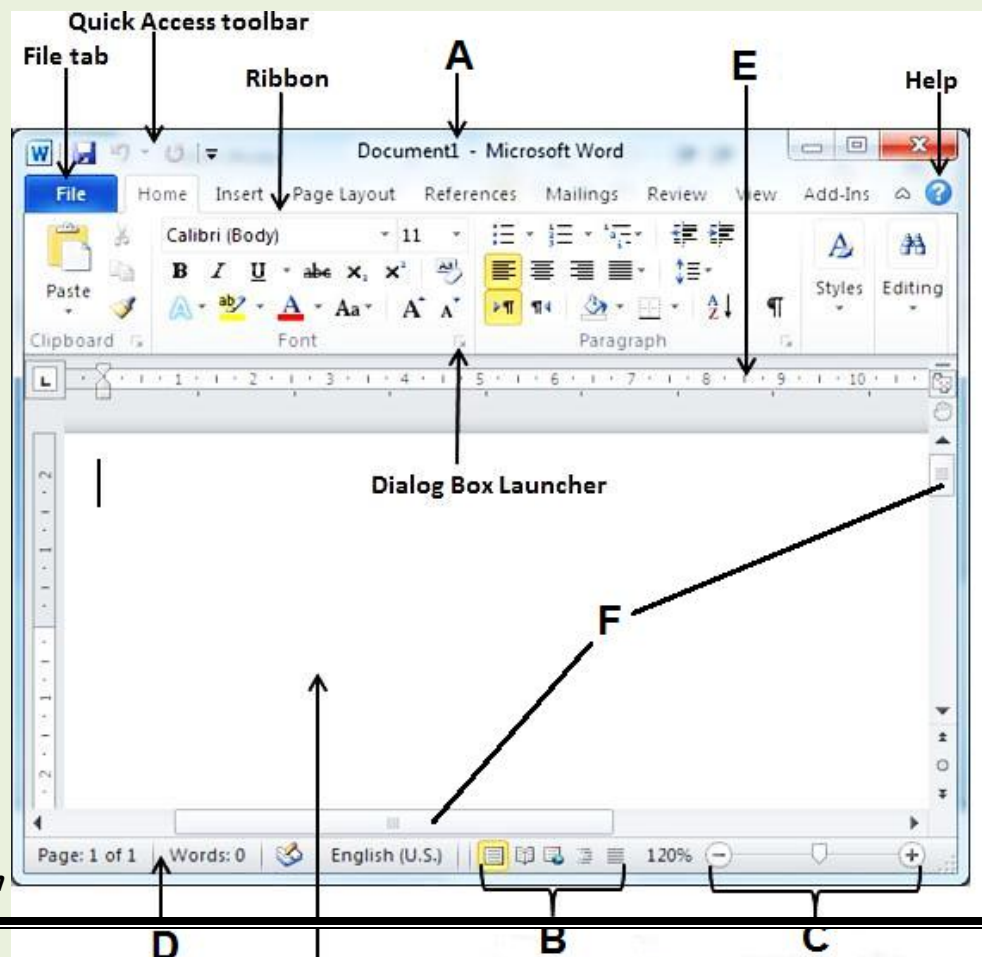
.....

19. The screen shot below shows a word processor's window:

(a).Name the elements labelled: **(3mks)**

B:

If you want something, then



SUB ICT SIMPLIFIED

C:

D:

E:.....

F:

(b) State the keyboard shortcut key combinations for executing the following commands during word processing: **(1mks)**

(i) Double Underline:

.....

(ii) Increasing font size:

.....

(c) Mention one **disadvantage** of using computer based word processors. **(1mk)**

.....

20. Draw two lines on the list to match the software to how it could be used. **(2mks)**

Software**Use**

Desktop publisher

producing a school magazine

Spreadsheet software

managing personal finance

END

•

SUBSIDIARY ICT**S850/1****THEORY****2 hours 30 minutes****UGANDA ADVANCED CERTIFICATE EXAMINATION****SUBSIDIARY ICT****Paper One (Theory)****Duration: 2 hours 30 minutes****Instructions to candidates**

- ✓ Attempt all questions
- ✓ Write the answers in the spaces provided only
- ✓ Be brief and precise when answering
- ✓ Poor handwriting and crossing of work may read to loss of marks

1. (a). (i). What is an e-mail? (1 mark)

.....

- b)How is an e-mail transmitted from your computer to its destination? (2 marks)

.....

- c)List three advantages of using an e-mail over ordinary mail (3 marks)

.....

2. (a). Write the following in full: (3 marks)

i. BCC:

ii. CC:

iii. SMTP:.....

- (b) Under what circumstance may one use BCC or CC during the process of sending an e-mail (2 marks)

BCC.....

SUB ICT SIMPLIFIED

.....CC.....

3. (a). (i). Define the term computer hardware. (1 mark)

.....

- (b) State two examples of hardware devices and state their use (2 marks)

.....

- c) Write short notes on the following: (2 marks)

(i) Bit

.....

(ii) Byte.....

.....

4. a) Distinguish between a storage medium and a storage device and give example of each (2 marks)

Storage medium.....

.....Storage

device.....

.....

- b) Explain how the magnetic technology works in storage devices and give two examples of magnetic storage device (3 marks)

.....

Examples

.....

5. (a) (i). Mention the two types of software programs needed to run a computer efficiently. (2 marks)

.....

SUB ICT SIMPLIFIED

b) Distinguish “customised” and “tailored” software and give an example to each. (3mks)

.....

.....

.....

.....

6. a) State three functions performed by utility software programs. (3 marks)

.....

.....

.....

b) What are graphical user interfaces? (2 marks)

.....

.....

.....

7. (a). (i). Distinguish primary and secondary memory.

Primary.....

.....Secondary.....

.....

b) State two examples of primary memory

.....

.....

c) What is the main function of RAM (1

mark).....

.....

8. a) What is data security (1 mark)

.....

.....

b) Give two ways how data can be accessed illegally on a computer (2 mark)

SUB ICT SIMPLIFIED

.....

.....

c) State two ways one can protect his/her data on a computer. (2 marks)

.....

.....

9. a) What is a browser? (1 mark)

.....

.....

b) Mention two examples of browsers currently in use. (2 marks)

.....

.....

c) Give two advantages of a website to a school. (2 marks)

.....

.....

.....

10. (a). Outline three dangers arising as a result of using computers. (3 marks)

.....

.....

.....

b) Suggest two ways of reducing these effects. (2 marks)

.....

.....

.....

11. a) Your school wants to maintain fees database in Ms. Excel. What advantages will your school have over manual records? (3 marks)

.....

.....

.....

b) Outline two advantages of a screen saver in computing. (2 marks)

SUB ICT SIMPLIFIED

.....

.....

.....

12. a)(i) Differentiate between a **cold boot** and a **warm boot** (2 marks)

.....

.....

.....

b) Name **two** kinds of information stored in the CMOS of a personal computer(2 marks)

- i.
- ii.

c) What is an operating system?

.....

.....

13. Suggest **one** suitable input device for each of the following tasks: (5 marks)

- a) To move the pointer in a GUI.....
- b) To input text/graphics from hard copies into electronic format.....
- c) To read the barcodes of products in a supermarket.....
- d) To record human voice into the computer.....
- e) To control a video game.....

14. a) What is a **computer network**? (1 marks)

.....

.....

b) Mention **two** network types you know (2 marks)

- i.
- ii.

c) List any **two** network media (2 marks)

- i.
- ii.

SUB ICT SIMPLIFIED

15. Below are specifications for **two** different types of computers. Study them and answer the questions that follow:

	Computer A	Computer B
RAM	256KB	128MB
HARD DISK	700MB	20GB
SPEED	800MHZ	1GHZ
COMPUTER WORD	32-bit	64-bit
NETWORK CARD	NO	YES
VIDEO CARD	YES	YES

- a. Which computer is faster in processing? **(01 Mark)**

.....

- b. Which computer has the highest amount of RAM? **(01 Mark)**

.....

- c. Which computer is more likely to be connected to a LAN **(01 Mark)**

.....

- d. Give your reason for the answer in (c) above **(02 Mark)**

.....

16. a) What is a **chart** in spreadsheets? **(02 Mark)**

.....

.....

- b) Define the following terms ad used in spreadsheet **(03 Marks)**

i) Label.....

.....

ii) Values.....

.....

SUB ICT SIMPLIFIED

iii)Function.....

17. a) What is a **computer virus**?

(1 mark)

.....

b) Give **two** examples of computer virus scanning software

(2 marks)

i).....

ii).....d)

Suggest **three** precautions that should be taken to guard against computer viruses

.....

18. a) (i) What is a **peripheral**?

(1 mark)

.....

ii) Give **two** examples of peripheral devices for a desktop computer

(2 marks)

1

2

b)(i) Name **two** input devices that are necessary for conducting a
 videoconference

(2 marks)

1

2.....

SUB ICT SIMPLIFIED

19. a) (i) What is a **search engine**?

(1 mark)

.....

.....

.....

(ii) Give **two** examples of search engines

(2 marks)

1

2

b) State the **two** commonly used LAN topologies

(2 marks)

1

2

20. a)(i) Differentiate between a **word processor** and **word processing**(2 marks)

Word processor.....

.....Word

processing

(b) State **three** merits of using a word processing program (3 marks)

i.

ii.

iii.

END

SUB ICT SIMPLIFIED

SUBSIDIARY ICT
S850/1
THEORY
1 hour 20 minutes

UGANDA ADVANCED CERTIFICATE EXAMINATION**SUBSIDIARY ICT****PAPER ONE (Theory)**

Duration: 1 hour 20 minutes

Instructions to candidates

- ✓ Attempt all questions
- ✓ Write the answers in the spaces provided only
- ✓ Be brief and precise when answering
- ✓ Poor handwriting and crossing of work may read to loss of marks

21. (a). (i). What is an e-mail? (1 mark)

.....

b) How is an e-mail transmitted from your computer to its destination? (2 marks)

.....

c) List three advantages of using an e-mail over ordinary mail (3 marks)

.....

22. (a). Write the following in full: (3 marks)

iv. BCC:

v. CC:

vi. SMTP:

(b) Under what circumstance may one use BCC or CC during the process of sending an e-mail
 (2 marks)

BCC.....
 CC.....

23. (a). (i). Define the term computer hardware. (1 mark)

SUB ICT SIMPLIFIED

.....

 (b) State two examples of hardware devices and state their use (2 marks)

.....

 c) Write short notes on the following: (2 marks)

(i) Bit

.....
 (ii) Byte.....

.....
 24. a) Distinguish between a storage medium and a storage device and give example of each (2 marks)

Storage medium.....

.....Storage

device.....

.....
 b) Explain how the magnetic technology works in storage devices and give two examples of magnetic storage device (3 marks)

.....
 Examples

.....
 25. (a) (i). Mention the two types of software programs needed to run a computer efficiently. (2 marks)

.....
 b) Distinguish "customised" and "tailored" software and give an example to each. (3mks)

SUB ICT SIMPLIFIED

.....

 26. a) State three functions performed by utility software programs. (3 marks)

.....

b) What are graphical user interfaces? (2 marks)

.....

27. (a). (i). Distinguish primary and secondary memory.

Primary.....

.....Secondary.....

.....

b) State two examples of primary memory

.....

c) What is the main function of RAM (1 mark)

.....

28. a) What is data security (1 mark)

.....

b) Give two ways how data can be accessed illegally on a computer (2 mark)

.....

c) State two ways one can protect his/her data on a computer. (2 marks)

SUB ICT SIMPLIFIED

.....
.....

29. a) What is a browser? (1 mark)

.....
.....

b) Mention two examples of browsers currently in use. (2 marks)

.....
.....

c) Give two advantages of a website to a school. (2 marks)

.....
.....

30. (a). Outline three dangers arising as a result of using computers. (3 marks)

.....
.....
.....

b) Suggest two ways of reducing these effects. (2 marks)

.....
.....
.....

End

SUB ICT SIMPLIFIED

Name:.....Centre/Index No:.....

Signature:.....

850/1**S.6****SUBSIDIARY ICT****Paper 1****2½ hours**

UGANDA ADVANCED CERTIFICATE OF EDUCATION

SUBSIDIARY ICT

PAPER I

INSTRUCTIONS TO CANDIDATES:

1. Attempt ALL questions in this paper
2. The maximum time allocated to this paper is 2 hours 30 Minutes
3. Write in dark blue or black pen in the spaces provided on the Question Paper. All answers should be written in the spaces provided.
4. No additional materials (answer sheets) are required.
5. The number of marks is given in brackets () at the end of each question or part question

1.(a) What is a keyboard?**[1mark]**

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*If you want something, then work for it***+256 750810100**

SUB ICT SIMPLIFIED

.....

(b) Name the parts of the computer system below: [3marks]



(c) What is the purpose of the cursor movement keys. [1 mark]

.....

2. You have been appointed a laboratory technician for your school. After taking up your job you find out that the computer laboratory lacks rules and regulations guiding students. The system administrator requested you to design five guidelines for the student. [5 marks]

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SUB ICT SIMPLIFIED

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3.(a)Windows Explorer is an application that provides detailed information about your files, folders, and drives. You can use it to see how your files are organized and to copy, move, and rename files, as well as perform other tasks pertaining to files, folders, and drives. With reference to the above operating system,explain one way of creating a folder on the desktop [3 marks]

.....

.....

(b) Define a file [1mark]

.....

.....

(c) What is a folder [1mark]

.....

.....

4. As a student of computer you have an assignment which you have to save on a storage medium after typing. You have been given the following storage mediums a flash disc, Compact disk ,digital versatile disk, a pen drive.

SUB ICT SIMPLIFIED

(a) Which of the storage medium would you use and why?

[2 marks]

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.....

(b) Write the path for saving an excel file (Ms Excel 2003) on the storage medium chosen

[3 marks]

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5 A shopkeeper uses a stand-alone computer for producing.

- ❖ Order forms for sending to suppliers
- ❖ Records of sales and purchases as evidence for the taxation authorities
- ❖ Leaflets advertising special offers

The computer has a hard drive, a CD- RW drive and a DVD drive. State a different use

by the shop for each of those three storage media. In case say why it's appropriate for

that use

[3 marks]

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SUB ICT SIMPLIFIED

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6.(a) Define a Computer virus.

[2 marks]

.....

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(b) After some time, computers tend to slow down, take long to boot or some times files disappear. As a computer student, how can you rectify these problems? Give any three and back up your answer.

[3 marks]

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7. (a) Which two general precautions should be taken before replacing any non-hot-Swappable laptop component? List any two.

[2 marks]

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(b) A technician wants to create a new partition on a new additional hard drive. Which tool should be used?

[1 mark]

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SUB ICT SIMPLIFIED

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(c) A network administrator has finished migrating from a peer-to-peer network to a new client-server network configuration. What are two advantages of the new configuration? Give any two. [2marks]

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8.(a) All default Windows XP tools and utilities are installed on a computer. The technician discovered that a few other Windows components are still needed. Which procedure can accomplish this task? [1 mark]

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(b) Distinguish between system software and application software. [4marks]

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SUB ICT SIMPLIFIED

9.(a) Computers are not only useful in classes where they are used to enhance the knowledge of learners but enhance other skills and jobs. Name and define five occupation associated with computers. [5marks]

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10.(a) Define the term ICT? [2 marks]

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.....

(b) New research into the potential impact of information and communication Technologies (ICTs) on society as regards Social, Economic and environmental

SUB ICT SIMPLIFIED

has shown negative and positive effects. With respect to the previous statement, identify any three impact of ict on the environment. [3marks]

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11.(a) Define the following terms

[1 mark@]

i) Internet

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ii) Intranet

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iii) Extranet

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(d) Give and explain any four advantages of e-mail over the ordinary postal services.

[4marks]

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SUB ICT SIMPLIFIED

12. Kasule, collects all kinds of music. He wants to record his collection into CD-ROMs, but he is not quite sure what kind of CD-ROM drive and CD-ROMs to buy. Kasule's girl friend, Namagembe is a writer. She does all the composition on her PC. She found out the other day her PC was infected by computer virus, and much of her creations were destroyed. Hoping the same disaster may happen again, she plans to do daily backup onto a CD-ROM. The situation is; that she does not know exactly what drive and disk she should buy. Knowing you're taking a computer course, they go to you for help.

- (a) All Kasule needs is to record the music onto the CD-ROMs once. What kind of drive and disk will you recommend? [3marks]

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- (b) Namagembe's case is slightly different. She needs to write her data on the CD-ROMs repeatedly. What drive and disk will you recommend? [2marks]

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- 13.(a)**What is a database management system? [1mark]

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SUB ICT SIMPLIFIED

(b) Your school hired a systems administrator he found out that the school lacked a school management system. He was required to explain why the school needed

it. As a computer student you are requested to give four reasons why the school should take on a school management information system. [4marks]

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14.Using the diagram below.

	A	B	C	D	E	F
1	ITEM	PRICE	QUANTITY	VALUE		
2	X	2000	180			
3	Y	3000	300			
4	Z	4000	500			
5	TOTAL					
6	MAXMUMPRICE					
7	AVERAGE QUANTITY					
8						
9						
10						
11						
12						
13						
14						

[1mark@]

(a) Write a formula to display the value (price by quantity) for each item in column D

.....

.....

(b) Calculate the total quantity in cell C5

.....

SUB ICT SIMPLIFIED

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(c) Write a formula to calculate the total value and display it in cell D5

.....

.....

(d) Write a formula to display the maximum price in cell B6.

.....

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(e) Write a formula to calculate the average quantity and display in Cell C7

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15.(a)List four top-level domains names.

[4marks]

.....

(b) Which device is used to connect two networks?

.....

[1mark]

.....

16.(a)What is the difference between Animation and Transition?

.....

[2marks]

.....

(b)Which view can be used to insert and test the: Animation effects and

Transition effects

[1mark]

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SUB ICT SIMPLIFIED

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(c) Briefly explain the usage of the *crop option* of the **Picture toolbar**. [2marks]

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17. Form six students are required to create e-mails and send their assessments to the teachers email. It was about Electronic mail which refers to as the sending and receiving of digital messages on a network. Explain any five features of e-mail.

[5marks]

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18.(a)Distinguish between customized software and off shelf(canned) software

[2marks]

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(c) Give three merits of customized software. [3marks]

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SUB ICT SIMPLIFIED

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19. Global Business Centre is interested in building a state of art modern electronic office in Kampala. You have been approached to explain the various benefits and dangers associated with electronic networking. Using your knowledge of information technology, explain three advantages and two disadvantages that Global Business Centre would derive from adopting an electronically networked business

Advantages

[3marks]

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.....

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Disadvantages

[2marks]

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20. Your father has been working in the ministry of public service for the last twenty

SUB ICT SIMPLIFIED

years. At the beginning of this year the permanent secretary requested them to acquire

Computer skills. Last week when he went to his office he found a computer on his

desk ,being computer illiterate he does not know the application of information

communication technology. You are tasked to advise him on the areas of application

of computers. Give any five areas.

[5 marks]

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ICT REVISION QUESTIONS

1. (a) What is the main reason for using BIOS by the computer
(b) Mention two types of buses that affect the speed of computer
(c) List four instances where a computer will require a warm boot
(d) What reason would you give for buying a new video card
2. (a) State four ways of keeping data secure for an organization
(b) Mention any three ways how data can be entered into the computer
(c) Outline four causes of fire accidents in a computer lab
(d) What is a computer peripheral? Give three examples
3. (a) Define the following terms as used in spreadsheets application
(i) A cell (ii) workbook, (iii) worksheet, (iv) range, (v) relative address (vi) absolute address
(b) State any five features of PowerPoint
(c) What is the difference between Freeware and system software
(d) What is meant by the following terms as used in database (i) Primary key, (ii) field size (iii) Table
(e) Give five examples of data types used in database
4. (a) Mention two practical uses of screen saver to a computer user
(b) Give the advantages and disadvantages of GUI
(c) What is a computer port?
(d) What type of ports where you can connect the following devices
(i) Keyboard (ii) PS/2 mouse (iii) Printer (iv) Monitor (v) Network device
5. (a) Distinguish between a webpage and website
(b) Write the following in full; (i) BCC (ii) CC (iii) URI (iv) URL
(v) DNS (vi) HTTP (vii) FTP (viii) RSS (ix) POST
(c) List any two ways the internet can be used to improve education
(d) Mentions the tasks carried out by a network administrator
6. (a) State reasons why it is necessary to use internet in business
(b) State the factors which influence the speed access to the network
(c) What is the difference between internet and intranet?
(d) What are the requirements to connect to the internet

SUB ICT SIMPLIFIED

7. (a)What is a printer
(b) Distinguish between impact and non impact printers and give examples of each
(c) What are the advantages of laser printers over dot matrix printers?
(d) Give the advantages of LCD over CRT
8. (a)Mention the uses of computers in the area of communication
(b) State any five demerits of using computers to our society
(c) What are utility programs?
(d) Explain five utility programs that you know
9. (a)What is a computer virus?
(b)Give examples of computer virus scanning software
(c)Give five ways how computer viruses are spread
(d) Suggest three precautions that should be taken to guard against computer virus
10. (a) What is the difference between data and information
(b) What is the function of the CPU in a computer system?
(c) Give three components of the CPU and state their roles
(d)Distinguish between primary and secondary storage.
11. (a)What is the difference between internet and intranet
(b)What are the requirements to connect to the internet?
(c)What are the advantages of e-mail over ordinary mail?
(d)Distinguish between e-mail and e-mail address
(e)Outline the limitations of using e-mail as means of communication
12. (a)What the advantages of a school having its own website
(b)What is a computer network?
(c) Mention four types network types you know
(d) List any two network media
(e) What is a browser? Give five examples of browsers
13. (a)What is computer literacy
(b) Give reasons why computers should be regularly serviced
(c) What is the difference between a file and folder?
(d) What is a file extension
(e) Identify the file extension of the following; (i) Word document (ii)Batch files (iii)Executable files (iv)Presentation files (v) Database file

SUB ICT SIMPLIFIED

- (vi) PFD files (vii) Spread sheets
14. (a) Define the term Blue tooth
 (b) Name three networking protocols
 (c) What is a network topology?
 (d) Give three network topologies you know.
15. (a) Define electronic publishing and give three examples of DTP software
 (b) The table below displays information about some KB employee's data. Study it and answer the questions below.

ID NUMBER	EMPLOYEE NAME	DOB	SEX	TELEPHONE NUMBER	AVAILABILITY
U01	ARINDA	23/11/68	F	07724311224	Y
U02	RUTH	03/03/79	F	07745823692	N
U03	NAMBA	12/01/85	M	07562310452	Y
	PERUTH				
	MAKUBA				
	ISAAC				

- (i) What software can use to store that data
 (ii) Name the data types for the following entries; ID NUMBER, EMPLOYEE NAME, DOB, SEX, TELEPHONE, AVAILABILITY
 (c) What is Boolean data
16. (a) Explain the following terms; (i) VOIP (ii) Blog (iii) GPS
 (iv) Peer-to-peer (v) Client/server (vi) Attenuation
 (b) What is the use of the following equipments in the computer lab; (i) Blower (ii) Antiglare screens (iii) Water proof covers
 (iv) Fire extinguishers (v) UPS (vi) Surge protector
17. (a) In reference to the following file path: **C:\user\desktop\Adul\exam.xls** identify the following
 (i) File name (ii) File type (iii) Profile folder (iv) Location
 (b) Study the URL and answer the questions that follow
http://www.ecdl/office_tutorials/ms_word/features_of_word
 (i) What does http stands for?
 (ii) The domain name is
 (iii) Which is the **Top-Level Domain** (TLD) in this URL
 (iv) What does **"office_tutorials"** represent in this URL?
 (v) What is the document name which is currently opened on this website?

SUB ICT SIMPLIFIED

18. (a) An e-mail is given as learningcomputer13@gmail.com identify each part of this address

(i) Learningcomputer13 (ii) @ (iii) Gmail (iv) .com

(b) What is green computing?

(c) What are the advantages of green computing

(d) The following is a mark sheet of senior four Ntandamo High school

	A	B	C	D	E
1	NAMES	HISTORY	ICT	CHEMISTRY	ART
2	Kabanda	34	76	75	85
3	Kainembabazi	67	56	45	74
4	Awori	98	45	45	42
5	Mawanda	45	63	12	42
6	Nakate	52	87	52	78

(i) Write a formula to calculate the following

(ii) Total marks of Kabanda

(iii) Average score of Nakate

(iv) The highest and lowest score of Awori

(v) Write the formula to count all the ICT students with marks greater than 60

(vi) Write the values in the range of B2:D5

Practical

19. How best to produce a professional word document

20. How best to do with data entry and manipulation in spread sheets

21. How to make the best presentation using Ms. PowerPoint

22. Creating and entering data into a database, creating tables, forms, queries and reports

23. Creating the best design using electronic publishing software like publisher.

SUB ICT SIMPLIFIED

ICT GRAND SEMINER

1.(a) Write down the formula to calculate the following

A	B	C	D	E	F	G	H	
1 item	Jan	Feb.	mar	total	average			
2								
3 Food	72	72	106					
4 Electricity	24	37	66					
5 Telephone	96	90	114					
6 stationary	48	38	64					
7								
8 total								

(i) Total expenses for electricity
=SUM(B4:D4) or =(B4+C4+D4)

(ii) Average expenses for stationary

=AVERAGE (B6:D6)

(iii) The highest expenses for telephone
=MAX(B5:D5)

(a) write down a formula to count all the item with the expense greater than 70 dollar
=COUNTIF(B3:D6,">70") OR +COUNTIF(B3:D6,">70")

2. The spread sheet table below was created using MS excel and shows part of sub ICT results for

Some students of s6 in a certain school in an end of term exams

M24	fx							
	A	B	C	D	E	F	G	H
1								
2	NAME	S850 /1	S850 /2	S850/1 OUT OF 40%	S850/2 OUT OF 60 %	TOTAL	GRADE	POSITION
3								
4	Ann	55	80	22	48	70	D1	2
5	Peter	57	59					
6	James	43	50					

SUB ICT SIMPLIFIED

7	Kezia	73	80					
8	David	59	59					
9	Mariam	66	70					
10	Suzie	49	40					
11	Gift	50	57					

(a) suggest two other spread sheet programs that could be used to create the records besides MS excel

- ✓ Lotus 1-2-3
- ✓ Visi Calc
- ✓ Open Office.orgCalc
- ✓ Jazz
- ✓ Symphony
- ✓ Vp planner

(b) Which function is suitable for grading students in column G

The IF function

(c) (i) write down a formula for creating position for students in column H basing on total

=RANK (F4,F\$4:F\$11,0)

(ii) State the tool you would employ to complete the positioning for all students

The auto fill/ copy hand outs

3. The table below shows part of a data base. Use it to answer questions that follow.

STUDENT ID	Name	Math	Eng	Scie	SST	Total marks
014/KSS/005	Sadic	80	60	78	46	
014/KSS/102	Debra	90	81	72	60	
014/KSS/009	Sharif	56	92	66	46	

(a) What is most appropriate primary key ? (1 mark)

STUDENT ID

(b) (i) Write down the criteria for students whose name ends with letter O and have a mark in math greater than 70 (01 mark)

Name :like "O", math >70.

(ii) Write down a criteria for students whose names begin with letter K to S and have a mark in science less than 68 (01 mark)

Name: Like "[K-S]", scie<68*

(c) The formula entered in design view for total marks is

Total : [Math]+[Eng]+[Scie]+[SST]

SUB ICT SIMPLIFIED

- (d) Write down an input mask for STUDENTID to contain KKK and incremental by one
"KKK1"

4. The table shows parts of database for employees of a certain company.

EMPLOYEES ID	FIRST NAME	LAST NAME	SEX	BASIC SALARY (S)	DEPART
K001/04	Sadick	Walusimbi	M	25000	Security
K007/088	Debra	Basemera	M	30000	Security
K013/098	Amina	Zarwango	M	20000	Messenger
K009/003	Dave	Ragii	F	40000	Council
K156/145	Arafat	Bukenya	F	58000	Council
K345/001	Jemimah	Namatovu	M	10000	Cleaner
K728/024	Bernard	Mubangizi	M	33000	Technical

- (a) Write the data type in the following fields.
- (i) *Text* **SEX**
- (ii) *ID number* **EMPLOYED**
- (a) How many records are there in the table
7 records (number of rows used for data entry)
- (b) What is the suitable primary key?
EMPLOYEE ID
- (c) Write down the query for employees in the security department and council department.
"council" or "security"

5. The manager of a bookshop uses database to store data about the books she sells. This is part of the database.

Title of the book	Rating	Number in stock	Next order due
Shakespear	G	15	20/12/2014
Vanity Fair	E	25	14/10/2014
Sandton	G	21	15/25/2014
Matrix	E	05	

- i) Suggest an appropriate field for the primary key

Title of Book

- ii) Suggest the appropriate data types for

- a) Book title

Text

- b) Rating

Text

- c) Number in stock

Number

- d) Next order due

Date/Time

6. (a) What are programming languages?

These are artificial languages designed to communicate instructions to a machine, particularly a computer.

- (b) **Mention the four classifications of programming languages.**

- ✓ *Machine language*
- ✓ *Assembler*
- ✓ *High level language*
- ✓ *Fourth generation language*

7. state the functions of the following utility programmes.

- (i). **Disk defragmentation**

This moves fragments of files to one location

- (ii). **File compression**

This compresses or uncompressed files to save storage space

SUB ICT SIMPLIFIED**(iii). Diagnostic**

This provides technical information showing identified problems in the system

(iv). Uninstaller utility

This removes application software and its associated entries within the file system.

(v). Network utility

These analyse the computer network connectivity, configure network settings, check data transfer or log events.

8. Distinguish between custom made software and off the shelf software.

Custom made software (special/ tailor made software) is software written to meet the needs of an organization that cannot be usually satisfied by another source of software

While

Off the shelf software (general purpose/ packed software) is software which may be used for a wide variety of purposes e.g. word processor , spread sheets, database and presentations.

9.Explain the following terms as used in presentation.**(i). Template**

This refers to predefined in built format and colour scheme to be applied to a presentation to give a fully design professional look.

(ii). Animation

These are visual effects applied to individual items in a slide as graphical images, titles or bulleted points. These help to stimulate action during a presentation.

(iii). Transition

This is the visual movement in a given slide show. It is the method of moving one slide off the screen and bringing on another slide during a slide show.

(iv). Master slide

This is a parent slide that contains text and object that will appear on all other slides such as headers & footers, date & time.

- (b) **Name any other presentation softwares besides MS powerpoint**
 ✓ *Open office presentation*
 ✓ *Lotus freelance*
 ✓ *corel presentation*
 ✓ *Harvard graphics*

10. what is meant by cell references?

This is the location/ address of a cell in a worksheet e.g B5, F9,DG43

- (b) **Distinguish between relative cell Reference and Absolute cell reference.**

Relative cell reference is a reference to cells relative to the formula

While

Absolute (fixed) cell reference is that always refers to cell in a specified location

- (c) **State the function of the following as used in spreadsheets**

- (i). **Auto fill / copy handle tool**

This is a feature that automatically fills in cell, create a row of dates and even copy formatting from one cell to the next.

- (ii). **Freezing and unfreezing panes**

This is a feature that enables heading of a large worksheet to be viewed as you scroll through a work sheet. In this case headings also do not disappear as you scroll through.

11. Distinguish between

- (i) **Validation rule and validation text**

Validation rule gives an expression that must be true whenever you add or change a value in a field "F" or "M" while validation text is a text that appears when a value violates the validation rule e.g. required to enter F or M for this field.

- (ii) **Data validation and data verification**

*Data validation involves checking data input against expected /predetermined data limits coded in the computer **while***

Data verification involves checking data manually by comparing input data with source data.

12.(a) Explain the following as used in presentation

- (i). **Hyperlink**

A hyper link is a connection from one slide to a web page, email address , slide or file.

Text or objects such as pictures and shapes can be formatted as a hyper link

(ii). Action button

Action buttons are built shapes that you add to a presentation and use as a hyperlink thus when one clicks or moves the button, the action can occur

(iii). Handouts

These are copies of a presentation that are distributed to an audience for note taking and reference

(c) State two application areas of presentation software.

- ✓ *They are used in e-learning and instructional design*
- ✓ *They are used in tutorials and instructions*
- ✓ *In businesses presentation*
- ✓ *In events and project documentations*

END OF SESSION ONE

SHORTCUTS	USE	SHORTCUT	USE
CTRL + A	Select All	ALT + W	view
CTRL + B	Bold	ALT + H	Home 188
CTRL + C	Copy	ALT + N	Insert
CTRL + D	Font	ALT + P	Page layout
CTRL + E	Center Align	ALT + S	Reference
CTRL + F	Find	ALT + M	Mailings
CTRL + G	Go to	ALT + R	Review
CTRL + H	Replace	ALT + F	File (office button)
CTRL + I	Italics	ALT + 1	Save
CTRL + J	Justify	ALT + 2	Undo
CTRL + K	Insert hyperlink	ALT + 3	Redo
CTRL + L	Left Align	ALT + CTRL + F	Insert footnote
CTRL + M	Indent Paragraph from left	ALT + CTRL + D	Insert endnote
CTRL + N	New document	ALT + CTRL + S	Split document window
CTRL + O	Open document	ALT + SHIFT + C	Remove document window split
CTRL + P	Print	ALT + CTRL + Y	Repeat find
CTRL + Q	Remove Paragraph formatting	ALT + CTRL + P	Print layout view
CTRL + R	Right Align	ALT + CTRL + O	Outline view
CTRL + S	Save	ALT + CTRL + N	Draft view
CTRL + T	Create a hanging indent	ALT + SHIFT + →	Promote paragraph
CTRL + U	Underline	ALT + SHIFT + ←	Demote paragraph
CTRL + V	Paste	ALT + CTRL + I	Switch in or out print preview
CTRL + W	Close	ALT + CTRL + M	Insert a comment
CTRL + X	Cut	ALT + SHIFT + O	Mark table of contents entry
CTRL + Y	Redo	CTRL + SHIFT + G	Open Word Count
CTRL + Z	Undo	ALT + SHIFT + X	Mark an index entry
CTRL + shift + >	Increase font size	ALT + CTRL + F	Insert footnote
CTRL + shift + <	Decrease font size	ALT + CTRL + D	Insert endnote
CTRL + Page up	Move to the previous edit location	ALT + F3	Create new building block
CTRL + Page Down	Move to the next edit location	ALT + SHIFT + R	Copy header or footer

SUB ICT SIMPLIFIED

CTRL + Home	Move to first preview page	ALT + CTRL + C	Insert copyright symbol
CTRL + End	Move to last preview page	ALT + CTRL + T	Insert trade mark symbol
CTRL + Backspace	Delete word to the left	CTRL + [	Decrease font size by 1 point
CTRL + Delete	Delete word to the right	CTRL + 1	Single space lines
CTRL + F3	Cut to the spike	CTRL + 2	Double space lines
CTRL + F9	Insert a field	CTRL + 5	Set 1.5 line spacing
CTRL + Enter	Insert a page break	CTRL + 0	Add/remove line space
CTRL + Shift + Enter	Insert column break	ALT + SHIFT + D	Insert a date field
CTRL + →	Move one word to the left	ALT + SHFT + T	Insert time field
CTRL + ←	Move one word to the left	CTRL + ALT + DEL	Warm bootingz
CTRL +]	Increase the font size by 1 point	SHIFT + F2	Copy text

PAPER TWO

sMICROSOFT ACCESS { DATABASE}

1. Create a table with the information below and name it "PAYROLL". Use the appropriate types.

ID	NAME	SEX	POST	DATE OF BIRTH	AMOUNT
C001	Nassolo Shamirah	F	Technician	03-May-1998	50,000
C002	Wampamba Ivan	M	Programmer	12-June-1992	25,000
C003	Kakai Anna Mary	F	Engineer	02-Nov-2006	20,000
C004	Ngabo Peter	M	Finance	22-May-1984	60,000
C005	Nabagulanyi Prossy	F	Technician	12-July-1997	32,000
C006	Zalwango Amina	F	Programmer	18-March-1996	90,000
C007	Bigirwa C Zamba	M	Finance	04-July-1988	45,000
C008	Ntambi Arnold	M	Engineer	15-May-1999	35,000
C009	Mpyangu Moreen	F	Technician	27-June-1984	20,000

- You are required to design a database to manage the above information. Name the database as "Employee Database".
- Create a form called "Employee entry form" and enter the above records.
- Insert a clip art and date in the Employee entry form.

SUB ICT SIMPLIFIED

- d) Create a query to show all the female employees who are programmers and name it "Programmers".
- e) Create a query showing male employees only.
- f) Create a query showing all employees who earn a salary between 45,000/= to 15,000/=.
- g) Create a query to show the employees who earn above 40,000/=.
- h) Give the table a background colour of yellow.
- i) Create a report showing all records. Include your name as footer.
- j) Format the report heading with font Microsoft sans serif (Ms sans serif).
- k) Arrange the names in ascending order.
- l) Print your work.

QUESTION B

Using any database software do the following.

Create a database and save it as "Students".

A) In the students database enter the data below and save your table as "Status".

Index	Name	District	Snap	Date of birth	Class
201	Ali	Kampala		15/Jun/1988	2
202	Afande	Masaka		23/Jan/1988	3
206	Joel	Masaka		01/Dec/1988	4
204	Rashid	Masaka		12/Aug/1988	2
203	Jacob	Kampala		24/Jul/1988	3

B) In the students database create another table for the following and save it as "Fees"

Index	Total Fees	Paid	Balance
201	£ 100000	£ 5000	
202	£ 100000	£ 100000	
206	£ 100000	£ 75000	
204	£ 100000	£ 0	
203	£ 100000	£ 75000	

- (a). Relate the two tables
- (b). Create query to:
 - Display names of people from Masaka who have paid more than £ 50000 and save it as "paid"
 - Display names of people born between 01/may/1988 and 30/Sept/1988 and save it as "age"
 - Calculate balances and save it as "balance"
- (c). Design a form for fees to enter paid, balance and index and save it as "formx" it should have your name as footer.

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- (d). Create a report which displays balance, names and index with name as the header and save it as “report”

DESKTOP PUBLISHING SOFTWARE (Publisher)

2. Using any publication software design a certificate that should be awarded to senior six candidates of your school who have excelled in the following computer applications.

1. Ms Word.
2. Ms Excel.
3. Ms Access.
4. Ms Power point.
5. Ms Publisher.

Additional Information:

- ± Heading should be the name of the school.
- ± Your name as owner of the certificate.
- ± Insert signatories for the Head teacher and HOD ICT.
- ± Use appropriate font colors, designs and styles.
- ± Insert a textural background of your choice.
- ± Insert a logo of your choice.
- ± Save your work as application-your name e.g. Application Amina.
- ± Print your work.

Question B

Using any Desktop Publishing Software, design Eight identical Business Cards having the following details.

Name of the business	:	LUNAX (INC) INVESTMENTS
Position held	:	Chief Executive Officer
P.O. Box	:	261 Kampala
Plot No.	:	26 KIMATHI AVENUE
E-mail	:	Your e-mail
Name	:	Your Name
Tel	:	Any Imaginary Telephone contact
Items Dealt in	:	Computer and Accessories

- (a). Include a Logo of your choice.
- (b). Use 2 background colors.
- (c). Use a boarder Art for your publication.

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- (d). Save your Presentation as your name.
- (e). Set the target sheet size to letter ("8.5×11").
- (f). Set side margins to 0.7.
- (g). Save and print your publication.

3. PRESENTATION SOFTWARE (MS PowerPoint)

Question 1

2. You are going to participate in a computer studies seminar and you have been told to prepare a presentation about the three categories of computers: Analog, digital and hybrid.

In your research session, you get the following notes:

Introduction

Computers can be categorized according to the process they used to represent data.

People communicate through speech by combining words into sentences.

Human speech is analog because it uses continuous (wave form) signals that vary in strength and quality.

Most computers are digital.

Analog computers

Analog Computer is a computing device that uses continuously changing values to represent information.

It generally deals with physical variables such as temperature, weight, voltage, pressure, speed, etc.

Examples of Analog computers include: Thermometers, Weighing scale, Voltmeters, barometers, Speedometers, etc.

Digital Computers

These are computers that use discrete (discontinuous) values, specially binary digits (0, 1) to represent data.

The state of being on is represented by 1 and off is represented by 0.

Digital computers are more accurate and work at a very fast rate.

Examples of Digital Computers Include: All Personal computers (PCs) , Laptops, Digital Watches, e.t.c

Hybrid Computers

A hybrid computer combines the desirable features of analog and digital computers.

A digital computer that accepts analog signals converts them to digital for processing.

Hybrid computers are mainly used for specialized tasks.

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Examples of Hybrid computers, include: Digital petrol pumps, Digital Speedometers devices used to measure the patients temperature, blood pressure In Hospitals' Intensive care units.

Use the following features ;

- Appropriate animation schemes to view the presentation
 - Appropriate transition for the show
 - Clip Art and Word Art where applicable
 - Appropriate design templates
 - Footer bearing your name
- (i) Create a Five slide presentation in which slide one is the title slide, which will have a suitable title, your name and an automatic date.
 - (ii) Set the following master styles; **Titles:** Dark blue, Right aligned, Font Tahoma **Bold**
Text styles: Black, Arial Narrow **Bold**
 - (iii) Use suitable background, slide transitions and text animations.
 - (iv) Your presentation should run automatically after 2 seconds
 - (v) Insert a Header with your Name and personal number and Footer with the and current Date
 - (vi) Include your name and index number on all pages except the title slide.
 - (vii) Put navigation tools with a hyperlink on each slide as Home, Next slide, Previous slide, and End slide
 - (viii) Set slides to have suitable backgrounds
 - (ix) Save as **Computer Categories**.ppt and print handout slides./

WORD PROCESSINGQuestion 1

Mrs. Kakai Anna is a secretary of Booster Trading Company Ltd and her company wants to invite shareholders for the Annual General Meeting (AGM). The contents of the letter are to be similar to all shareholders except for each shareholder's personal details and titles.

Mrs. Kakai Anna has prepared the letter below to be sent to various shareholders.

BOOSTER TRADING COMPANY LIMITED

P.O. Box 5678, WAKISO.

24th/11/2020

TO: [title] [shareholder's name]

[Address]

[Telephone contact]

Dear Sir/Madam

RE: ANNUAL GENERAL MEETING SCHEDULED FOR FRIDAY 30TH NOVEMBER 2013.

You are cordially invited to attend the AGM without fail on the date indicated above starting at 10:00am. In this meeting we shall elect new office bearers. Each shareholder is expected to pay non-refundable amount of [fee] to cater for elections.

Agenda for the meeting is attached hereon.

See you there.

Yours faithfully,

MUDONDO HOPE

SECRETARY

C.C Managing Director

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Below is the table showing a list of various shareholders to receive the above letter.

Title	Shareholder's name	Telephone contact	Address	Fee
Mr. Dr.	Nsubuga Rashid	254 752-456 678	570, Nairobi	380,000/=
Mr. Rev.	Mulondo Steven	256 774-678 908	320, Kampala	165,000/=
Rev. Dr.	Mukiibi Kenneth	256 772-456 234	321, Luwero	245,000/=
Mrs.	Nassolo Shamirah	256 313-678 098	567, Wakiso	120,000/=
Mr. Br.	Ntambi Arnold	256 752-453 576	765, Kalule	200,000/=

Required;

- | | |
|--|--|
| <p>(i). Create the table and save it as Shareholde-Your Name in your folder.</p> <p>(ii). of your name and index no. on the letter.</p> <p>(iii). an appropriate Word Processing feature, link the table to the letter.</p> <p>(iv). letters and save your file in the folder as shareholders' letters.</p> <p>(v). right margins to 5 inches.</p> <p>(vi). Watermark of the name of your school in the entire document.</p> | <p>Type the letter</p> <p>Include a footer</p> <p>Using</p> <p>Print all the 5</p> <p>Set the left and</p> <p>Insert a printed</p> |
|--|--|

END

SUB ICT SIMPLIFIED

Sub-ICT Simplified is designed to meet the needs of Secondary 'A' level students preparing for UNEB.

It fully covers form **5 & 6 topics** in the **NCDC** curriculum for the Advanced Level (A Level). It can also be used by teachers as their Revision Notes because of its simplified topics and revision questions.

This book consists of the following **chapters/ topics**: Introduction to Computers, Computer Systems, Computer Software, Data communication and Networking, Internet and the WWW, Computer Ethics and data security.

The book also contains **past papers** to enable students know the areas where most UNEB Questions come from.

This book is also designed to guide teachers in question setting and also guide students in their revision and question attempting.

Mukeere Tonny Birungi is knowledgeable in the field of **Computer Science** with a lot of ICT and Computer Studies resources. And he has been teaching computer studies and ICT for the past **Eight** years. He is currently working as a **Digital Ambassador** at **Mount of Olives College Kakiri**.

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By Mukeere Tonny Birungi

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